

O‘ZBEKISTON RESPUBLIKASI
OLIY TA’LIM, FAN VA INNOVATSIYALAR
VAZIRLIGI
FARG‘ONA DAVLAT UNIVERSITETI

SONLI USULLAR
(O‘QUV QO‘LLANMA)

I QISM

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Mazkur qo'llanma universitetning amaliy matematika yo'nalishlari talabalari o'quv rejalariga kiritilgan sonli usullar fani dasturlariga asosan berilgan mavzularni o'z ichiga olib, amaliy matematika talabalari uchun mo'ljallangan. Unda sonli usullar fanining tenglamalarni, tenglamalar sistemasini taqribiy yechish, matritsaning xos son va unga mos xos funksiyasini topish, funksiyalarni interpolyasiyalash, aniq integrallarni taqribiy hisoblash bo'limlariga oid mavzularining nazariy qismi berilib, ularga doir misollar yechilgan hamda C# dasturlash tilida dasturlari keltirilgan bo'lib, mustaqil yechish uchun misol va masalalar berilgan.

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Qo'llanma Oliy ta'lim, fan va innovatsiyalar vazirligi tomonidan oliy o'quv yurtlarning 60540200–"Amaliy matematika"yo'nalishi talabalari uchun o'quv qo'llanma sifatida tansiya etilgan.

No

SO‘Z BOSHI

Hozirgi kun talabiga javob beradigan mutaxassislar tayyorlash, ularning nazariy va amaliy masalalarni chuqur o‘zlashtirishlariga yordam beradigan darsliklarni, o‘quv qo‘llanmalarni o‘zbek tilida lotin alifbosida yozish muhim masalalardan biridir.

Bu o‘quv qo‘llanma amaliy matematika yo‘nalishining o‘quv rejasiga kiritilgan sonli usullar fani mashg‘ulotlariga bag‘ishlangan.

O‘quv qo‘llanmada algebraik va transendent tenglamalarni, chiziqli va chiziqsiz tenglamalar sistemasini taqribiy yechish, matrisaning xos son va ularga mos xos vektorlarini taqribiy topish, funksiyalarni interpolatsiyalash masalasini, aniq integrallarni taqribiy hisoblashga bag‘ishlangan. Algebraik va transendent tenglamalarni, chiziqli va chiziqsiz tenglamalar sistemasini taqribiy yechish, matrisaning xos son va ularga mos xos vektorlarini taqribiy yechish usuli, funksiyalami interpolatsiyalash masalasi, aniq integrallarni taqribiy hisoblash mavzulariga oid nazariy va amaliy misollar orqali ko‘rsatilgan.

Mazkur o‘quv qo‘llanma Oliy o‘quv yurtlari amaliy matematika yo‘nalishi talabalariga "Sonli usullar" fani dasturlariga asosan olib borilgan nazariy va amaliy mashg‘ulotlar asos qilib olindi.

Shuningdek, o‘quv qo‘llanmani yozishda bir qator ustozlarimizni darslik va o‘quv qo‘llanmalari, ma‘ruza matnlaridan foydalanildi.

Qo‘llanma universitetlarning amaliy matematika yo‘nalishi talabalarining sonli usullar fanidan olgan bilimlarini mustahkamlashga va fanni chuqur o‘zlashtirishlariga yordam beradi degan umiddamiz.

I BOB.

XATOLIKLAR NAZARIYASI

1.1-§. Aniq va taqribiy sonlar haqida tushuncha

Kundalik hayotimizda va texnikada uchraydigan ko'plab masalalarni yechishda turli sonlar bilan ish ko'rishga to'g'ri keladi. Bular aniq yoki taqribiy sonlar bo'lishi mumkin. Aniq sonlar biror kattalikning aniq, qiymatini ifodalaydi. Taqribiy sonlar esa biror kattalikning aniq qiymatiga juda yaqin bo'lgan sonni ifodalaydi. Taqribiy sonning aniq songa yaqinlik darajasi hisoblash yoki o'lchash jarayonida yo'l qo'yilgan xatolik bilan ifodalanadi.

Masalan, kitobda 738 ta varoq, auditoriyada 30 ta talaba, uchburchakda 3 ta qirra, telefon apparatida 10 ta raqam, 738, 30, 3, 10 aniq sonlar. Yer bo'lagining perimetri 210 m, yerning radiusi 6000 km, Qalamning og'irligi 8 g. Bu o'lchamlar, ya'ni 210, 6000, 8 taqribiy sonlar. Bu kattaliklarning taqribiy bo'lishlariga sabab, o'lchov asboblarning takomillashmaganligidir. Mutlaq aniq o'lchaydigan o'lchov asboblari yo'q bo'lib, ulardan foydalanganda ma'lum xatoliklarga yo'l qo'yiladi.

Bundan tashqari, yer aniq shar shaklida bo'lmaganligi tufayli, uning radiusi taqribiy olingan. Uchinchi misolda esa qalamlar har xil bo'lganligi uchun ularning og'irligi turlicha. 8 g deb o'rtacha qalamning og'irligi olingan.

Amaliyotda taqribiy a son deb, aniq qiymatli A son dan biroz farq qiladigan va hisoblash jarayonida uning o'rnida ishlatiladigan songa aytiladi.

Qisqalik uchun bundan keyin aniq qiymatli son o'rniga aniq son, kattalikning taqribiy qiymati o'rniga taqribiy son deb yozamiz.

Amaliy masalalarni yechish asosan quyidagi ketma-ket qadamlardan iborat:

- 1) yechilayotgan masalani matematik ifodalar orqali yozish;
- 2) qo'yilgan matematik masalani yechish.

Tabiatda uchraydigan masalalarni doim ham aniq matematik tilda ifodalash mumkin bo'lmaganligi tufayli masala ma'lum darajada ideallashtirilgan model vositasida yoziladi, ya'ni xatolikka yo'l qo'yiladi (birinchi qadamda).

Masalaning tarkibiga kirgan ba'zi parametrlar tajribadan olinganligi tufayli, bunda ham xatolikka yo'l qo'yiladi. Bularning yig'indisi esa boshlang'ich informatsiya xatoligini keltirib chiqaradi.

Juda ko'p xollarda matematik masalaning (ikkinchi qadam) aniq yechimini (analitik) topishning iloji bo'lmaydi. Shuning uchun amaliyotda taqribiy matematik usullar qo'llaniladi. Aniq yechimning o'rniga taqribiy yechimni qabul qilish (majburiy ravishda) yana xatolikni keltirib chiqaradi. Masalani yechish jarayonida boshlang'ich shartlarni va hisoblash natijalarini yaxlitlashda ham xatolikka yo'l qo'yiladi, bunga **hisoblash xatoliklari** deyiladi.

Taqribiy sonlar bilan ish ko'rilayotganda quyidagilarga amal qilish lozim:

1. taqribiy sonlarning aniqligi haqida ma'lumotga ega bo'lish;
2. boshlang'ich qiymatlarning aniqlik darajasini bilgan holda natijaning aniqligini baholash;
3. boshlang'ich qiymatlarning aniqlik darajasini shunday tashlash kerakki, natija belgilangan aniqlikda bo'lsin.

Xatolar manbai. Ko'pincha matematik masalalarni sonli yechishda biz doimo aniq yechimga ega bo'la olmasdan, balki yechimni u yoki bu darajadagi aniqlikda topamiz. Demak, aniq yechim bilan taqribiy yechim orasidagi xatolik qanday qilib kelib qoladi degan savol tug'ilishi tabiiydir. Bu savolga javob berish uchun xatoliklarning hosil bo'lish sabablarini o'rganish lozim.

1. Matematikada tabiat hodisalarining miqdoriy nisbati u

yoki bu funksiyalarni bir-birlari bilan bog'laydigan tenglamalar yordamida tasvirlanadi va bu funksiyalarning bir qismi ma'lum bo'lib (dastlabki ma'lumotlar), boshqalarni topishga to'g'ri keladi. Tabiiyki, topilishi kerak bo'lgan miqdorlar (masalaning yechimi) dastlabki ma'lumotlarning funksiyasi bo'ladi. Kerakli yechimni ajratib olish uchun dastlabki ma'lumotlarga aniq qiymatlar berish kerak. Bu dastlabki ma'lumotlar, odatda, tajribadan olinadi (masalan, yorug'lik tezligi, Plank doimiysi, Avogadro soni va h.k.) yoki boshqa biror masalani yechishdan hosil bo'ladi. Har ikkala holda ham biz dastlabki ma'lumotlarning aniq qiymatiga emas, balki uning taqribiy qiymatiga ega bo'lamiz. Shuning uchun agar dastlabki ma'lumotlarning har bir qiymati uchun tenglamani aniq, yechganimizda ham, baribir (dastlabki ma'lumotlardagi qiymatlar taqribiy bo'lganligi uchun) taqribiy natijaga ega bo'lamiz va natijaning aniqligi dastlabki ma'lumotlarning aniqligiga bog'liq bo'ladi.

Aniq, yechim bilan taqribiy yechim orasidagi farq xato deyiladi. Dastlabki ma'lumotlarning noaniqligi natijasida hosil bo'lgan xato yo'qotilmas xato deyiladi. Bu xato masalani yechayotgan matematikga bog'liq. Bo'lmasdan, unga berilgan ma'lumotlarning aniqligiga bog'liqdir. Lekin matematik dastlabki ma'lumotlar hatosining kattaligini bilishi va shunga qarab natijaning yo'qotilmas hatosini baholashi kerak. Agar dastlabki ma'lumotlarning aniqligi katta bo'lmasa, aniqligi juda katta bo'lgan metodni qo'llash o'rinsizdir. Chunki aniqligi katta bo'lgan metod ko'p mehnatni (hisoblashni) talab qiladi, lekin natijaning hatosi bari bir yo'qotilmas xatodan kam bo'lmaydi.

2. Ba'zi matematik ifodalar tabiat hodisasining ideallashtirilgan modelini tasvirlaydi. Shuning uchun tabiat hodisalarining aniq matematik ifodasini (formulasini, tenglamasini) berib bo'lmaydi, buning natijasida xato kelib chiqadi. Yoki biror masala aniq matematik formada yozilgan bo'lsa va uni shu ko'rinishda

yechish mumkin bo'lmasa, bunday holda bu masala unga yaqinroq va yechish mumkin bo'lgan masalaga almashtirilishi kerak. Buning natijasida kelib chiqadigan xato **metod xatosi** deyiladi.

3. Biz doimo π , e , $\ln 2$ va shunga o'xshash irratsional sonlarning taqribiy qiymatlarini olamiz, bundan tashqari, hisoblash jarayonida oraliq natijalarda ko'p xonali sonlar hosil bo'ladi, ularni yaxlitlab olishga to'g'ri keladi. Ya'ni masalalarni yechishda hisoblashni aniq olib bormaganligimiz natijasida ham xatoga yo'l qo'yamiz, bu xato **hisoblash xatosi** deyiladi.

Shunday qilib, to'liq xato yuqorida aytilgan yo'qotilmas xato, metod xatosi va hisoblash xatolarining yig'indisidan iboratdir. Ravshanki, biror aniq masalani yechayotganda yuqorida aytilgan xatolarning ayrimlari qatnashmasligi yoki uning ta'siri deyarli bo'lmasligi mumkin. Lekin, umuman olganda, xato to'liq tahlil qilinishi uchun bu xatolarning hammasi hisobga olinishi kerak.

1.2-§. Hisoblash xatosi

Masalani qo'lda yoki hisoblash mashinasida yechayotganda biz barcha haqiqiy sonlar bilan ish ko'rmadan, sonlarning ma'lum diskret to'plami bilan ish ko'ramizki, u yoki bu sanoq sistemasida ma'lum miqdordagi xonalar bilan olingan sonlar shu to'plamda yotadi. Bu to'plam

$$\pm (a_1 q^n + a_2 q^{n-1} + \dots + a_m q^{n-m+1}) \quad (1)$$

ko'rinishdagi sonlardan iborat bo'lib, by yerda natural son q - sanoq sistemasining asosidir; $a_1, a_2, \dots, a_m$ - butun sonlar bo'lib, $0 \leq a_i \leq q - 1$ shartni qanoatlantiradi; m bu to'plamdagi sonlar xonasining miqdori, butun n son esa $|n| \leq n_0$ shartni qanoatlantiradi. Qo'lda hisoblayotganda, asosan, o'nlik sanoq sistemi ($q = 10$) bilan ish ko'riladi.

Odatda, arifmetik amallarni bajarayotganda ko'p xonali sonlar hosil bo'ladi (masalan, ko'paytirishda xonalarning soni ikkilanadi, bo'lishda esa xonalarning soni nihoyatda kattalashib ketishi ham mumkin). Natijada hosil bo'lgan son qaralayotgan to'plamdan chiqib ketmasligi uchun m - xonasigacha yaxlitlanadi, ya'ni shu to'plamdagi boshqa son bilan almashtiriladi, tabiiyki yaxlitlanadigan son unga eng yaqin son bilan almashtirilishi, ya'ni yaxlitlash xatosi eng kichik bo'lishi kerak.

Agar biz juft raqam qoidasini qo'llab 5,780475 sonini ketma-ket yaxlitlasak, quyidagi 5,78048; 5,7805; 5,780; 5,78; 5,8; 6 sonlar kelib chiqadi.

Ko'pincha biror natijani olish uchun berilgan metodda ko'rsatilgan bir qator amallarni bajarishga to'g'ri keladi. Agar natijani katta aniqlik bilan topish talab qilinsa, bu qator yanada o'zayib ketadi.

Faraz qilaylik, A aniq son, a - uning taqribiy qiymati bo'lsin. Agar $a < A$ bo'lsa, a kami bilan olingan taqribiy son deyiladi. Agar $a > A$ bo'lsa, a ortig'i bilan olingan taqribiy son deyiladi.

1-ta'rif. Taqribiy sonning xatoligi deb A va a o'rasidagi ayirmaga aytiladi.

Xatolikni Δ deb belgilasak, u holda quyidagicha bo'ladi:

$$\Delta = A - a \quad (2)$$

$$A = \Delta + a$$

2-ta'rif. Taqribiy sonning absolyut xatoligi deb A va a orasidagi ayirmaning moduliga aytiladi.

Absalyut xatolikni Δa deb belgilasak, u holda quyidagicha bo'ladi:

$$\Delta a = |A - a|. \quad (3)$$

Amaliyotda ko'p hollarda 0,01 gacha aniqlik bilan, 1 sm gacha aniqlik bilan va h.k.lar uchraydi. Bu esa absolyut xatolikning

0,01; 1 sm va h.k. ga teng ekanligini bildiradi.

3-ta'rif. Taqribiy a sonning nisbiy xatoligi $\delta(a)$ deb absolyut xatolik Δa ning A ning moduliga nisbatiga aytiladi:

$$\delta(a) = \frac{\Delta a}{|A|} \quad (4)$$

yoki

$$\delta(a) = \frac{\Delta a}{|a|} \quad (5)$$

(4) va (5) formulalarni 100 ga ko'paytirsak, nisbiy xatolik foiz (%) hisobida chikadi.

1-misol. L uzunlikdagi kesmani 0,01 sm aniqlikda o'lchandi va $l = 21,4$ sm natija olindi. Nisbiy xatolikni aniqlang.

Yechish. Bu yerda absolyut xatolik $\Delta l = 0,01$ sm. (2) formulaga asosan $L = 21,4 \pm 0,01$ ya'ni $21,39 \leq l \leq 21,41$.

Absolyut xatolik o'lchash yoki hisoblashni faqat miqdoriy tomondan ifodalaydi va sifat tomonlarini tavsiflamaydi. Shu munosabat bilan nisbiy xatolik tushunchasi kiritiladi.

2-misol. $a = 35,148 \pm 0,00074$ taqribiy sonning nisbiy xatosi (foizlarda) topilsin

Yechish. Bu yerda $\Delta a = 0,00074$, 35,148 ni (4) ga asosan

$$\delta(a) = \frac{0,00074}{35,148} = 0,000022 \approx 0,003\%$$

3-misol. Nisbiy xatoligi $\delta(a) = 0,01\%$ bo'lgan $a = 4,123$ taqribiy sonning absolyut xatoligi Δa topilsin.

Yechish. Foizni o'nli kasr orqali ifodalab va (5) formulaga asosan:

$$\Delta a = |a| \cdot \delta(a) = 4,123 \cdot 0,0001 = 0,0005; A = 4,123 \pm 0,0005$$

4-misol. Jismning og'irligini o'lchashda $m = 23,4 \pm 0,2$ g natija olingan. Nisbiy xatolik topilsin.

Yechish. Bu yerda $\Delta m = 0,2$ u holda

$$\delta(m) = \frac{0,2}{23,4} \cdot 100\% = 0,9\%$$

4-tarif. Agar $\Delta a \leq \frac{1}{2}10^{m-n+1}$ tengsizlik bajarilsa, u holda taqribiy

$$a = \alpha_m \cdot 10^m + \alpha_{m+1} \cdot 10^{m-1} + \dots + \alpha_{m+n-1} \cdot 10^{m-n+1} + \dots$$

sonning birinchi n ta ma'noli raqami **ishonchli raqamlar** deyiladi.

Taqribiy a sonning ishonchli raqamlari soni bilan uning nisbiy xatoligi orasida quyidagi

$$\delta a \leq \frac{1}{\alpha_m} \left(\frac{1}{10} \right)^{n-1}$$

bog'lanish mavjud.

Taqribiy a sonining limit nisbiy xatoligi deb

$$\delta(a) = \frac{1}{\alpha_m} \left(\frac{1}{10} \right)^{n-1}$$

ni olish mumkin, bu yerda $\alpha_m \neq 0$ bo'lib, a taqribiy sonning birinchi ma'noli raqami.

Agar taqribiy a sonning ishonchli raqamlar soni ikkidan katta bo'lsa, amaliyotda

$$\delta(a) = \frac{1}{2\alpha_m} \left(\frac{1}{10} \right)^{n-1}$$

deb olish o'rinli.

Taqribiy sonlarni qo'shganda yoki ayirganda ularning absolyut xatoliklari qo'shiladi:

$$\Delta(a \pm b) = \Delta a + \Delta b$$

bu yerda a va b - taqribiy sonlar.

Taqribiy sonni taqribiy songa bo'lganda yoki ko'paytirganda ularning nisbiy xatoliklari qo'shiladi:

$$\delta(a \cdot b) = \delta(a) + \delta(b); \quad \delta\left(\frac{a}{b}\right) = \delta(a) + \delta(b)$$

Taqribiy son darajaga oshirilganda, uning nisbiy xatoligi shu daraja ko'rsatkichiga ko'paytiriladi:

$$\delta(a^n) = n \cdot \delta(a)$$

1.3-§. Funksiya xatoligi.

Faraz qilaylik, uzluksiz differensiallanuvchi

$$y = f(x_1, x_2, \dots, x_n)$$

funksiya argumentlarining taqribiy qiymatlari $\bar{x}_i$ va ularning absolyut xatoliklari $\Delta\bar{x}_i, i = 1, 2, \dots, n$ ma'lum bo'lsin. Berilgan funksiyaning limit absolyut $\Delta\bar{u}$ va limit nisbiy $\delta\bar{u}$ xatoligini topishni ko'raylik.

Bu masalani hal etish uchun quyidagi shartlar o'rinli bo'lishligini talab etamiz:

1. Qaralayotgan sohada f uzluksiz differensiallanuvchi bo'lib, xususiyl hosilalari sekin o'zgaruvchi bo'lsin.

2. Argumentlarning absolyut xatoliklari aytarli katta emas, nisbiy xatoliklari yetarlicha kichik bo'lsin.

U holda Lagranj formulasiga ko'ra

$$y - \bar{y} = f(x_1, x_2, \dots, x_n) - f(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n) =$$

$$= \sum_{i=1}^n \frac{\partial f(\xi)}{\partial x_i} \cdot (x_i - \bar{x}_i)$$

bu yerda $\xi = (\xi_1, \xi_2, \dots, \xi_n)$ esa $(x_1, x_2, \dots, x_n)$ va $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n)$ nuqtalarni birlashtiruvchi kesmaga tegishli qandaydir nuqta.

Funksiyaga qo'yilgan birinchi shartlarga asosan, $\frac{\partial f(\xi)}{\partial x_i}$ ni $\frac{\partial f(\bar{x})}{\partial x_i}$ ga almashtirish mumkin:

$$y - \bar{y} = \sum_{i=1}^n \frac{\partial f(\bar{x})}{\partial x_i} \cdot (x_i - \bar{x}_i)$$

Bundan

$$\Delta(\bar{y}) = \sum_{i=1}^n \left| \frac{\partial f(\bar{x})}{\partial x_i} \right| \cdot \Delta(\bar{x}_i).$$

Endi nisbiy xatoligini chiqaramiz:

$$\delta(\bar{y}) = \frac{\Delta(\bar{y})}{|f(\bar{x})|} = \sum_{i=1}^n \left| \frac{\frac{\partial f(\bar{x})}{\partial x_i}}{f(\bar{x})} \right| \cdot \Delta(\bar{x}_i).$$

Bu formulaning quyidagicha ko'rinishi ham ishlatishadi:

$$\delta(\bar{y}) = \sum_{i=1}^n \left| \frac{\frac{\partial f(\bar{x})}{\partial x_i}}{f(\bar{x})} \right| \cdot \Delta(\bar{x}_i) = \sum_{i=1}^n |x_i| \cdot \left| \frac{\frac{\partial f(\bar{x})}{\partial x_i}}{f(\bar{x})} \right| \cdot \delta(\bar{x}_i)$$

5-misol. Agar shar diametri $d = 3,7 \text{ sm} \pm 0,5 \text{ sm}$ bo'lsa, shar hajmi

$$V = \frac{1}{6} \pi d^3$$

ning limit absolut va limit nisbiy xatoligi qanday bo'ladi?

Yechish.

$$\frac{\partial V}{\partial \pi} = \frac{1}{6}d^3 = 8,44, \quad \frac{\partial V}{\partial d} = \frac{1}{2}\pi d^2 = 21,5.$$

$$\begin{aligned} \Delta(V) &= \left| \frac{\partial V}{\partial \pi} \right| \cdot |\Delta\pi| + \left| \frac{\partial V}{\partial d} \right| \cdot |\Delta d| = 8,44 \cdot 0,0016 + 21,5 \cdot 0,05 = \\ &= 0,013 + 1,075 = 1,088, \Delta(V) \cong 1,1 \text{ sm}^3. \end{aligned}$$

Demak,

$$V = \frac{1}{6}\pi d^3 = 27,4 \text{ sm}^3 \pm 1,1 \text{ sm}^3,$$

$$\delta(V) = \frac{1,088}{27,4} = 0,0397 \cong 0,04, \delta(V) = 4\%.$$

Javob: $\Delta(V) = 1,1 \text{ sm}^3, \delta(V) = 4\%$.

Bob mavzularini mustahkamlashga doir misollar.

Quyidagilarni hisoblang:

- a) Qaysi tenglik aniqroq ekanligini aniqlang.
- b) Sonning shubhali raqamlarini yaqinlashtirib, aniq raqamlarni qoldiring. Natijaning absolyut xatosini aniqlang.
- c) Barcha raqamlari aniq deb hisoblangan yaqinlashtirilgan sonning chegaraviy absolyut va nisbiy xatolarini toping.

$$1. \text{ a) } 14/17 = 0.824, \sqrt{53} = 7.28; \text{ b) } 23.3748, \delta = 0.27\%; \\ \text{ c) } 0.645.$$

$$2. \text{ a) } 7/3 = 2.33, \sqrt{58} = 7.62; \text{ b) } 13.5726 \pm 0.0072; \\ \text{ c) } 4.8556.$$

$$3. \text{ a) } 27/31 = 0.871, \sqrt{42} = 6.48; \text{ b) } 0.088748, \delta = 0.56\%; \\ \text{ c) } 71.385.$$

4. a) $23/9 = 2.56$, $\sqrt{87} = 9.33$; b) 4.57633 ± 0.00042 ;
c) 6.8346 .
5. a) $6/7 = 0.857$, $\sqrt{41} = 6.40$; b) 46.7843 , $\delta = 0.32\%$;
c) 7.38 .
6. a) $12/7 = 1.71$, $\sqrt{47} = 6.86$; b) 0.38725 ± 0.00112 ;
c) 0.00646.
7. a) $21/13 = 1.62$, $\sqrt{63} = 7.94$; b) 45.7832 , $\delta = 0.18\%$;
c) 3.6765 .
8. a) $16/7 = 2.29$, $\sqrt{11} = 3.32$; b) 0.75244 ± 0.00013 ;
c) 5.374 .
9. a) $18/7 = 2.57$, $\sqrt{22} = 4.69$; b) 46.453 , $\delta = 0.15\%$;
c) 6.125 .
10. a) $17/9 = 1.89$, $\sqrt{17} = 4.12$; b) 0.66385 ± 0.00042 ;
c) 24.6 .
11. a) $51/11 = 4.64$, $\sqrt{35} = 5.92$; b) 0.66385 , $\delta = 0.34\%$;
c) 0.543 .
12. a) $19/12 = 1.58$, $\sqrt{12} = 3.46$; b) 4.88445 ± 0.00052 ;
c) 4.633 .
13. a) $13/7 = 1.857$, $\sqrt{7} = 2.65$; b) 2.8867 , $\delta = 0.43\%$;
c) 63.749 .
14. a) $49/13 = 3.77$, $\sqrt{14} = 3.74$; b) 5.6483 ± 0.0017 ;
c) 0.00858 .
15. a) $5/3 = 1.667$, $\sqrt{38} = 6.16$; b) 3.7542 , $\delta = 0.32\%$;
c) 0.389 .

16. a) $17/11 = 1.545$, $\sqrt{18} = 4.243$; b) 0.8647 ± 0.0013 ;
c) 0.864 .
17. a) $7/22 = 0.318$, $\sqrt{13} = 3.61$; b) 0.3944 , $\delta = 0.15\%$;
c) 21.7 .
18. a) $13/17 = 0.765$, $\sqrt{31} = 5.57$; b) 3.6878 ± 0.0013 ;
c) 8.74 .
19. a) $50/19 = 2.63$, $\sqrt{27} = 5.20$; b) 0.85638 , $\delta = 0.22\%$;
c) 231.57 .
20. a) $21/29 = 0.724$, $\sqrt{44} = 6.63$; b) 13.6853 ± 0.0023 ;
c) 2.16 .
21. a) $17/19 = 0.895$, $\sqrt{52} = 7.21$; b) 7.521 , $\delta = 0.12\%$;
c) 0.5748 .
22. a) $6/11 = 0.545$, $\sqrt{83} = 9.11$; b) 3.7832 ± 0.0043 ;
c) 2.678 .
23. a) $16/19 = 0.842$, $\sqrt{55} = 7.416$; b) 17.356 , $\delta = 0.11\%$;
c) 0.5718 .
24. a) $23/15 = 1.53$, $\sqrt{98} = 9.899$; b) 8.7432 ± 0.0023 ;
c) 0.578 .
25. a) $12/11 = 1.091$, $\sqrt{68} = 8.246$; b) 0.5532 ± 0.0014 ;
c) 3.4479 .
26. a) $6/7 = 0.857$, $\sqrt{48} = 6.928$; b) 14.5841 , $\delta = 0.17\%$;
c) 0.421 .
27. a) $15/7 = 2.14$, $\sqrt{10} = 3.16$; b) 4.5012 ± 0.0013 ;
c) 1.4229 .

28. a) $4/17 = 0.235$, $\sqrt{105} = 10.25$; b) 1.1341 , $\delta = 0.12\%$;
c) 2.401 .

29. a) $7/15 = 0.467$, $\sqrt{30} = 5.48$; b) 6.7702 ± 0.0015 ;
c) 11.1239 .

30. a) $21/29 = 0.724$, $\sqrt{44} = 6.63$; b) 13.6853 ± 0.0023 ;
c) 2.16 .

Bitta variant uchun yechish namunasi

$9/19 = 0.474$, $\sqrt{103} = 10.149$. Ushbu ifodalarning ko'proq o'nlik raqamlar bilan qiymatlarini toping:

$$a = 9/19 = 0.47368 \dots, b = \sqrt{103} = 10.14889 \dots$$

Absolyut xatolarni hisoblaymiz:

$$\Delta_a = |0.47368 - 0.474| = 3.2 \cdot 10^{-4} \leq 0.0004,$$

$$\Delta_b = |10.14889 - 10.149| = 1.1 \cdot 10^{-4} \leq 0.0002.$$

Absolyut nisbiy xatolar quyidagicha:

$$\delta_a = \frac{\Delta_a}{|a|} = \frac{0.0004}{0.474} \approx 8 \cdot 10^{-4} = 0.08\%$$

$$\delta_b = \frac{\Delta_b}{|b|} = \frac{0.0002}{10.149} \approx 2 \cdot 10^{-5} = 0.002\%$$

Hisoblashlardan ko'rinadiki δ_b , δ_a dan ancha kichik, demak, $\sqrt{103} = 10.149$ tenglik ancha aniqroq.

b) Quyidagi yaqinlashtirilgan sonlar berilgan bo'lsin: $c = 72.354 \pm 0.021$. Absolyut xatolik $\Delta_c = 0.021 < 0.1$, c sonini o'nlikkacha yaqinlashtirib, aniq raqamlar bilan 72.4 ko'rinishida yozib olamiz.

Quyidagi yaqinlashtirilgan son berilgan bo'lsin: $d = 5.272$, va u nisbiy xatolik $\delta_d = 0.1\%$ bilan ma'lum. Chunki

$$\Delta_d = |d| \cdot \delta_d = 5.272 \cdot 0.001 = 0.005272 < 0.01,$$

d sonini yuzlikkacha yaqinlashtirib, aniq raqamlar bilan 5.27 ko'rinishida yozib olamiz.

c) Quyidagi yaqinlashtirilgan son berilgan bo'lsin: $f = 14.278$. Chunki barcha raqamlari aniq, demak, chegaraviy mutlaq xatolik $\Delta_f \leq 0.001$. Shunda chegaraviy nisbiy xatolik quyidagicha bo'ladi:

$$\delta_f = \frac{\Delta_f}{|f|} = \frac{0.001}{14.278} \approx 7 \cdot 10^{-5} \approx 10^{-4} = 0.01\%$$

II BOB.

ALGEBRAIK VA TRANSENDENT TENGLAMALARNI YECHISH USULLARI

2.1-§. Umumiy mulohazalar. Ildizlarni ajratish.

Algebraik va transtsendent tenglamalarni xamda bunday tenglamalar sistemasini yechish analizning muxim masalalaridan biridir. Fizika, mexanika, texnika va umuman tabiatshunoslikning xilma-xil masalalari algebraik va transtsendent tenglamalarni yechishga olib keladi. Masalan, mexanik sistema tebranishi chastotalarining kvadratlari matritsalar xarakteristik tenglamalarining ildizlari bo'ladi, bunday tenglama esa n -darajali algebraik tenglamadir.

Ikkinchi tomondan, matematikaning extiyojlari xam bunday tenglamalarni yechishni taqozo etadi. Masalan, noma'lumlarni yo'qotish yo'li bilan murakkab algebraik va geometrik munosabatlar ikkinchi yoki yuqori darajali algebraik tenglamalarga keltiriladi.

Ma'lumki, darajasi to'rt dan yuqori bo'lgan algebraik xamda transendent tenglamalarni yechish uchun aniq metodlar mavjud emas. Shuning uchun ham bunday tenglamalarning taqribiy yechimlarini yetarlicha aniqlik bilan topish imkonini beradigan metodlar kerak. Bu bobda shu metodlarning keng qo'llanadiganlarini keltiramiz.

Bir noma'lumli ixtiyoriy tenglamani umumiy holda quyidagi ko'rinishda yozish mumkin:

$$f(x) = 0 \tag{1}$$

Biz quyidagi (1) tenglamaning yechish bilan tanishamiz, bunda $f(x)$ funksiya $[a, b]$ kesmada aniqlnagan va uzluksiz deb qaraladi.

1-ta’rif. Agar $f(x)$ funksiya ko‘phaddan iborat bo‘lsa, ya’ni

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n \quad n = 0, 1, 2, \dots$$

bo‘lsa,

$$a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n = 0$$

ko‘rinishdagi tenglama **algebraik tenglama** deyiladi.

2-ta’rif. Agar $f(x)$ funksiya elementar funksiyalar (logarifmik, ko‘rsatkichli, trigonometrik va hokazo) yoki maxsus funksiyalardan iborat bo‘lsa, u holda (1) ko‘rinishdagi tenglama **transendent tenglama** deyiladi.

3-ta’rif. (1) tenglamani ayniyatga aylantiruvchi

$$f(x^*) = 0$$

ixtiyoriy x^* qiymat tenglamaning ildizi (yechimi) deyiladi.

(1) ko‘rinishdagi tenglamani aniq yechimini topish formulalari faqat qisqa sinf tenglamalari uchun, masalan, kvadrat, bikvadrat, ayrim trigonometrik, logarifmik va ko‘rsatkichli tenglamalar uchun ma’lum xolos.

Lekin, ko‘pincha amaliyotda elementar almashtirishlar yordamida yechish imkoniyati bo‘lmagan tenglamalar uchraydiki, ularni sodda amaliyotlar ko‘magida yechish imkoniyati yo‘q. Ammo, ularni berilgan aniqlikda taqribiy yechish mumkin. Yechimni taqribiy topish masalasi ikki bosqichga bo‘linadi, ya’ni:

1. tenglamaning yechimi yotgan biror oraliqni ajratish;
2. berilgan aniqlikda yechimni topish.

Yechim yotgan oraliqni ajratishda quyidagi teoremlarni bilish muhimdir.

1-teorema. Agar $f(x)$ funksiya $[a, b]$ kesmada uzluksiz va oraliqning chetki nuqtalarida har xil ishorali, ya’ni $f(a)f(b) < 0$ bo‘lsa, u holda $[a, b]$ kesmada (1) tenglamaning hech bo‘lmaganda bitta yechimi mavjud.

2-teorema. Agar $f(x)$ funksiya $[a, b]$ oraliqda uzluksiz bo'lib, $f(a)f(b) < 0$ shart bajarilsa hamda $f'(x)$ hosila $[a, b]$ oraliqda o'z ishorasini o'zgartirmasa, u holda $[a, b]$ oraliqda (1) tenglamaning yagona yechimi mavjud.

3-teorema. Agar

$$A = \max_{1 \leq k \leq n} \left| \frac{a_k}{a_0} \right|, A_1 = \max_{0 \leq k \leq n-1} \left| \frac{a_k}{a_n} \right|$$

bo'lsa, u holda (1) tenglamaning barcha ildizlari

$$r = \frac{1}{1 + A_1} < |x| < 1 + A = R$$

xalqa ichida yotadi.

4-teorema (Lagranj teoremasi). Agar (1) tenglamaning manfiy koeffitsiyentlaridan eng birinchisi (chapdan o'ngga tomon hisoblanayotganda) a_k bo'lib, B manfiy koeffitsiyentlarning absolyut qiymatlari bo'yicha eng kattasi bo'lsa, u holda musbat ildizlarning yuqori chegarasi

$$R = 1 + \sqrt[k]{\frac{B}{a_0}}$$

son bilan ifodalanadi.

5-teorema (Nyuton teoremasi). Agar $x = c > 0$ uchun $f(x)$ ko'phad va uning barcha $f'(x), f''(x), \dots, f^{(n)}(x)$ xosilalari nomanfiy bo'lsa:

$$f^{(k)}(c) \geq 0 \quad (k = 0, 1, \dots, n),$$

u holda $R = c$ ni (1) tenglamaning musbat ildizlari uchun yuqori chegara deb hisoblash mumkin.

Yechim yotgan oraliqni ajratishda grafik usuldan ham keng foydalaniladi. Bunda (1) tenglamaning yechimi $y = f(x)$ funksiyaning absissalar o'qini kesib o'tuvchi x^* qiymat bo'la oladi.

Agar $y = f(x)$ funksiya grafigini chizish qiyinchilik tug'dirsa, u holda $y = f(x)$ funksiya $h_1(x) = h_2(x)$ ko'rinishda almashtirib, $y_1 = h_1(x)$ va $y_2 = h_2(x)$ funksiyalar grafigini chizish lozim. Bu funksiyalar kesishgan nuqtaning absissasi (1) tenglamaning yechimidir.

Misol. Quyidagi tenglama haqiqiy ildizlarining chegarasini toping.

$$f(x) = x^4 - 5x^2 + 8x - 8 = 0.$$

Yechish. 3-teoremani qo'llaymiz, bu yerda $a_0 = 1$, $A = 8$. Demak, $R = 1 + 8 = 9$, ya'ni tenglamaning ildizlari $(-9; 9)$ oraliqda yotar ekan.

Endi Logranj teoremasini qo'llaymiz: $a_0 = 1$, $k = 2$, $B = 8$. Bu qiymatlarni teorema formulasiga qo'yib, musbat ildizlarining yuqori chegarasi uchun

$$R = 1 + \sqrt{\frac{8}{1}} = 1 + 2\sqrt{2} < 3,84$$

ni hosil qilamiz. Keyin tenglamada x ni $-x$ ga almashtirsak,

$$f_1(x) \equiv x^4 - 5x^2 - 8x - 8 = 0$$

tenglama kelib chiqadi. Bu tenglama musbat ildizining yuqori chegarasi uchun ham $R < 3,84$ tengsizlik kelib chiqadi. Yani Lagranj teoremasiga ko'ra tenglamaning ildizlari $(-3,84; 3,84)$ oraliqda joylashgan ekan.

Mavzuni mustahkamlash uchun misollar.

Tenglamalar ildizlari yotgan oraliqlarni toping.

1. $x^3 - 1,5x^2 + 0,58x - 0,057 = 0$
2. $x^3 - 2,5x^2 - x + 2 = 0$

3. $x^3 + x^2 - 4x - 4 = 0$
4. $x^4 - 4x^3 + 5,5x^2 - 3x + 0,3 = 0$
5. $x^4 - 7,99x^3 - 24,10x^2 + 47,81x + 80,21 = 0$
6. $x^4 + 2,83x^3 - 4,5x^2 - 64x - 20 = 0$
7. $x^5 - 3x^3 - 14x - 8 = 0$
8. $x^5 - x - 0,2 = 0$
9. $x^5 + 3,2x^2 - 0,2163923 = 0$
10. $x^4 - 31,2x - 25,8944 = 0$
11. $x^3 - 0,83x^2 + 5,2x - 81,2868 = 0$
12. $x^4 - 0,79x - 59,67 = 0$
13. $x^5 - 8,2x - 2077,8273 = 0$
14. $x^5 - 5x^4 - 4x^3 + 20x^0 - 5x + 25 = 0$
15. $x^3 - 0,4x + 0,08 = 0$
16. $x^4 + x^3 - 6x^2 + 20x - 16 = 0$
17. $x^5 + 11x^4 - 101x^3 + 11x^2 - 10 = 0$
18. $x^4 + 47,89x^3 + 797,3x^2 + 5349x - 12300 = 0$
19. $x^4 + 10x^3 - 1 = 0$
20. $x^5 + 1,1x - 1 = 0$
21. $x^4 - 4x^3 - 40x^2 - 56x - 20 = 0$
22. $x^3 + 7,05x^2 - x - 101,76 = 0$

$$23. x^4 - 7,18x^3 + 8,244539 = 0$$

$$24. x^4 + 3x^3 - x + 6 = 0$$

$$25. x^2 - x^3 + 1,51593 = 0$$

$$26. 2x^3 - 3x^2 - 12x - 5 = 0$$

$$27. x^3 + 3x^2 - 24x - 10 = 0$$

$$28. x^3 - 3x^2 + 3 = 0$$

$$29. 2x^3 + 9x^2 - 21 = 0$$

$$30. x^3 + 3x^2 - 2 = 0$$

2.2-§. Algebraik va transcendent tenglamalarni oddiy iteratsiya usuli bilan yechish.

Bizga

$$f(x) = 0 \quad (1)$$

ko'rinishdagi tenglama berilgan bo'lsin. (1) ni unga ekvivalent bo'lgan

$$x = \psi(x) \quad (2)$$

tenglama bilan almashtiramiz. (1) ko'rinishdagi tenglamani hamma vaqt (2) ko'rinishga keltirish mumkin, masalan (1) ni

$$x = x + kf(x), \quad (k \neq 0) \quad (3)$$

ko'rinishda yozib olish mumkin, bunda $\psi(x) = x + kf(x)$.

Tenglama yechimi yotgan biror oraliqda x_0 sonni olib quyidagi sonli ketma-ketlikni tuzamiz:

$$x_1 = \psi(x_0), \quad x_2 = \psi(x_1), \dots, x_n = \psi(x_{n-1}) \quad (4)$$

(4) ko‘rinishdagi ketma-ketlik yordamida $x_n (n = 1, 2, 3, \dots)$ sonni topish oddiy iteratsiya (ketma-ket yaqinlashish) usuli deyiladi.

Teorema. Agar tenglama yechimi yotgan biror $[a; b]$ oraliqda x_0 va har galdagi x_n ketma-ketlik uchun uzluksiz $\psi'(x)$ - hosila mavjud bo‘lib,

$$|\psi'(x)| \leq q < 1$$

shart o‘rinli bo‘lsa, u holda (4) iteratsiya jarayoni yaqinlashuvchi bo‘ladi, ya’ni n ning ortishi bilan (1) tenglamaning x^* - haqiqiy yechimidan deyarli kam farq qiladigan taqribiy yechimni hosil qilish mumkin. Yaqinlashish tezligi quyidagi tengsizlik orqali ifodalanadi:

$$|x^* - x| \leq \frac{q}{1 - q} |x_n - x_{n-1}| \quad (5)$$

(5) tengsizlikdan ma’lumki, oddiy iteratsiya usulining yaqinlashish tezligi bevosita q miqdorga bog‘liq, ya’ni q miqdor qanchalik kichik bo‘lsa, (4) ketma-ketlik shunchalik tez yechimga yaqinlashadi. Agar $q \leq 0,5$ bo‘lsa, (5) munosabat o‘rniga yetarlicha oddiy

$$|x_n - x_{n-1}| < \varepsilon$$

tengsizlikdan foydalanish mumkin. Shu bois, (1) tenglamani (2) ko‘rinishga keltirish uchun $y(x)$ ni shunday almashtirish lozimki, $y'(x)$ hosila, yechim atrofida absolyut qiymati bo‘yicha yetarlicha kichik bo‘lishi kerak, bunga odatda (3) da k parametrni tanlash orqali erishish mumkin.

Misol. $f(x) = x^3 - 20x + 12 = 0$ tenglama ildizlarini ajrating va ajratilgan ildizlarini iteratsiya usuli bilan aniqlaydigan jarayonni ko‘rsating.

Yechish. Ildizlar chegarasini Lagranj teoremasiga ko‘ra aniqlaymiz:

$$a_0 = 1, \quad k = 2, \quad a_2 = -20, \quad R < 1 + \sqrt{\frac{20}{1}} < 5,5,$$

Tenglamada $x = -x$ almashtirish o'tkazib, hosil bo'lgan tenglamaga Lagranj teoremasini qo'llasak, $a_0 = 1$, $k = 2$, $B = 20$, $R < 1 + \sqrt{\frac{20}{1}} < 5,5$, $k = 2$, $B = 20$, $R + 1 < 5,5$ bo'ladi. Demak, berilgan tenglamaning haqiqiy ildizlari $(-5, 5; 5, 5)$ oraliqda yotadi. Quyidagini aniqlaymiz:

x	$-5,5$	-5	-4	0	1	4	5	$5,5$
$\text{sign } f(x)$	$-$	$-$	$+$	$+$	$-$	$-$	$+$	$+$

Ya'ni berilgan tenglama 3 ta haqiqiy ildizga ega bo'lib, ular $(-5; -4)$, $(0; 1)$ va $(4, 5)$ oraliqlarda joylashgan.

A) $(-5; -4)$ oraliqdagi ildiz uchun berilgan tenglamani

$$x = \sqrt[3]{20x - 12} = \varphi(x)$$

ko'rinishga keltiramiz, chunki

$$\begin{aligned} \max_{[-5; -4]} |\varphi'(x)| &= \max_{[-5; -4]} \left| \frac{20}{3\sqrt[3]{(20x - 12)^2}} \right| < \\ &< \frac{20}{3\sqrt[3]{92^2}} < \frac{20}{3 \cdot 4,5^2} = \frac{80}{243} < \frac{1}{3} \end{aligned}$$

bo'ladi. Hisoblash jarayoni

$$x_{n+1} = \sqrt[3]{20x_n - 12}, \quad n = 0, 1, \dots$$

formula bilan tashkil etiladi, bu yerda x_0 ixtiyoriy bo'lib, $[-5, -4]$ oraliqqa tegishli.

B) $(0; 1)$ oraliqdagi ildiz uchun

$$x = \frac{x^3 + 12}{20} = \varphi(x)$$

deb olamiz, chunki

$$\max_{[0,1]} |\varphi'(x)| = \max_{[0,1]} \left| \frac{3x^3}{20} \right| \leq \frac{3}{20} < 1$$

boladi. Hisoblash jarayoni

$$x_{n+1} = \frac{x_n^3 + 12}{20}, \quad n = 0, 1, 2, \dots$$

formula bilan tashkil etiladi va x_0 sifatida $[0; 1]$ ga tegishli ixtiyoriy qiymatni olish mumkin.

D) $(4, 5)$ oraliqdagi ildiz uchun ham hisoblash jarayoni

$$x_{n+1} = \sqrt[3]{20x_n - 12}, \quad n = 0, 1, 2, \dots$$

formula bilan amalga oshiriladi, x_0 ixtiyoriy bo'lib, $[4, 5]$ oraliqqa tegishli.

$$x_1 = \sqrt[3]{20x_0 - 12} = \sqrt[3]{20 * 4 - 12} = 4,081655$$

$$x_2 = \sqrt[3]{20x_1 - 12} = \sqrt[3]{20 * 4,081655 - 12} = 4,114072$$

.....

Algebraik va transcendent tenglamani oddiy iteratsiya usuli bilan yechishning C# dasturlash tilida dasturini keltiramiz.

```
public static void Main(string[] args)
{
    Console.WriteLine("Quyi chegarani kiriting");
    Console.Write("a=");
    double a=Double.Parse(Console.ReadLine());
    Console.WriteLine("Yuqori chegarani kiriting");
    Console.Write("b=");
    double b=Double.Parse(Console.ReadLine());
```

```

Console.WriteLine("Aniqlikni kiriting");
Console.Write("e=");
double e=Double.Parse(Console.ReadLine());
double [] x=new double[100];
Console.WriteLine("Bosh. yaqin. kiriting");
Console.Write("x[0]=");
x[0]=Double.Parse(Console.ReadLine());
int i=0;
do
{x[i+1]=F(x) i++;}
while (Math.Abs(x[i]-x[i-1])>e);
Console.WriteLine("Tenglamaning taqribiy yechimi
x="+x[i]);
Console.WriteLine("Tenglamaning taq. yech. qadam.
soni N="+i);
Console.Write("Press any key to continue . . . ");
Console.ReadKey(true);
}

```

Izoh. $F(x)$ funksiya o‘rniga $f(x)$ funksiyani $x=\varphi(x)$ ko‘rinishga o‘tkazil- gandan keyingi $\varphi(x)$ funksiya yoziladi.

Mavzuni mustahkamlash uchun misollar.

Grafik yoki analitik ravishda berilgan tenglama ildizi yotgan oraliqni ajrating va ketma-ket yaqinlashish usuli yordamida 10^{-3} aniqlikda yechimni toping hamda yechimni topish uchun zarur bo‘lgan iteratsiya sonini aniqlang.

$$1. x^4 + 2x^3 - 4x^2 - 12 = 0$$

$$2. x^3 - 0.2x^2 - 4x - 7 = 0$$

$$3. 2x^4 - 3x^3 - 4x^2 - 5 = 0$$

4. $e^x + 4x + 1 = 0$

5. $3^x + 2x - 1 = 0$

6. $2^x - \frac{1}{(x-2)^2} = 0$

7. $\frac{1}{x} - \lg(x+1) = 0$

8. $x - 4 \lg(x) + 2 = 0$

9. $x^2 - 2 \lg(x) + 1 = 0$

10. $\lg(x^2 + 4) + 5x = 0$

11. $\log_3(x-1) - 2/x = 0$

12. $x - 4 \lg(x) - 1 = 0$

13. $\sin(x+1) - x^2 = 0$

14. $\cos x + \sqrt{x+1} = 0$

15. $\cos(x) - 3x + 1 = 0$

16. $\sin(2x) - 2x + 1/4 = 0$

17. $x - 5 \sin(x) - 1/2 = 0$

18. $\cos(2x+1) - 3x + 1 = 0$

19. $x - 2 \sin(x+1) = 0$

20. $\sin(x + \pi/2) + 2 \cos x + x = 0$

21. $e^x + \cos(x + \pi/2) + 2 = 0$

22. $\arcsin(x) - 0.2x - 0.1 = 0$

$$23. \arccos(x) - \sqrt{1 - 0.3x^2} = 0$$

$$24. \operatorname{arctg}(x) - 0.5x + 1.5 = 0$$

$$25. 3x^3 + \operatorname{arctg}(x - 1) = 0$$

$$26. 3x^3 \operatorname{arctg}(x) - 1 = 0$$

$$27. \operatorname{arcctg}(x) - 3x - 0.1 = 0$$

$$28. \operatorname{arctg}(\sqrt{x + 1 - 2x + 3}) = 0$$

$$29. \operatorname{sh}(x) - \lg(x + 2) = 0$$

$$30. \operatorname{arctg}(\sqrt[3]{x + 1}) - e^x + x = 0$$

2.3-§. Algebraik va transcendent tenglamalarni urinmalar (Nyuton) usuli bilan yechish.

Agar

$$f(x) = 0$$

tenglamani yechishda boshlang'ich yaqinlashish yetarlicha yaxshi tanlansa, u holda yechim aniqligini oshishi jihatidan boshqa taqribiy usullar orasida urinmalar (Nyuton) usuli ustunligi bilan e'tiborga molikdir.

Aytaylik, $f(x)$ funksiya biror $[a, b]$ oraliqda aniqlangan va ikki marta differensiallanuvchi bo'lsin va oraliqning chetki nuqtalarida har xil ishorali qiymatlar qabul qilsin (yechim mavjudligi sharti), ya'ni

$$f(a)f(b) < 0$$

$f'(x)$ va $f''(x)$ hosilalar $[a, b]$ oraliqda ishorasini o'zgartirmasin. $y = f(x)$ funksiya grafigiga $f(x)$ va $f''(x)$ lar hosila ishoralari bir xil bo'lgan $[a, b]$ oraliqning chetki nuqtasidan urinma o'tkazamiz.

Urinma tenglamasi

agar $f(a)f''(a) > 0$, bo'lsa,

$$y - f(a) = f'(a)(x - a),$$

agar $f(b)f''(b) > 0$ ($f(a)f'(a) < 0$) bo'lsa

$$y - f(b) = f'(b)(x - b)$$

bo'ladi.

Urinmaning absissa o'qini kesib o'tuvchi x_1 nuqtasini topamiz. $y = 0$ deb olsak, u holda

$$\text{agar } f(a)f''(a) > 0 \text{ bo'lsa, } x_1 = a - f(a)/f'(a),$$

yoki

$$\text{agar } f(b)f''(b) > 0 \text{ bo'lsa, } x_1 = b - f(b)/f'(b)$$

birinchi yaqinlashishga ega bo'lamiz. Bu yaqinlashishdan foydalanib, navbatdagi yaqinlashishlarni topish mumkin. Navbatdagi yaqinlashishlar umumiy holda quyidagi formuladan topiladi:

$$x_n = x_{n-1} - f(x_{n-1})/f'(x_{n-1}), n = 2, 3, \dots$$

Hisoblash jarayoni

$$|x_n - x_{n-1}| < \varepsilon$$

tengsizlik bajarilgunga qadar davom ettiriladi, bunda

$$\varepsilon = \sqrt{2m_1/m_2},$$

$m_1 = |f'(x)|$ - birinchi hosilaning absolyut qiymati jihatida bo'yicha ($|f'(x)|$) $[a, b]$ oraliqdagi eng kichik qiymati: $m_2 = |f''(x)|$ -

ikkinchi hosilaning $[a, b]$ oraliqdagi absolyut qiymati bo'yicha eng katta qiymati: ε - yechimning absolyut xatoligi.

Misol. $f(x) = x^3 - 20x + 12 = 0$ tenglamani Nyuton usuli bilan aniqlaydigan jarayonni ko'rsating.

Yechish. Tenglama 3 ta haqiqiy ildizga ega bo'lib, ular $(-5; -4)$, $(0; 1)$ va $(4; 5)$ oraliqlarda joylashgan.

$$f'(x) = 3 \cdot x^2 - 20, \quad f''(x) = 6x$$

A) $(-5, -4)$ oraliqdagi ildiz uchun $f(x_0) \cdot f'(x_0) > 0$ shartni tek shiramiz:

$$f(-5) < 0, f(-4) > 0, \quad f''(-5) < 0, \quad f''(-4) < 0.$$

$f(-5) \cdot f''(-5) > 0$ bo'lganligi uchun $x_0 = -5$ bo'ladi.

B) Xuddi shuningdek, $(0; 1)$ oraliq uchun $x_0 = 0$,

D) $(4; 5)$ oraliq uchun $x_0 = 5$ bo'lishligini aniqlaymiz va uch holda ham

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, n = 0, 1, \dots$$

iteratsion jarayon bilan ildizning taqribiy qiymati berilgan aniqlikda hisoblanadi.

$$x_1 = x_0 - \frac{x_0^3 - 20x_0 + 12}{3x_0^2 - 20} =$$

$$5 - \frac{5^3 - 20 \cdot 5 + 12}{3 \cdot 5^2 - 20} = 5 - \frac{37}{55} = 4,3(27)$$

.....

Algebraik va transcendent tenglamani urinmalar (Nyuton) usuli bilan yechishning C# dasturlash tilida dasturini keltiramiz.

```
public static void Main(string[] args)
{
    Console.WriteLine("Quyidagilarni kiriting");
    Console.Write("a=");
    double a=Double.Parse(Console.ReadLine());
    Console.WriteLine("Yuqoridagilarni kiriting");
    Console.Write("b=");
    double b=Double.Parse(Console.ReadLine());
    Console.WriteLine("Aniqlikni kiriting");
    Console.Write("e=");
    double e=Double.Parse(Console.ReadLine());
    double [] x=new double[100];
    Console.WriteLine("Boshlangich yaqinlashishni
        kiriting");
    Console.Write("x[0]=");
    x[0]=Double.Parse(Console.ReadLine());
    int i=0;
    do
    {
        x[i+1]=x[i]-f(x[i])/f'(x[i]); i++;
    }
    while (Math.Abs(x[i]-x[i-1])>e);
    Console.WriteLine("Tenglamaning taqribiy yechimi
        x="+x[i]);
    Console.WriteLine("Tenglamaning taqribiy yechimi
        soni N="+i);
    Console.Write("Press any key to continue . . . ");
    Console.ReadKey(true);
}
```

Izoh. $f(x)$ va $f'(x)$ lar o'rniga tenglamada berilgan $f(x)$ va $f'(x)$ funksiyalarning C# dagi ifodasi yoziladi.

Mavzuni mustahkamlash uchun misollar.

Grafik yoki analitik ravishda berilgan tenglama ildizi yotgan oraliqni ajrating va Nyuton (urinmalar) usuli yordamida 10^{-3} aniqlikda yechimni toping hamda yechimni topish uchun zarur bo'lgan iteratsiya sonini toping.

1. $\ln x + x - 2 = 0.$
2. $\ln x + x^2 - 8 = 0.$
3. $\ln x + 2x^2 - 6 = 0.$
4. $2 \ln x - x^2 + 5 = 0.$
5. $2 \ln x + 2x - 3 = 0.$
6. $\sin x + x^2 - 1 = 0.$
7. $\sin x + 2x^2 - 5 = 0.$
8. $\sin x - x + 3 = 0.$
9. $3 \sin x - x^2 + 4 = 0.$
10. $3 \sin x + x - 2 = 0.$
11. $\cos x + x^3 - 2 = 0.$
12. $\cos x + 2x - 3 = 0.$
13. $\cos x - x^3 + 2 = 0.$
14. $\ln x - x^2 + 5 = 0.$
15. $\ln x + 2x - 3 = 0.$
16. $2 \ln x + x - 2 = 0.$

17. $2 \ln x + x^2 - 3 = 0.$

18. $2 \ln x + 2x^2 - 6 = 0.$

19. $\sin x - x^2 + 4 = 0.$

20. $\sin x + x - 2 = 0.$

21. $3 \sin x + x^2 - 1 = 0.$

22. $\sin x + 2x^2 - 5 = 0.$

23. $3 \sin x - x + 3 = 0.$

24. $\cos x + x - 2 = 0.$

25. $\cos x + 3x - 6 = 0.$

26. $2 \ln x + 2x^2 - 3 = 0.$

27. $2 \ln x + x - 7 = 0.$

28. $2 \ln x + 0.01x^2 = 0.$

29. $\ln x + 0.25x - 3 = 0.$

30. $\ln x + x^2 - 3 = 0.$

2.4-§. Algebraik va transcendent tenglamalarni yarim (teng ikkiga) bo'lish usuli bilan yechish.

Bizda

$$f(x) = 0$$

ko'rinishdagi tenglama berilgan hamda $f(x)$ funksiya $[a; b]$ oraliqda aniqlangan va uzluksiz funksiya bo'lib, oraliqning chetki nuqtalari uchun

$$f(a)f(b) < 0$$

shart o'rinli bo'lsin.

Endi yarim(teng ikkiga) bo'lish usulining mohiyatini keltiramiz, buning uchun $[a; b]$ oraliqni teng ikkiga bo'lamiz va bu nuqtani x_0 bilan belgilaymiz, ya'ni

$$x_0 = \frac{a+b}{2}.$$

Agar $f(x_0) = 0$ tenglik o'rinli bo'lsa, u holda, x_0 berilgan tenglamaning yechimi bo'ladi. Aks holda, ya'ni $f(x_0) \neq 0$ bo'lsa, u holda $[a; x_0]$ va $[x_0; b]$ oraliqlarning birortasida $f(x)$ funksiya har xil ishorali qiymat qabul qiladi. Aynan shu oraliqni $[a_1; b_1]$ bilan belgilaymiz.

Agar yetarlicha kichik ε musbat son uchun $[a_1 - b_1] < \varepsilon$ tengsizlik o'rinli bo'lsa, u holda $[a_1; b_1]$ oraliqqa tegishli ixtiyoriy sonni berilgan tenglamaning taqribiy yechimi sifatida qabul qilish mumkin. Agar bu tengsizlik bajarilmasa, u holda $a = a_1$, $b = b_1$ deb olib hosil qilingan yangi $[a; b]$ oraliqni yana ikkiga bo'lib, yuqoridagi yarim bo'lish jarayonini davom ettirsak, ma'lum bir qadamdan so'ng $[a; b]$ oraliq tanlangan yetarlicha kichik ε sonidan ham kichik bo'ladi. Natijada taqribiy yechim sifatida hosil qilingan $[a; b]$ oraliqqa tegishli bo'lgan ixtiyoriy sonni olish mumkin.

Misol. $f(x) = x^3 - 20x + 12 = 0$ tenglamani oraliqlarni teng ikkiga bo'lish usuli bilan aniqlaydigan jarayonni ko'rsating.

Yechish. Tenglama 3 ta haqiqiy ildizga ega bo'lib, ular $(-5; -4)$, $(0; 1)$ va $(4; 5)$ oraliqlarda joylashgan. Biz $(0; 1)$ oraliqda ko'rib chiqamiz.

$$x_0 = \frac{a+b}{2} = \frac{0+1}{2} = \frac{1}{2}.$$

$$f(a) > 0, \quad f(1) < 0, \quad f(x_0) = f\left(\frac{1}{2}\right) > 0.$$

Demak

$$f\left(\frac{1}{2}\right)f(1) < 0$$

ekanligidan tenglamani yechmi $\left(\frac{1}{2}; 1\right)$ oraliqda joylashgan.

$$x_1 = \frac{x_0 + b}{2} = \frac{\frac{1}{2} + 1}{2} = \frac{3}{4}.$$

$$f(x_0) = f\left(\frac{1}{2}\right) > 0, \quad f(1) < 0, \quad f(x_1) = f\left(\frac{3}{4}\right) > 0.$$

Demak

$$f\left(\frac{1}{2}\right)f\left(\frac{3}{4}\right) < 0$$

ekanligidan tenglamani yechmi $\left(\frac{1}{2}; \frac{3}{4}\right)$ oraliqda joylashgan.

Ushbu jarayonni ketma-ket huddi shunday davom ettirib,

$$|x_i - x_{i-1}| < \epsilon$$

bo'lgunicha davom ettiramiz. Hisoblashlar davomida berilgan tengsizlik bajarilsa tenglamaning yechimi ϵ aniqliqda $x = x_i$ bo'ladi.

Algebraik va transcendent tenglamani yarim (teng ikkiga) bo'lish bilan yechishning $C\#$ dasturlash tilida dasturini keltiramiz.

```
class Program
{
    public static void Main(string[] args)
    {
        Console.WriteLine("Quyida chegarani kiriting");
        Console.Write("a=");
```

```
double a=Double.Parse(Console.ReadLine());
Console.WriteLine("Yuqori chegarani kiriting");
Console.Write("b=");
double b=Double.Parse(Console.ReadLine());
Console.WriteLine("Aniqlikni kiriting");
Console.Write("e=");
double e=Double.Parse(Console.ReadLine());
double [] x=new double[100];
int i=0;
n1: double c=(a+b)/2;
double f=f(c);
double fa=f(a);
if (Math.Abs(b-a)>e)
{if (fa*f<0)
{b=c; i=i+1;goto n1;}
else
a=c;i=i+1; goto n1;}
Console.WriteLine("Tenglamaning taqribiy yechimi
x="+c);
Console.WriteLine("Tenglamaning taqribiy
yechimi qadamlari soni N="+i);
Console.Write("Press any key to continue . . . ");
Console.ReadKey(true);
```

Izoh. $f(x)$ ning o'rniga tenglamada berilgan $f(x)$ funksiya-ning $C\#$ dagi ifodasi yoziladi.

Mavzuni mustahkamlash uchun misollar.

Grafik yoki analitik ravishda berilgan tenglama ildizi yotgan oraliqni ajrating va yarim (teng ikkiga) bo'lish usuli yordamida 0.001 aniqlikda yechimni toping.

1. $x^4 + 3x^2 - 4x - 1 = 0$

2. $x^3 - 2x^2 + x + 3 = 0$
3. $x^3 - 2x^2 + x + 1 = 0$
4. $x^3 - e^x + 3x = 0$
5. $x^2 - 3e^x - 2 = 0$
6. $2x^4 - 3x + e^x = 0$
7. $3^x - \ln(3^x + 1) = 0$
8. $2^x + \lg(x + 1) = 0$
9. $\ln(x + 1) - xe^x = 0$
10. $3 \ln(x^2 + 1) + 0.5x = 0$
11. $4x^2 + 0.5e^x - 3 = 0$
12. $2e^x + 3x + 1 = 0$
13. $\sin(x + 0.5) - x^2 = 0$
14. $\sin(e^x + 1) - 1 = 0$
15. $\log_3(x + 1) - \sin(x) = 0$
16. $\sin(2x) - 2x \cos(x + 1) = 0$
17. $x^2 - \sin(x^2 + 2.4x) = 0$
18. $x^2 + \sqrt{x + 1} \sin(x) = 0$
19. $\sin(x + \pi/3) - 2x \operatorname{tg}(x) = 0$
20. $0.5x + \sin(x + 1) = 0$
21. $3x - 4x^2 + 2 \sin(x) = 0$

22. $\sin(x) - 3^x + 1 = 0$

23. $\cos^2(x) + 2^x - 1 = 0$

24. $\sin 2(x) - 2x + 1/4 = 0$

25. $\arcsin(x) + 0.4x - 1 = 0$

26. $\operatorname{arctg}(x+1) - \sqrt{x+1} = 0$

27. $\arccos(x^2+1) + \ln(x) = 0$

28. $x^2 - \operatorname{tg}(\pi x) - 1 = 0$

29. $\cos(\ln x) - \sin(\ln(x)) - 2\ln(x) = 0$

30. $x + \operatorname{tg}(x/2) - \operatorname{ctg}(x/2) = 0$

CHIZIQLI ALGEBRAIK TENGLAMALAR SISTEMASINI YECHISH USULLARI

Bizga n noma'lumli n ta chiziqli algebraik tenglamalar sistemasi berilgan bo'lsin:

[illegible]

Sistemaning noma'lumlari oldidagi koeffitsiyentlardan quyidagi jadvalni tuzamiz:

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} \dots & a_{1n} \\ a_{21} & a_{22} \dots & a_{2n} \\ \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} \dots & a_{nn} \end{pmatrix} \quad i = \overline{1, n}; j = \overline{1, n} \quad (2)$$

1-ta'rif. (2) jadval n ta satr va n ta ustundan iborat bo'lib, u odatda kvadrat matritsa deb ataladi.

Matritsa tushunchasidan foydalanib, (1) sistemani quyidagicha matritsaviy ko'rinishda ifodalash mumkin:

bu yerda x izlanuvchi n o'lchovli vektor, B sistemaning o'ng tomonidagi ma'lum ozod hadlardan iborat n o'lchovli vektor, ya'ni:

$$x = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix} \quad B = \begin{Bmatrix} a_{1n+1} \\ a_{2n+1} \\ \vdots \\ a_{nn+1} \end{Bmatrix}$$

Agar A matritsa xosmas matritsa bo'lsa, ya'ni (2) matritsa-niing determinanti (aniqlovchisi) noldan farqli bo'lsa, ya'ni

$$\Delta = \det a \neq 0$$

u holda (1) sistema yagona yechimga ega bo'ladi.

Chiziqli algebraik tenglamalar sistemasini yechish usullarini odatda ikki, ya'ni aniq va iteratsiya usullari guruhiga ajratiladi.

2-ta'rif. Chekli arifmetik operatsiyasilarni ma'lum formulalar va aniq hisoblashlar yordamida yordamida bajarib, talab qilingan aniq yechimni topish chiziqli algebraik tenglamalar sistemasini yechishning aniq usuli deyiladi.

Aniq usullar guruhiga Gauss usuli, Kramer usuli, Jordan-Gaussning bosh elementlar usuli, kvadratik ildiz usuli, optimal (eng qulay, maqbul) yo'qotish usuli va hokazolar mansubdir.

3-ta'rif. Hisoblashlar garchand xatoliqsiz bajarilsada, sistemaning yechimini berilgan aniqlikda taqribiy hisoblashlar usullari iteratsiya usuli deyiladi.

Bu holda sistemaning yechimi nazariy jihatdan cheksiz hisoblashlar jarayonining natijasi sifatida topiladi. Iteratsiya usullari guruhiga oddiy iteratsiya (ketma-ket yaqinlashishlar) usuli, Zeydel usuli, relaksatsiya usuli va hokazolar mansubdir.

3.2-§. Chiziqli algebraik tenglamalar sistemasini Gauss usuli bilan yechish.

Gauss usuli mohiyatini (1) ko'rinishidagi chiziqli algebraik tenglamalar sistemasi uchun keltiramiz. Aytaylik, bu sistemada $a_{11} \neq 0$ bo'lsin (odatda a_{11} sistema birinchi satrining hal qiluvchi elementi deb yuritiladi), aks holda tenglamalar o'rnini almashtirish lozim. Sistemaning birinchi tenglamasini a_{11} ga bo'lib, ya'ni quyidagi tenglamani hosil qilamiz:

$$x_1 + a_{12}^{(1)}x_2 + \dots + a_{1n}^{(1)}x_n = a_{1n+1}^{(1)}$$

x_1 noma'lum (1) sistemaning har bir tenglamasidan yo'qotish uchun (3) tenglamaga ketma-ket a_{21}, a_{31} va hokazo a_{n1} larni ko'paytirib, mos holda (1) sistemaning ikkinchi, uchinchi va hokazo n -tenglamalaridan hadma-had ayiramiz.

Natijada

$$\left\{ \begin{array}{l} x_1 + a_{12}^{(1)}x_2 + \dots + a_{1n}^{(1)}x_n = a_{n+1}^{(1)} \\ a_{22}^{(1)}x_2 + \dots + a_{2n}^{(1)}x_n = a_{2n+1}^{(1)} \\ a_{32}^{(1)}x_2 + \dots + a_{3n}^{(1)}x_n = a_{3n+1}^{(1)} \\ \vdots \\ a_{m2}^{(1)}x_2 + \dots + a_{mn}^{(1)}x_n = a_{m+1}^{(1)} \end{array} \right. \quad (4)$$

ko‘rinishdagi sistemaga ega bo‘lamiz.

Yuqoridagi kabi, (4) sistemaning ikkinchi tenglamasini $a_{22} \neq 0$ ga bo'lib, yangi hosil qilingan tenglamaga ketma-ket a_{21} , a_{31} va hokazo a_{n1} larni ko'paytirib, mos holda (4) sistemaning uchinchi, to'rtinchi va hokazo n -tenglamalaridan ayirsak, bu tenglamalardan x_2 noma'lum yo'qotiladi. Bu jarayonni ketma-ket davom ettirib, (1) sistemaga ekvivalent bo'lgan quyidagi uchburchakli sistemani hosil qilamiz:

$$\begin{cases} x_1 + a_{12}^{(1)}x_2 + \dots + a_{1n}^{(1)}x_n = a_{1n+1}^{(1)} \\ x_2 + a_{23}^{(2)}x_3 + \dots + a_{2n}^{(2)}x_n = a_{2n+1}^{(2)} \\ x_3 + \dots + a_{3n}^{(3)}x_n = a_{3n+1}^{(3)} \\ \vdots \\ x_n = a_{nn+1}^{(1)} \end{cases} \quad (5)$$

sistemaning noma'lumlari oldidagi koeffitsiyentlari quyidagi formula yordamida hisoblanadi:

$$a_{kj}^{(k)} = a_{kj}^{(k-1)} / a_{kk}^{(k-1)}; a_{ij}^{(k)} = a_{ij}^{(k-1)} - a_{ik}^{(k-1)} a_{kj}^{(k)}$$

bundan $k + 1 \leq i \leq n, k + 1 \leq j \leq n + 1, k = \overline{1, n}$.

(5) uchburchakli sistemaning koeffitsiyentlarini topish jarayoni Gauss usulining to'g'ri yoli, sistemaning talab qilingan yechimini

$$x_n = a_{nn+1}^{(n)}; \quad x_i = a_{in+1}^{(i)} - \sum_{j=i+1}^n a_{ij}^{(i)} x_j, \quad i = n-1, n-2, \dots, 1$$

formulalar yordamida topish jarayoniga Gauss usulining teskari yoli deyiladi.

Misol. Berilgan chiziqli algebraik tenglamalar sistemasini Gauss usuli yordamida yeching.

$$\begin{cases} 3,5x_1 + 7,21x_2 - 0,35x_3 + 0x_4 = 2,5 \\ -7,5x_1 + 1,42x_2 + 2,11x_3 - 1,14x_4 = -3,5 \\ 2,1x_1 + 3,8x_2 + 0x_3 - 8,91x_4 = 5,55 \\ 0x_1 + 2,31x_2 + 9,27x_3 - 2,2x_4 = -4,12 \end{cases}$$

Yechish. Tenglamalar sistemasining yechimi mavjud va yagonaligini tekshirish maqsadida sistema koeffitsiyentlaridan tuzilgan asosiy matrisaning determinantlarini hisoblaymiz:

$$\begin{aligned} \det A &= \begin{vmatrix} 3,5 & 7,21 & -0,35 & 0 \\ -7,5 & 1,42 & 2,11 & -1,14 \\ 2,1 & 3,8 & 0 & -8,91 \\ 0 & 2,31 & 9,27 & -2,2 \end{vmatrix} = \\ &= (-1)^{1+1} 3,5 \begin{vmatrix} 1,42 & 2,11 & -1,14 \\ 3,8 & 0 & -8,91 \\ 2,71 & 9,27 & -2,2 \end{vmatrix} + \\ &+ (-1)^{2+1} (-7,5) \begin{vmatrix} 7,21 & 0,35 & 0 \\ 3,8 & 0 & -8,91 \\ 2,31 & 9,27 & -2,2 \end{vmatrix} \end{aligned}$$

$$+(-1)^{3+1}2,1 \begin{vmatrix} 7,21 & -0,35 & 0 \\ 1,42 & 2,11 & -1,14 \\ 2,31 & 9,27 & -2,2 \end{vmatrix} =$$

$$= 3,5 \cdot 51,34 + 7,5 \cdot 606,93 + 2,1 \cdot 16,172 = 4756,625.$$

Shunday qilib, $\det A = 4765,625 \neq 0$. Demak, berilgan sistemaning yechimi mavjud va yagona. Hisoblashlarda xatoliklar kam bo'lishi, ya'ni yengillik bo'lishi uchun x_1 noma'lum koeffitsiyentlaridan moduli bo'yicha eng kattasi joylashgan satrni dastlabki satr bilan o'rnini almashtiramiz, ya'ni

$$\begin{cases} -7,5x_1 + 1,42x_2 + 2,11x_3 - 1,14x_4 = -3,5 \\ 3,5x_1 + 76,21x_2 - 0,35x_3 = 2,5 \\ 2,1x_1 + 3,8x_2 - 8,91x_4 = 5,55 \\ 2,31x_2 + 9,27x_3 - 2,2x_4 = -4,12 \end{cases}$$

oxirgi sistemaning birinchi tenglamasidan (ya'ni 7,5 ga tenglikni xar ikkala tomonini bo'lib)

$$x_1 - 0,189x_2 - 0,281x_3 + 0,152x_4 = 0,48$$

tenglamani hosil qilamiz.

Tenglamani dastlab 3,5 ga va so'ngra 2,1 ga ko'paytirib, tenglamalar sistemasini mos ravishda ikkinchi va uchinchi tenglamasini hadma-had ayrib

$$\begin{cases} 7,872x_2 + 0,634x_3 - 0,152x_4 = 0,48 \\ 4,197x_2 + 0,59x_3 - 9,229x_4 = 4,542 \\ 2,31x_2 + 9,27x_3 - 2,2x_4 = -4,12 \end{cases}$$

sistemaga ega bo'lamiz. x_2 noma'lum oldidagi koeffitsientlar orasidagi eng kattasi sistemaning birinchi tenglamasi bo'lganligi sababli tenglamalar o'rnini almashtirishga xojat yo'q.

Oxirga sistemaning birinchi tenglamasidan

$$x_2 + 0,08x_3 - 0,068x_4 = 0,104$$

tenglamaga ega bo'lamiz. Yuqoridagi kabi oxirgi sistemani dastlab 4, 197 va so'ngra 2, 31 ga ko'paytirib, mos holda mazkur sistemaning ikkichi va uchinchi tenglamasidan hadma-had ayrisak,

$$\begin{cases} 0,254x_3 - 8,944x_1 = 4,106 \\ 9,085x_3 - 2,043x_4 = -4,36 \end{cases}$$

sistemaga ega bo'lamiz. Xatoliklar kam bo'lishi uchun oxirgi tenglamalar sistemasidagi tenglamalarni o'rnini almashtiramiz:

$$\begin{cases} 9,085x_3 - 2,043x_4 = -4,36 \\ 0,254x_3 - 8,944x_4 = 4,106 \end{cases}$$

ushbu tenglamalar sistemasining birinchi tenglamasidan quyidagiga ega bo'lamiz

$$x_3 - 0,225x_4 = -0,48$$

Bu tenglamani 0,254 ga ko'paytirib yuqoridagi tenglamalar sistemasining ikkinchi tenglamasini hadma-had ayiramiz va

$$x_4 = -0,446$$

ni aniqlaymiz. Topilgan x_4 yuqoridagi tenglamaga, so'ngra topilgan x_3 va x_4 ning son qiymatini yuqoridagi tenglamaga qo'yib x_2 , x_3 , x_4 ni topamiz. Shu tartibda topilgan x_2 , x_3 , x_4 qiymatlar orqali x_1 ham aniqlanadi, ya'ni

$$\begin{aligned} x_3 &= -0,48 + 0,225x_4 = \\ &= -0,48 - 0,225(-0,446) = -0,587 \\ x_2 &= 0,104 - 0,08x_3 + 0,068x_4 = 0,119 \\ x_1 &= 0,48 + 0,189x_2 + 0,281x_3 - 0,152x_4 = \\ &= 0,48 + 0,189 \cdot 0,119 + \\ &+ 0,281 \cdot (-0,587) + 0,152 \cdot -0,446 = 0,409 \end{aligned}$$

Mavzuni mustahkamlash uchun misollar.

Chiziqli algebraik tenglamalar sistemasini Gauss usuli yordamida 0.001 aniqlikda yeching.

$$1. \begin{cases} 3.2x_1 - 11.5x_2 + 3.8x_3 = 2.8 \\ 0.8x_1 + 1.3x_2 - 6.4x_3 = -6.5 \\ 2.4x_1 + 7.2x_2 - 1.2x_3 = 4.5 \end{cases}$$

$$2. \begin{cases} 2.4x_1 + 3.7x_2 - 8.3x_3 = 2.3 \\ 1.8x_1 + 4.3x_2 + 1.2x_3 = -1.2 \\ 3.4x_1 - 2.3x_2 + 5.2x_3 = 3.5 \end{cases}$$

$$3. \begin{cases} 5.4x_1 - 2.4x_2 + 3.8x_3 = 5.5 \\ 2.5x_1 + 6.8x_2 - 1.1x_3 = 4.3 \\ 2.7x_1 - 0.6x_2 + 1.5x_3 = -3.5 \end{cases}$$

$$4. \begin{cases} 2.4x_1 + 2.5x_2 - 2.9x_3 = 4.5 \\ 0.8x_1 + 3.5x_2 - 1.4x_3 = 3.2 \\ 1.5x_1 - 2.3x_2 + 8.6x_3 = -5.5 \end{cases}$$

$$5. \begin{cases} 0.9x_1 + 2.7x_2 - 3.8x_3 = 2.4 \\ 2.5x_1 + 5.8x_2 - 0.5x_3 = 3.5 \\ 4.5x_1 - 2.1x_2 + 3.2x_3 = -1.2 \end{cases}$$

$$6. \begin{cases} 1.5x_1 + 2.3x_2 - 3.7x_3 = 4.5 \\ 2.8x_1 + 3.4x_2 + 5.8x_3 = -3.2 \\ 1.2x_1 + 7.3x_2 - 2.3x_3 = 5.6 \end{cases}$$

$$7. \begin{cases} 6.3x_1 + 5.2x_2 - 0.6x_3 = 1.5 \\ 3.4x_1 - 2.3x_2 + 3.4x_3 = 2.7 \\ 0.8x_1 + 1.4x_2 + 3.5x_3 = -2.3 \end{cases}$$

$$8. \begin{cases} 3.7x_1 - 2.3x_2 + 4.5x_3 = 2.4 \\ 2.5x_1 + 4.7x_2 - 7.8x_3 = 3.5 \\ 1.6x_1 + 5.3x_2 + 1.3x_3 = -2.4 \end{cases}$$

$$9. \begin{cases} 4.1x_1 + 5.2x_2 - 5.8x_3 = 7.1 \\ 3.8x_1 - 3.1x_2 + 4.1x_3 = 5.3 \\ 7.8x_1 + 5.3x_2 - 6.3x_3 = 5.8 \end{cases}$$

$$10. \begin{cases} 3.7x_1 + 3.1x_2 + 4.1x_3 = 5.1 \\ 3.9x_1 + 4.5x_2 - 4.8x_3 = 4.9 \\ -2.1x_1 - 3.7x_2 + 1.8x_3 = 2.7 \end{cases}$$

$$11. \begin{cases} 7.1x_1 + 6.8x_2 + 6.1x_3 = 7.1 \\ 5.1x_1 + 4.8x_2 + 5.3x_3 = 6.1 \\ 8.2x_1 + 7.8x_2 + 7.1x_3 = 5.8 \end{cases}$$

$$12. \begin{cases} 3.3x_1 + 3.7x_2 + 4.2x_3 = 5.8 \\ 2.7x_1 + 2.3x_2 - 2.9x_3 = 6.1 \\ 4.1x_1 + 4.8x_2 - 5.1x_3 = 6.9 \end{cases}$$

$$13. \begin{cases} 2.8x_1 + 3.8x_2 - 3.2x_3 = 4.5 \\ 2.5x_1 - 2.8x_2 + 3.3x_3 = 7.1 \\ 6.5x_1 - 7.3x_2 + 4.8x_3 = 6.3 \end{cases}$$

$$14. \begin{cases} 1.7x_1 - 2.2x_2 + 2.9x_3 = 1.9 \\ 2.1x_1 + 1.9x_2 - 2.3x_3 = 2.8 \\ 4.2x_1 + 3.9x_2 - 3.1x_3 = 5.1 \end{cases}$$

$$15. \begin{cases} 3.8x_1 + 4.2x_2 - 2.3x_3 = 4.8 \\ -2.1x_1 + 4.1x_2 - 6.1x_3 = 3.3 \\ 1.8x_1 + 1.2x_2 - 2.2x_3 = 5.8 \end{cases}$$

$$16. \begin{cases} 7.7x_1 + 5.2x_2 + 4.9x_3 = 1.8 \\ 3.1x_1 + 1.2x_2 + 1.8x_3 = 2.3 \\ 4.5x_1 + 3.2x_2 + 2.8x_3 = 3.4 \end{cases}$$

$$17. \begin{cases} 5.4x_1 - 6.2x_2 - 0.5x_3 = 0.6 \\ 3.4x_1 + 2.3x_2 + 0.8x_3 = -0.8 \\ 2.4x_1 - 1.1x_2 + 3.8x_3 = 1.8 \end{cases}$$

$$18. \begin{cases} 3.8x_1 + 6.7x_2 - 1.2x_3 = 5.2 \\ 6.4x_1 + 1.3x_2 - 2.7x_3 = 3.8 \\ 2.4x_1 - 4.5x_2 + 3.5x_3 = -0.8 \end{cases}$$

$$19. \begin{cases} 4.4x_1 - 3.5x_2 + 7.4x_3 = 2.5 \\ 3.1x_1 - 0.6x_2 - 2.3x_3 = -1.5 \\ 0.8x_1 + 7.4x_2 - 0.5x_3 = 6.4 \end{cases}$$

$$20. \begin{cases} 2.8x_1 + 0.9x_2 - 1.5x_3 = 3.5 \\ 4.4x_1 - 28x_2 + 6.7x_3 = 2.6 \\ 5.1x_1 + 3.7x_2 - 1.4x_3 = -0.6 \end{cases}$$

$$21. \begin{cases} 5.6x_1 + 2.7x_2 - 1.7x_3 = 1.9 \\ 3.4x_1 - 3.6x_2 - 6.6x_3 = -2.4 \\ 0.8x_1 + 1.3x_2 + 3.7x_3 = 1.2 \end{cases}$$

$$22. \begin{cases} 3.6x_1 + 1.8x_2 - 4.7x_3 = 3.8 \\ 2.7x_1 - 3.6x_2 + 1.9x_3 = 0.4 \\ 1.5x_1 + 4.5x_2 + 3.3x_3 = -1.6 \end{cases}$$

$$23. \begin{cases} 5.4x_1 - 2.3x_2 + 3.4x_3 = -3.5 \\ 4.2x_1 + 1.7x_2 - 2.3x_3 = 2.7 \\ 3.4x_1 + 2.4x_2 + 7.3x_3 = 1.9 \end{cases}$$

$$24. \begin{cases} 3.2x_1 - 2.5x_2 + 3.7x_3 = 6.5 \\ 0.4x_1 + 0.6x_2 + 1.7x_3 = -0.4 \\ 1.6x_1 + 2.3x_2 - 1.5x_3 = 4.2 \end{cases}$$

$$25. \begin{cases} 7.6x_1 + 5.8x_2 + 4.7x_3 = 10.1 \\ 3.8x_1 + 4.2x_2 + 2.5x_3 = 9.7 \\ 2.9x_1 + 2.2x_2 + 3.8x_3 = 7.4 \end{cases}$$

$$26. \begin{cases} 3.3x_1 + 2.1x_2 + 2.9x_3 = 0.8 \\ 4.1x_1 + 3.7x_2 + 4.8x_3 = 5.3 \\ 2.5x_1 + 1.9x_2 + 1.1x_3 = 3.2 \end{cases}$$

$$27. \begin{cases} 9.1x_1 + 5.6x_2 + 7.8x_3 = 9.8 \\ 3.8x_1 + 5.1x_2 + 2.8x_3 = 6.3 \\ 4.1x_1 + 5.7x_2 + 1.2x_3 = 5.8 \end{cases}$$

$$28. \begin{cases} 3.1x_1 + 28x_2 + 1.9x_3 = 0.3 \\ 1.9x_1 + 3.2x_2 + 2.1x_3 = 2.2 \\ 7.5x_1 + 3.8x_2 + 4.8x_3 = 5.4 \end{cases}$$

$$29. \begin{cases} 1.7x_1 + 2.9x_2 + 1.8x_3 = 0.8 \\ 2.1x_1 + 3.3x_2 + 1.9x_3 = 1.3 \\ 4.2x_1 - 1.7x_2 + 1.3x_3 = 2.8 \end{cases}$$

$$30. \begin{cases} 2.7x_1 + 3.3x_2 + 1.3x_3 = 2.1 \\ 3.5x_1 - 1.7x_2 + 2.8x_3 = 1.7 \\ 4.1x_1 + 5.8x_2 - 1.8x_3 = 0.8 \end{cases}$$

3.3-§. Chiziqli tenglamalar sistemasini iteratsiya va Zeydel usuli bilan yechish

Faraz qilaylik,

$$Ax = b$$

chiziqli algebraik tenglamalar sistemasi biror usul bilan

$$x = Bx + c$$

ko'rinishiga keltirilgan bo'lsin. Ixtiyoriy $x^{(0)}$ vektor olib uni boshlang'ich yaqinlashish deylik. Agar keyingi yaqinlashishlar

$$x^{(k)} = Bx^{(k)} + c, \quad k = 0, 1, \dots$$

rekurrent formulalar yordamida aniqlansa, bunday metod oddiy iteratsiya metodi deyiladi.

Agar (3) ketma-ketlikning limiti x^* mavjud bo'lsa, bu limit (2) sistemaning (shu bilan (1) sistemaning ham) yechimi bo'ladi.

Haqiqatan ham (3) da limitga o'tsak, $x^* = Bx^* + c$ kelib chiqadi. Oddiy iteratsiya metodining yaqinlashishini quyidagi teorema ko'rsatadi.

Teorema. (3) oddiy iteratsion jarayon ixtiyoriy boshlang'ich $x^{(0)}$ da yaqinlashuvchi bo'lishi uchun B matritsaning barcha xos sonlarining modullari birdan kichik bo'lishi zarur va yetarli.

Teorema. (3) iteratsion jarayon yaqinlashuvchi bo'lishligi uchun B matritsaning biron normasi birdan kichik bo'lishi yetarli.

Natija. (3) iteratsion jarayon yaqinlashuvchi bo'lishi uchun B matritsa elementlari quyidagi

$$\max_{i \leq s \leq n} \sum_{j=1}^n |b_{sj}| < 1, \quad \max_{1 \leq n} \sum_{i=1}^n |b_{vi}| < 1 \quad \sqrt{\sum_{i,j=1}^n |b_{ij}|^2} < 1$$

tengsizliklarning birortasini qanoatlantirishi yetarlidir. (1) tenglamalar sistemasini (2) ko'rinishga keltirish masalasiga to'xtalib o'tamiz.

Agar $a_{ii} \neq 0$ bo'lib, quyidagi

$$\max_i \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| < |a_{ii}|, \quad \max_j \sum_{\substack{i=1 \\ i \neq j}}^n |a_{ij}| < |a_{jj}|$$

tengsizliklarning birortasi bajarilsa, u holda (2) quyidagi ko'rinishda bo'ladi:

$$\begin{cases} x_1 = 0 \cdot x_1 - \frac{a_{12}}{a_{11}}x_2 - \frac{a_{13}}{a_{11}}x_3 - \dots - \frac{a_{1n}}{a_{11}}x_n + \frac{b_1}{a_{11}} \\ x_2 = \frac{a_{21}}{a_{22}}x_1 - 0 \cdot x_2 - \frac{a_{23}}{a_{22}}x_3 - \dots - \frac{a_{2v}}{a_{22}}x_n + \frac{b_2}{a_{22}} \\ \dots \\ x_n = \frac{a_{n1}}{a_{nn}}x_1 - \frac{a_{n2}}{a_{nv}}x_3 - \frac{a_{n3}}{a_{nn}}x_3 - \dots - 0 \cdot x_n + \frac{b_n}{a_{nn}} \end{cases}$$

ya'ni

$$B = \begin{bmatrix} 0 - \frac{a_{12}}{a_{11}} - \frac{a_{13}}{a_{11}} & \dots & -\frac{a_{1n}}{a_{11}} \\ \frac{a_{21}}{a_{22}} - 0 - \frac{a_{23}}{a_{22}} & \dots & -\frac{a_{2v}}{a_{22}} \\ \dots & \dots & \dots \\ \frac{a_{n1}}{a_{nn}} - \frac{a_{n2}}{a_{nn}} - \frac{a_{n3}}{a_{nn}} & \dots & 0 \end{bmatrix}$$

bo'lib, ixtiyoriy $x^{(0)}$ da (3) iteratsion jarayon yaqinlashuvchi bo'ladi.

Agar (7), (8) tengsizliklarning hech qaysisi bajarilmasa, (1) tenglamalar sistemi ustida shunday chiziqli almashtirishlar qilish kerakki, yuqoridagi tengsizliklarning birortasi bajarilsin. Xususan, berilgan tenglamalardan shundaylari ajratib olinadiki, bu

$$\begin{aligned}
 & \dots\dots\dots \\
 x_i^{(k+1)} &= \beta_i + \sum_{j=1}^n a_{ij}^{(k)} x_j^{(k)} \\
 k &= 0, 1, \dots (a_{ii} = 0)
 \end{aligned} \tag{3}$$

ko‘rinishida bo‘ladi.

Zeydel usuli bilan ketma-ket yaqinlashish (iteratsiya) usuli-ning boshqacha ko‘rinishi bo‘lib, unda $(k + 1)$ - yaqinlashishni hisoblashda endigina topilgan $x_1, x_2, \dots, x_{i-1}$ noma‘lumning $(k + 1)$ - yaqinlashishi (iteratsiya usulida k yaqinlashish) e‘tiborga olinadi.

x_i noma‘lumning k -yaqinlashishi ma‘lum bo‘lsin, u holda $(k + 1)$ -yaqinlashish quyidagi formula bilan topiladi:

$$\begin{aligned}
 x_1^{(k+1)} &= \beta_1 + \sum_{j=1}^n a_{1j}^{(k)} x_j^{(k)} \\
 x_2^{(k+1)} &= \beta_2 + a_{21} x_1^{(k)} + \sum_{j=2}^n a_{2j}^{(k)} x_j^{(k)} \\
 & \dots\dots\dots \\
 x_i^{(k+1)} &= \beta_i + \sum_{j=1}^n a_{ij}^{(k)} x_j^{(k+1)} + \sum_{j=i}^n a_{ij}^{(k)} x_j^{(k)} \\
 x_n^{(k+1)} &= \beta_n + \sum_{i=1}^n a_{ni} x_i^{(k+1)} + a_{nn} x_n^{(k)}
 \end{aligned} \tag{4}$$

Agar keltirilgan (2) sistema uchun

$$\sum_{j=1}^n |a_{ij}| \leq 1, \quad (i = \overline{1, n}) \quad \text{yoki} \quad \sum_{i=1}^n |a_{ij}| \leq 1 \quad (j = \overline{1, n})$$

shartlarning birortasi o‘rinli bo‘lsa, u holda (4) iteratsiya jarayoni boshlang‘ich yaqinlashishning qanday tanlanishidan qat’iy nazar berilgan tenglamaning yagona yechimiga yaqinlashadi.

(1) berilgan sistema uchun iteratsiya usulining yaqinlashish sharti

$$|a_{ii}| \geq \sum_{i \neq j} a_{ij} \quad (j = \overline{1, n}) \quad (5)$$

tengsizlikning bajarilishidir, ya’ni asosiy diagonal elementlari-ning moduli qolgan elementlar modullarining yig‘indisidan katta yoki teng bo‘lishidir.

Eslatma. (5) shart faqat yetarli shart bo‘la oladi, lekin zaruriy shart bo‘lmasligi ham mumkin, ba’zan (5) shart buzilganda ham iteratsiya jarayoni yaqinlashuvchi bo‘lishi mumkin.

Har bir iteratsiya jarayoni qadamidan keyin yaqinlashish aniqligini tekshirib borish lozim. Buning uchun

$$\Delta = \max_i \left\{ \left| x_i^{(k+1)} - x_i^{(k)} \right| \right\}$$

ni hisoblash kifoya, agar Δ oldindan tanlangan aniqlikdan kichik bo‘lsa, iteratsiya jarayoni to‘xtatiladi (ya’ni $\Delta \leq \varepsilon$ bo‘lsa).

Misol. Ushbu

$$\begin{cases} 20.9x_1 + 1.2x_2 + 2.1x_3 + 0.9x_4 = 21.70 \\ 1.2x_1 + 21.2x_2 + 1.5x_3 + 2.5x_4 = 27.46 \\ 2.1x_1 + 1.5x_2 + 19.8x_3 + 1.3x_4 = 28.76 \\ 0.9x_1 + 2.5x_2 + 1.3x_3 + 32.1x_4 = 49.72 \end{cases}$$

chiziqli tenglamalar sistemasini oddiy iteratsiya usuli yordamida $\varepsilon = 10^{-3}$ aniqlikda yeching.

Yechish. Iteratsiya jarayonini qurish uchun tenglamalar sis-

temasini qulay shaklga keltiraylik:

$$\begin{cases} x_1 = \frac{1}{20.9} (21.70 - 1.2x_2 - 2.1x_3 - 0.9x_4) \\ x_2 = \frac{1}{21.2} (27.46 - 1.2x_1 - 1.5x_3 - 2.5x_4) \\ x_3 = \frac{1}{19.8} (28.76 - 2.1x_1 - 1.5x_2 - 1.3x_4) \\ x_4 = \frac{1}{32.1} (49.72 - 0.9x_1 - 2.5x_2 - 1.3x_3) \end{cases}$$

$$\sum_{j=1}^4 |c_{1j}| = |c_{11}| + |c_{12}| + |c_{13}| + |c_{14}| = \frac{1.2 + 2.1 + 0.9}{20.9} \approx 0.20$$

$$\sum_{j=1}^4 |c_{2j}| \approx 0.24, \quad \sum_{j=1}^4 |c_{3j}| \approx 0.25, \quad \sum_{j=1}^4 |c_{4j}| \approx 0.15$$

$$\|C\|_{\infty} = \max_{1 \leq i \leq 4} \sum_{j=1}^4 |c_{ij}| = \max\{0.20; 0.24; 0.25; 0.15\} = 0.25$$

Demak, yaqinlashish koeffitsiyenti $q = \|C\|_{\infty} = 0.25 < 1$. q ning bunday qiymatida iteratsion jarayonning yaqinlashish tezligi yuqori bo'ladi, chunki $q/(1 - q) = 1/3$.

Endi nolinchii yaqinlashish sifatida ixtiyoriy sonlarni tanlab olamiz (Biz hammasini β_i deb oldik):

$$x^{(0)} = \begin{pmatrix} 1.038 \\ 1.295 \\ 1.453 \\ 1.549 \end{pmatrix}.$$

Birinchii yaqinlashishni hisoblaymiz.

$$x_1^{(1)} = \frac{1}{20.9} \left(21.70 - 1.2x_2^{(0)} - 2.1x_3^{(0)} - 0.9x_4^{(0)} \right) = 1.0382,$$

$$x_2^{(1)} = \frac{1}{21.2} \left(27.46 - 1.2x_1^{(0)} - 1.5x_3^{(0)} - 2.5x_4^{(0)} \right) = 0.95,$$

$$x_3^{(1)} = \frac{1}{19.8} \left(28.76 - 2.1x_1^{(0)} - 1.5x_2^{(0)} - 1.3x_4^{(0)} \right) = 1.14,$$

$$x_4^{(1)} = \frac{1}{32.1} \left(49.72 - 0.9x_1^{(0)} - 2.5x_2^{(0)} - 1.3x_3^{(0)} \right) = 1.36.$$

Demak,

$$\mathbf{x}^{(1)} = \begin{pmatrix} 0.75 \\ 0.95 \\ 1.14 \\ 1.36 \end{pmatrix}$$

Aniqlikni hisoblaymiz.

$$\frac{q}{1-q} \left\| x_i^{(k)} - x_i^{(k-1)} \right\| = \frac{1}{5} \left\| x_i^{(k)} - x_i^{(k-1)} \right\|$$

Ikkinch iteratsiya

$$\mathbf{x}^{(2)} = \begin{pmatrix} 0.8106 \\ 1.0118 \\ 1.2117 \\ 1.4077 \end{pmatrix}.$$

Uchinchi iteratsiya

$$\mathbf{x}^{(3)} = \begin{pmatrix} 0.7978 \\ 0.9977 \\ 1.1975 \\ 1.3983 \end{pmatrix}.$$

Shunday hisoblashlarni ketma-ket bajaramiz.

Mavzuni mustahkamlash uchun misollar.

Chiziqli algebraik tenglamalar sistemasini oddiy iteratsiya va Zeydel usuli yordamida 0.001 aniqlikda yeching.

$$1. \begin{cases} x_1 = 0.32x_1 - 0.23x_2 + 0.41x_3 - 0.06x_4 + 0.67, \\ x_2 = 0.18x_1 + 0.12x_2 - 0.33x_3 - 0.88, \\ x_3 = 0.12x_1 + 0.32x_2 - 0.05x_3 + 0.67x_4 - 0.18, \\ x_4 = 0.05x_1 - 0.11x_2 + 0.09x_3 - 0.12x_4 + 1.44. \end{cases}$$

$$2. \begin{cases} x_1 = 0.34x_2 + 0.23x_2 - 0.06x_4 + 1.42, \\ x_2 = 0.11x_1 - 0.23x_2 - 0.18x_3 + 0.36x_4 - 0.66, \\ x_3 = 0.23x_1 - 0.12x_2 + 0.15x_3 - 0.35x_4 + 1.08, \\ x_4 = 0.12x_1 + 0.11x_2 - 0.47x_3 + 0.17x_4 + 1.72. \end{cases}$$

$$3. \begin{cases} x_1 = 0.06x_1 + 0.17x_2 + 0.34x_3 + 0.16x_4 + 2.43, \\ x_2 = 0.32x_1 + 0.23x_2 - 0.35x_4 - 1.12, \\ x_3 = 0.16x_1 - 0.08x_2 - 0.12x_4 + 0.43, \\ x_4 = 0.09x_1 + 0.21x_2 - 0.13x_3 + 0.83. \end{cases}$$

$$4. \begin{cases} x_1 = 0.08x_2 - 0.23x_3 + 0.36x_4 + 1.37 \\ x_2 = 0.16x_1 - 0.22x_2 + 0.18x_3 - 2.38 \\ x_3 = 0.15x_1 + 0.12x_2 + 0.35x_3 - 0.17x_4 + 0.38 \\ x_4 = 0.25x_1 + 0.21x_2 - 0.19x_3 + 0.03x_4 + 0.64 \end{cases}$$

$$5. \begin{cases} x_1 = 0.32x_1 - 0.13x_2 - 0.08x_3 + 0.16x_4 + 2.42 \\ x_2 = 0.17x_1 - 0.22x_2 + 0.13x_3 - 0.21x_4 + 1.48 \\ x_3 = 0.05x_1 - 0.08x_2 + 0.34x_4 - 0.16 \\ x_4 = 0.12x_1 + 0.11x_2 - 0.19x_3 + 0.06x_4 + 1.64 \end{cases}$$

$$6. \begin{cases} x_1 = 0.12x_1 + 0.23x_2 - 0.33x_3 + 0.07x_4 + 0.17 \\ x_2 = 0.45x_1 - 0.23x_3 + 0.07x_4 - 0.38 \\ x_3 = 0.12x_1 - 0.08x_3 + 0.77x_4 + 0.88 \\ x_4 = 0.08x_1 + 0.09x_2 + 0.39x_3 + 0.22x_4 - 1.74 \end{cases}$$

$$7. \begin{cases} x_1 = 0.03x_1 - 0.05x_2 + 0.21x_3 - 0.36x_4 + 0.43 \\ x_2 = 0.28x_1 + 0.52x_2 - 0.83x_3 + 0.07x_4 - 1.8 \\ x_3 = 0.33x_1 + 0.12x_2 - 0.08x_3 - 0.07x_4 - 0.8 \\ x_4 = 0.08x_1 + 0.17x_2 + 0.29x_3 + 0.32x_4 + 1.7 \end{cases}$$

$$\begin{aligned}
 8. \quad & \begin{cases} x_1 = 0.17x_2 - 0.31x_3 + 0.16x_4 - 1.2 \\ x_2 = 0.18x_1 + 0.43x_3 - 0.08x_4 + 0.38 \\ x_3 = 0.22x_1 + 0.18x_2 + 0.25x_3 + 0.07x_4 + 0.48 \\ x_4 = 0.08x_1 + 0.07x_2 + 0.71x_3 + 0.04x_4 - 1.24 \end{cases} \\
 9. \quad & \begin{cases} x_1 = 0.02x_1 + 0.03x_2 - 0.62x_3 + 0.06x_4 - 1.6 \\ x_2 = 0.03x_1 + 0.28x_2 + 0.33x_3 - 0.07x_4 + 1.1 \\ x_3 = 0.09x_1 + 0.12x_2 + 0.42x_3 + 0.27x_4 - 1.7 \\ x_4 = 0.19x_1 - 0.21x_2 + 0.08x_3 + 0.37x_4 + 1.4 \end{cases} \\
 10. \quad & \begin{cases} x_1 = 0.52x_1 + 0.03x_2 + 0.11x_4 - 0.22 \\ x_2 = 0.07x_1 - 0.38x_2 - 0.03x_3 + 0.41x_4 + 1.8 \\ x_3 = 0.04x_1 + 0.42x_2 + 0.15x_3 - 0.07x_4 - 1.1 \\ x_4 = 0.15x_1 + 0.17x_2 - 0.13x_3 + 0.19x_4 + 0.34 \end{cases} \\
 11. \quad & \begin{cases} x_1 = 0.23x_1 - 0.17x_3 + 0.06x_4 + 0.21 \\ x_2 = 0.52x_2 + 0.12x_3 + 0.17x_4 - 1.18 \\ x_3 = 0.16x_1 - 0.18x_2 + 0.21x_4 - 0.88 \\ x_4 = 0.11x_1 + 0.21x_2 + 0.03x_3 + 0.02x_4 + 0.74 \end{cases} \\
 12. \quad & \begin{cases} x_1 = 0.15x_1 + 0.05x_2 - 0.08x_3 + 0.16x_4 - 0.47 \\ x_1 = 0.32x_1 - 0.42x_2 - 0.13x_3 + 0.11x_4 + 1.24 \\ x_3 = 0.17x_1 + 0.06x_2 - 0.08x_3 + 0.12x_4 + 1.18 \\ x_4 = 0.21x_1 - 0.16x_2 + 0.39x_3 - 0.84 \end{cases} \\
 13. \quad & \begin{cases} x_1 = 0.17x_1 + 0.26x_2 - 0.11x_3 - 0.16x_4 - 1.41 \\ x_2 = 0.13x_1 - 0.12x_2 + 0.09x_3 - 0.06x_4 + 0.48 \\ x_3 = 0.12x_1 + 0.05x_2 - 0.03x_3 + 0.12x_4 - 2.38 \\ x_4 = 0.13x_1 + 0.18x_2 + 0.24x_3 + 0.45x_4 + 0.72 \end{cases} \\
 14. \quad & \begin{cases} x_1 = 0.23x_1 + 0.21x_2 + 0.06x_3 - 0.34x_4 + 1.47 \\ x_2 = 0.05x_1 + 0.33x_3 + 0.12x_4 - 0.57 \\ x_3 = 0.35x_1 - 0.27x_2 - 0.05x_4 + 0.68 \\ x_4 = 0.12x_1 - 0.43x_2 + 0.34x_3 - 0.22x_4 - 2.14 \end{cases} \\
 15. \quad & \begin{cases} x_1 = 0.15x_1 + 0.23x_2 + 0.11x_3 + 0.18x_4 - 1.41 \\ x_2 = 0.12x_1 - 0.14x_2 + 0.08x_3 + 0.09x_4 - 0.83 \\ x_3 = 0.16x_1 + 0.24x_2 - 0.35x_4 + 1.18 \\ x_4 = 0.23x_1 - 0.08x_2 + 0.59x_3 + 0.22x_4 + 0.64 \end{cases}
 \end{aligned}$$

$$\begin{aligned}
 16. \quad & \begin{cases} x_1 = 0.22x_1 - 0.14x_2 + 0.06x_3 - 0.16x_4 + 1.27 \\ x_2 = 0.12x_1 + 0.32x_3 - 0.18x_4 = 0.78 \\ x_3 = 0.08x_1 - 0.12x_2 + 0.23x_3 + 0.32x_4 - 0.58 \\ x_4 = 0.25x_1 + 0.21x_2 + 0.19x_3 + 1.51 \end{cases} \\
 17. \quad & \begin{cases} x_1 = 0.12x_1 - 0.23x_2 + 0.25x_3 - 0.16x_4 + 1.07 \\ x_1 = 0.14x_1 + 0.34x_2 - 0.18x_3 + 0.24x_4 - 0.89 \\ x_1 = 0.33x_1 + 0.03x_2 + 0.45x_3 - 0.37x_4 + 1.18 \\ x_4 = 0.12x_1 - 0.05x_2 + 0.12x_3 - 1.14 \end{cases} \\
 18. \quad & \begin{cases} x_1 = 0.08x_2 - 0.03x_3 - 0.06x_4 - 1.7 \\ x_2 = 0.54x_1 + 0.24x_3 - 0.08x_4 + 0.81 \\ x_3 = 0.33x_1 - 0.37x_2 + 0.25x_4 - 0.98 \\ x_4 = 0.11x_1 + 0.03x_2 + 0.52x_3 + 0.14 \end{cases} \\
 19. \quad & \begin{cases} x_1 = 0.05x_1 - 0.06x_2 - 0.12x_3 + 0.13x_4 - 2.17 \\ x_2 = 0.04x_1 - 0.14x_2 + 0.68x_3 + 0.11x_4 + 1.8 \\ x_1 = 0.34x_1 + 0.08x_2 - 0.05x_3 + 0.47x_4 - 2.18 \\ x_4 = 0.11x_1 + 0.15x_2 - 0.03x_4 - 0.8 \end{cases} \\
 20. \quad & \begin{cases} x_1 = 0.07x_1 - 0.08x_2 + 0.11x_3 - 0.18x_4 - 0.51 \\ x_2 = 0.14x_1 - 0.42x_2 + 0.21x_4 + 1.18 \\ x_3 = 0.13x_1 + 0.31x_2 - 0.19x_4 - 1.02 \\ x_4 = 0.08x_1 - 0.33x_2 + 0.28x_4 - 0.28 \end{cases} \\
 21. \quad & \begin{cases} x_1 = 0.21x_2 - 0.12x_3 + 0.31x_4 + 2.7 \\ x_2 = 0.34x_1 - 0.11x_3 + 0.21x_4 - 1.5 \\ x_3 = 0.11x_1 + 0.23x_2 - 0.42x_4 + 1.28 \\ x_4 = 0.17x_1 - 0.21x_2 + 0.31x_3 - 0.18 \end{cases} \\
 22. \quad & \begin{cases} x_1 = 0.15x_1 - 0.08x_2 + 0.32x_3 - 0.23x_4 - 0.87 \\ x_2 = -0.21x_1 + 0.04x_2 + 0.11x_3 + 0.31x_4 - 0.68 \\ x_3 = 0.51x_1 - 0.06x_2 + 0.07x_3 - 0.17x_4 + 1.78 \\ x_4 = 0.31x_1 - 0.45x_2 - 1.28 \end{cases} \\
 23. \quad & \begin{cases} x_1 = 0.12x_1 + 0.26x_2 - 0.19x_3 - 0.17x_4 + 1.23 \\ x_2 = -0.18x_1 - 0.44x_3 + 0.18x_4 - 0.31 \\ x_3 = 0.12x_1 + 0.13x_2 - 0.35x_3 + 0.07x_4 - 0.48 \\ x_4 = 0.31x_1 - 0.05x_2 + 0.05x_3 - 0.27x_4 - 0.16 \end{cases}
 \end{aligned}$$

$$\begin{aligned}
 24. \quad & \begin{cases} x_1 = 0.14x_1 + 0.31x_2 - 0.17x_3 + 0.21x_4 - 1.73 \\ x_2 = -0.218x_1 + 0.32x_3 + 0.19x_4 + 0.61 \\ x_3 = 0.32x_1 - 0.15x_2 + 0.05x_3 - 0.17x_4 - 0.88 \\ x_4 = 0.11x_1 + 0.25x_2 - 0.13x_3 + 0.96 \end{cases} \\
 25. \quad & \begin{cases} x_1 = 0.13x_1 + 0.22x_2 - 0.41x_3 - 0.04x_4 + 2.13 \\ x_2 = 0.23x_1 - 0.31x_3 + 0.11x_4 - 0.18 \\ x_3 = 0.06x_1 + 0.13x_2 - 0.21x_4 + 1.48 \\ x_4 = 0.71x_1 - 0.04x_2 - 0.05x_3 + 2.46 \end{cases} \\
 26. \quad & \begin{cases} x_1 = 0.17x_1 - 0.24x_2 - 0.12x_3 + 0.14x_4 - 1.33 \\ x_2 = 0.11x_1 + 0.22x_2 - 0.37x_3 + 0.27x_4 + 0.81 \\ x_3 = 0.05x_1 - 0.12x_2 + 0.13x_3 - 0.17x_4 - 1.16 \\ x_4 = 0.12x_1 + 0.04x_2 + 0.05x_3 + 0.57 \end{cases} \\
 27. \quad & \begin{cases} x_1 = 0.33x_1 - 0.28x_2 + 0.02x_3 + 1.43 \\ x_2 = 0.31x_1 - 0.21x_2 - 0.35x_3 + 1.41 \\ x_3 = 0.12x_1 + 0.08x_2 - 0.13x_3 - 0.15x_4 - 1.06 \\ x_4 = 0.14x_1 - 0.34x_2 - 0.15x_3 - 1.47 \end{cases} \\
 28. \quad & \begin{cases} x_1 = 0.26x_1 - 0.14x_2 + 0.02x_3 + 0.24x_4 + 1.73 \\ x_2 = 0.15x_1 + 0.12x_2 - 0.17x_3 + 0.26x_4 - 0.65 \\ x_3 = 0.35x_1 + 0.22x_2 - 0.03x_3 - 0.27x_4 + 2.26 \\ x_4 = 0.12x_1 + 0.24x_2 - 0.15x_3 + 0.23x_4 - 1.17 \end{cases} \\
 29. \quad & \begin{cases} x_1 = 0.16x_1 + 0.13x_2 - 0.32x_3 - 0.16x_4 - 0.64 \\ x_2 = 0.34x_1 - 0.08x_2 + 0.15x_3 - 0.18x_4 + 1.42 \\ x_3 = 0.15x_1 + 0.32x_2 + 0.13x_3 - 0.25x_4 - 2.06 \\ x_4 = 0.11x_1 - 0.26x_2 - 0.08x_3 + 0.24x_4 + 0.83 \end{cases} \\
 30. \quad & \begin{cases} x_1 = 0.23x_1 - 0.04x_2 + 0.22x_3 - 0.14x_4 + 1.24 \\ x_2 = 0.34x_1 - 0.23x_2 + 0.07x_3 - 0.88 \\ x_3 = 0.26x_1 + 0.31x_2 - 0.02x_3 + 0.29 \\ x_4 = 0.07x_1 - 0.21x_2 + 0.27x_3 - 0.13x_4 - 1.19 \end{cases}
 \end{aligned}$$

CHIZIQLI BO‘LMAGAN ALGEBRAIK TENGLAMALAR SISTEMASINI TAQRIBIY HISOBLASH

Berilgan ushbu

[illegible]

tenglamalar sistemasi

[illegible]

kanonik shaklga keltiriladi. So‘ng $\vec{x}^{(0)} = (x_1^{(0)}, x_2^{(0)}, \dots, x_n^{(0)})$ boshlang‘ich yaqinlashishlar tanlanadi, keyingi yaqinlashishlar esa

$$\begin{cases} x_1^{(k+1)} = \varphi_1(x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)}) \\ x_2^{(k+1)} = \varphi_2(x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)}) \\ \vdots \\ x_n^{(k+1)} = \varphi_n(x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)}) \end{cases} \quad (3)$$

formulalar bo'yicha izlanadi. Agar φ_{ii} ($ii = \overline{1, n}$) funksiyalar

$$\max_i |\vec{x} - \vec{x}^{(0)}| \leq \delta$$

sohada aniqlangan, uzluksiz differensiallanuvchi va

$$q_m = \max_{1 \leq i \leq x_n} \left\{ \max_{|x - x^{(0)}| \leq \delta} \sum_{j=1}^n \left| \frac{\partial \varphi_1}{\partial x_i} \right| \right\} < 1 \quad (4)$$

tengsizlikni qanoatlantirsa, hamda $(x_1^{(0)}, x_2^{(0)}, \dots, x_n^{(0)})$ boshlang'ich yaqinlashishlar uchun

$$\left| x_i^{(0)} - \varphi_i(x_1^{(0)}, x_2^{(0)}, \dots, x_n^{(0)}) \right| \leq \eta \quad (i = \overline{1, n}), \quad \frac{\eta}{1 - q_m} \leq \delta$$

shartlar bajarilsa, tenglamalar sistemasi shu sohada yagona $\xi = (\xi_1, \xi_2, \dots, \xi_n)$ yechimga ega bo'ladi va (3) ketma-ketliklar bu yechimga intiladi va intilish tezligi

$$\left| \xi_i - x_i^{(k)} \right| \leq \frac{\eta}{1 - q_m} q_m^n \quad (ii = \overline{1, n}) \quad (5)$$

tengsizlik bilan baholanadi. q ning qiymati m masofada (kubik masofada, normada) hisoblanadi. U s (oktaedrik) yoki l (sferik) masofada ham berilishi mumkin.

Misol.

$$\begin{cases} f_1(x, y) = 2x^2 - x(y + 5) + 1 = 0 \\ f_2(x, y) = x + 3 \lg x - y^2 = 0 \end{cases}$$

sistemaning musbat ildizlari $\varepsilon = 0,5 \cdot 10^{-4}$ aniqlikda topilsin.

Yechish. Sistemaning ildizlarini grafik yordamida taqriban aniqlaymiz. Kompyuter yordamida quyidagi natijani beradi, programma va chizmadan ko'rinishicha shu ildiz $3,4 < x < 3,6$,

$2, 1 < y < 2, 3$ to'g'ri to'rtburchak ichida yotadi. (x_0, y_0) boshlang'ich yaqinlashish sifatida $(3, 5; 2, 2)$ nuqtani olamiz. Sistemani kanonik shaklga keltiramiz:

$$x = \sqrt{0,5(x(y+5)-1)} \equiv \varphi_1(x, y)$$

$$y = \sqrt{x+3\lg x} \equiv \varphi_2(x, y).$$

Endi $\{|x-3, 5| \leq 0, 1, |y-2, 2| \leq 0, 1\}$ sohada (4) shartning bajarilishini tekshiramiz ($M = 0,43429$ – o'tish moduli):

$$\frac{\partial \varphi_1}{\partial x} = \frac{y+5}{4\sqrt{0,5(x(y+5)-1)}},$$

$$\frac{\partial \varphi_1}{\partial y} = \frac{x}{4\sqrt{0,5(x(y+5)-1)}}$$

$$\frac{\partial \varphi_2}{\partial x} = \frac{x}{2\sqrt{x+3\lg x}}, \quad \frac{\partial \varphi_2}{\partial y} = 0;$$

$$\max_{x,y} \left| \frac{\partial \varphi_1}{\partial x} \right| = \frac{(2,3+5)\sqrt{2}}{4\sqrt{3,4(2,1+5)}} < 0,54,$$

$$\max_{x,y} \left| \frac{\partial \varphi_1}{\partial y} \right| = \frac{3,6\sqrt{2}}{4\sqrt{3,4(2,1+5)-1}} < 0,27$$

$$\max_{x,y} \left| \frac{\partial \varphi_2}{\partial x} \right| = \frac{1 + \frac{3M}{3,4}}{4\sqrt{3,43\lg 3,4}} < 0,27, \quad \frac{\partial \varphi_2}{\partial y} = 0$$

$$\max \left(\left| \frac{\partial \varphi_1}{\partial x} \right| + \left| \frac{\partial \varphi_1}{\partial y} \right| \right) < 0,81,$$

$$\max \left(\left| \frac{\partial \varphi_2}{\partial x} \right| + \left| \frac{\partial \varphi_2}{\partial y} \right| \right) < 0,27.$$

Demak, $q = 0,81 < 1$. Iteratsiya jarayoni yaqinlashadi. Ketma-ket yaqinlashishlarni

$$x_{k+1} = \sqrt{0,5(x_k(y_k + 5) - 1)}$$

va

$$y_{k+1} = \sqrt{x_k + 3 \lg(x_k)} \quad (k = 0, 1, 2, \dots)$$

formulalar bo'yicha hisoblaymiz:

k	x	y	k	x	y
0	3,5	2,2	4	3,48580	2,26084
1	3,47851	2,26544	5	3,48639	2,26113
2	3,48374	2,25891	6	3,48677	2,26131
3	3,48483	2,26050			

Javob: $x = 3,4866 \pm 0,2 \cdot 10^{-4}$, $y = 2,2612 \pm 0,1 \cdot 10^{-4}$.

Mavzuni mustahkamlash uchun misollar.

1. $\begin{cases} x^2 + y^2 - 4 = 0 \\ x - y^2 - 1 = 0 \end{cases}$
2. $\begin{cases} x^2 + y^2 - 1 = 0 \\ x^3 - y = 0 \end{cases}$
3. $\begin{cases} (x^2 + y^2)^2 - 4(x^2 - y^2)^2 = 0 \\ x^2 + y^2 - 1 = 0 \end{cases}$
4. $\begin{cases} x + y + xy - 7 = 0 \\ x^2 + y^2 + xy - 13 = 0 \end{cases}$
5. $\begin{cases} 3x^2 + 5xy - 2y^2 - 20 = 0 \\ x^2 + y^2 + xy - 7 = 0 \end{cases}$
6. $\begin{cases} 2x^2 + xy - y^2 - 20 = 0 \\ x^2 - 4xy + 7y^2 - 13 = 0 \end{cases}$

$$7. \begin{cases} x^2 - y^2 + 3y = 0 \\ x^2 + 3xy + 2y^2 + 2x + 4y = 0 \end{cases}$$

$$8. \begin{cases} (x + y)(x^2 - y^2) - 16 = 0 \\ (x - y)(x^2 + y^2) - 40 = 0 \end{cases}$$

$$9. \begin{cases} (x + y)(x + 2y)(x + 3y) - 60 = 0 \\ (x + y)(2x + y)(3x + y) - 105 = 0 \end{cases}$$

$$10. \begin{cases} x^4 + 6x^2y^2 + y^4 - 136 = 0 \\ x^3y + xy^3 - 30 = 0 \end{cases}$$

$$11. \begin{cases} 10x^2 + 5y^2 - 2xy - 38x - 6y + 41 = 0 \\ 3x^2 - 2y^2 + 5xy - 17x - 6y + 20 = 0 \end{cases}$$

$$12. \begin{cases} x^3 + y^3 - 19 = 0 \\ (xy + 8)(x + y) - 2 = 0 \end{cases}$$

$$13. \begin{cases} x^2y^2 - 2x + y^2 = 0 \\ 2x^2 - 4x + 3 + y^3 = 0 \end{cases}$$

$$14. \begin{cases} x^3 + x^3y^3 + y^3 - 17 = 0 \\ x + xy + y - 5 = 0 \end{cases}$$

$$15. \begin{cases} (x^2 + y^2)(x + y) - 15xy = 0 \\ (x^4 + y^4)(x^2 + y^2) - 85x^2y^2 = 0 \end{cases}$$

$$16. \begin{cases} \sin(x + 1) - y = 1.2, \\ 2x + \cos y = 2 \end{cases}$$

$$17. \begin{cases} \operatorname{tg}(xy + 0.4) = x^2, \\ 0.6x^2 + 2y^2 = 1 \quad (x > 0, y > 0). \end{cases}$$

$$18. \begin{cases} \cos(x - 1) + y = 0.5 \\ x - \cos(y) = 3 \end{cases}$$

$$19. \begin{cases} \sin(x+y) = 1.6x \\ x^2 + y^2 = 1 (x > 0, y > 0) \end{cases}$$

$$20. \begin{cases} \sin x + 2y = 2, \\ x + \cos(y-1) = 0.7 \end{cases}$$

$$21. \begin{cases} \operatorname{tg}(xy + 0.1) = x^2 \\ x^2 + 2y^2 = 1. \end{cases}$$

$$22. \begin{cases} \cos(x) + y = 1.5, \\ \sin(y - 0.5) = 2x - 1 \end{cases}$$

$$23. \begin{cases} \sin(x+y) - 1.2x = 0.2 \\ x^2 + y^2 = 1. \end{cases}$$

$$24. \begin{cases} \sin(x + 0.5) - y = 1 \\ \cos(y - 2) + x = 0 \end{cases}$$

$$25. \begin{cases} \operatorname{tg}(xy + 0.3) = x^2, \\ 0,9x^2 + 2y^2 = 1. \end{cases}$$

$$26. \begin{cases} \cos(x + 0.5) + y = 0.8 \\ \sin(y) - 2x = 1.6 \end{cases}$$

$$27. \begin{cases} \sin(x-1) + y = 1.3, \\ x - \sin(y+1) = 0.8 \end{cases}$$

$$28. \begin{cases} \operatorname{tg}(xy) = x^2, \\ 0.8x^2 = 1 - 2y^2. \end{cases}$$

$$29. \begin{cases} \cos(x+1) = 2y, \\ x + \sin(y) = -0.4. \end{cases}$$

$$30. \begin{cases} \sin(x+y) = 1.5x + 0.1, \\ x^2 + y^2 = 1. \end{cases}$$

4.2-§. Chiziqli bo'lmagan algebraik tenglamalar sistemasini oddiy Nyuton usuli bilan taqribiy hisoblash

Ushbu

$$f_1(x_1, x_1, \dots, x_n) = 0 \quad (i = \overline{1, n}) \quad (6)$$

chiziqli bo'lmagan tenglama yechimining biror $\vec{x}^{(m)} = (x_1^{(m)}, x_2^{(m)}, \dots, x_n^{(m)})$ yaqinlashishi topilgan bo'lsin.

Aniq yechim $\vec{x} = \vec{x}^{(m)} + \vec{\varepsilon}^{(m)}$ ko'rinishida yozilishi mumkin, bunda $\vec{\varepsilon}^{(m)} = (\varepsilon_1^{(m)}, \varepsilon_2^{(m)}, \dots, \varepsilon_n^{(m)})$ tuzatma (taqribiy ildiz xatosi).

Agar $f(\vec{x})$ funksiya $\vec{x}$ va $\vec{x}^{(m)}$ ni o'z ichiga olgan biror qavariq sohada uzluksiz differensiallanuvchi bo'lsa, u holda $f(\vec{x}^{(m)} + \vec{\varepsilon}^{(m)}) = \vec{0}$ tenglamani $\vec{\varepsilon}$ vektor darajalari bo'yicha yoyib (va bunda chiziqli hadlar bilan chegaralanib),

$$\vec{f}(\vec{X}^{(m)}) + \vec{f}'(\vec{x}^{(m)})\vec{\varepsilon}^{(m)} = \vec{0}$$

chiziqli tenglamani olamiz, undagi $\vec{f}'(\vec{x}^{(m)})$ hosila $f_1, \dots, f_n$ funksiyalar sistemasining $I(\vec{x}^{(n)})$ Y akobi matrisasini tashkil qiladi:

$$I(\vec{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \dots & \dots & \dots & \dots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}.$$

$I(\vec{x}^{(m)})$ matritsa xosmas deb faraz qilingan holda

$$\vec{\varepsilon}^{(m)} = -I^{-1}(\vec{x}^{(m)})\vec{f}(\vec{x}^{(m)}),$$

nihojat, $\vec{\varepsilon}^{(m)} = \vec{x}^{(m+1)} - \vec{x}^{(m)}$ bo'lgani uchun ushbu

$$\vec{x}^{(m+1)} = \vec{x}^{(m)} - I^{-1} \vec{x}^{(m)} \vec{f}'(\vec{x}^{(m)}) \quad (m = 0, 1, 2, \dots) \quad (7)$$

rekurrent formulani hosil qilamiz. $\vec{x}^{(0)}$ boshlang'ich yaqinlashish sifatida izlanayotgan ildizning biror dag'al qiymati olinishi mumkin.

Xususan, ikkinchi tartibli

$$\begin{cases} f(x, y) = 0, \\ g(x, y) = 0 \end{cases}$$

sistemani Nyuton usuli bo'yicha yechish uchun

$$x_{n+1} = x_n - A_n / I_n,$$

$$y_{n+1} = y_n - B_n / I_n$$

formulalardan foydalanamiz, bunda

$$A_n = \begin{vmatrix} f(x_n, y_n) & f'_y(x_n, y_n) \\ g(x_n, y_n) & g'_y(x_n, y_n) \end{vmatrix}, B_n = \begin{vmatrix} f'_x(x_n, y_n) & f(x_n, y_n) \\ g'_x(x_n, y_n) & g(x_n, y_n) \end{vmatrix},$$

$$I_n = \begin{vmatrix} f'_x(x_n, y_n) & f'_y(x_n, y_n) \\ g'_x(x_n, y_n) & g'_y(x_n, y_n) \end{vmatrix} \neq 0.$$

Misol.

$$\begin{cases} f(x, y) = x^2 y^2 - 2x^3 - 5y^3 - 2,401 = 0, \\ g(x, y) = x^4 - 8y + 3,474 = 0 \end{cases}$$

sistema yechimi $\varepsilon = 10^{-2}$ aniqlikda topilsin.

Yechish.

$$f'_x = 2xy^2 - 6x^2, \quad f'_y = 2x^2y - 15y^2, \quad g'_x = 4x^3, \quad g'_y = -8.$$

Boshlang'ich yaqinlashish sifatida $(0, 8; 0, 5)$ ni olamiz. Keyingi hisoblashlar natijalari quyidagi jadvalda joylashtirilgan:

n	x_n	I_n	f'_x	$g'_x g'_y$	$A_n B_n$	I_n
0	0,8	-0,0259	-3,44	2,048	-0,154	33,8891
	0,5	-0,1164	-3,11	-8	0,45367	
1	0,8045	0,0228	-3,5027	2,0831	-0,1827	34,1085
	0,4866	0,00007	-2,9219	-8	-0,0479	
2	0,811	0,0178	-3,5591	2,1329	-0,1363	34,7235
	0,4881	0,0022	-2,9306	-8	-0,0461	
3	0,8148	0,00006				
	0,4907	0,00131				

Mavzuni mustahkamlash uchun misollar.

$$1. \begin{cases} \sqrt{-1-x} - \sqrt{2y-x} - 1 = 0 \\ \sqrt{1-2y} + \sqrt{2y-x} - 4 \end{cases}$$

$$2. \begin{cases} \operatorname{tg}(xy) = x^2 \\ 0.8x^2 + 2y^2 = 1 \end{cases}$$

$$3. \begin{cases} \sin(x-y) - xy = -1 \\ x^2 - y^2 = 3/4 \end{cases}$$

$$4. \begin{cases} y - \sin x - 1.6x = 0 \\ x - \cos 1.4y = 0 \end{cases}$$

$$5. \begin{cases} x + \operatorname{tg}(xy) = 0 \\ (y^2 - 7.5)^2 + \ln x = 0 \end{cases}$$

$$6. \begin{cases} \sin(1.1x + y) - x + 1.4 = 0 \\ y^2 - \frac{1}{x} = 0 \end{cases}$$

$$7. \begin{cases} 0.6x^2 + 2y^2 = 1 \\ \cos(y-1) + x = 0.5 \end{cases}$$

$$8. \begin{cases} \sin x - \sin 2y = 0 \\ \cos x - \sin y = 0 \end{cases}$$

$$9. \begin{cases} \cos x + y = 0 \\ 2x - \sin(y - 0.5) = 0 \end{cases}$$

$$10. \begin{cases} \cos(x - y) - 2 \cos(x + y) = 0 \\ \cos x \cos y - \frac{3}{4} = 0 \end{cases}$$

$$11. \begin{cases} 0.16x + 2y + x^2y = 0 \\ \cos y + 1.1x = 0 \end{cases}$$

$$12. \begin{cases} 2^{2x} - 3^y + 17 = 0 \\ 2^x + 3^{y/2} + 1 = 0 \end{cases}$$

$$13. \begin{cases} \cos(y - 1) + x - 0.5 = 0 \\ 2x - \sin(y - 0.5) = 1 \end{cases}$$

$$14. \begin{cases} 10^{-3}e^{40x-1} - y = 0 \\ 10^{-3}e^{40y-1} + 2x = 0 \end{cases}$$

$$15. \begin{cases} \cos(x + 0.5) - y = 2, \\ \sin(y) - 2x = 1 \end{cases}$$

$$16. \begin{cases} \operatorname{tg}(xy) - x^2 = 0 \\ 0.7x^2 + 2y^2 = 1 \end{cases}$$

$$17. \begin{cases} \cos(x + 2) - y = 1.5, \\ \cos(y - 2) + x = 0.5 \end{cases}$$

$$18. \begin{cases} \sin(y + 1) - x = 1.2, \\ 2y + \cos x = 2. \end{cases}$$

$$19. \begin{cases} \operatorname{tg}(xy + 0.2) = x^2, \\ 0.6x^2 + 2y^2 = 1. \end{cases}$$

$$20. \begin{cases} \cos(y-1) + x = 0.5, \\ \cos x - y + 3 = 0. \end{cases}$$

$$21. \begin{cases} \sin(x+y) - 1.5x = -0.1 \\ x^2 + y^2 = 1. \end{cases}$$

$$22. \begin{cases} \sin y + 2x = 2, \\ y + \cos(x-1) = 0.7 \end{cases}$$

$$23. \begin{cases} \operatorname{tg}(xy + 0.4) = x^2 \\ 0.8x^2 + 2y^2 = 1. \end{cases}$$

$$24. \begin{cases} x + \cos y = 1.5, \\ 2y - \sin(x - 0.5) = 1. \end{cases}$$

$$25. \begin{cases} \sin(x+y) - 1.2x = -0.1 \\ x^2 + y^2 = 1. \end{cases}$$

$$26. \begin{cases} \sin(y + 0.5) - x = 1, \\ y + \cos(x - 2) = 0. \end{cases}$$

$$27. \begin{cases} \operatorname{tg}(xy + 0.1) = x^2 \\ 0.9x^2 + 2y^2 = 1 \end{cases}$$

$$28. \begin{cases} x + \cos(y + 0.5) = 0.8 \\ 2y - \sin x = -1.6 \end{cases}$$

$$29. \begin{cases} \sin(x+y) = 1.4x \\ x^2 + y^2 = 1. \end{cases}$$

$$30. \begin{cases} \sin(y-1) + x = 1.3, \\ y - \sin(x+1) = 0.8 \end{cases}$$

V BOB.

MATRITSALARNING XOS SONLARI VA XOS VEKTORLARI TOPISH

5.1-§. Xarakteristik ko'phad.

$A = [a_{ij}]$ - haqiqiy elementlarga ega bo'lgan n -tartibli kvadrat matritsa va λ biror-bir noma'lum son bo'lsin. U holda $(A - \lambda E)$ matrisa, A matritsaning xarakteristik matritsasi deyiladi, bunda $(E - n)$ - tartibli birlik matrisa. Xarakteristik matritsa:

$$(A - \lambda E) = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{1n} \\ a_{21} & a_{22} - \lambda & a_{23} - \lambda & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & a_{nn} - \lambda \end{vmatrix} \quad (1)$$

ko'rinishga ega. Bu matritsaning determinanti xarakteristik determinant deyiladi va u quyidagicha yoziladi:

$$\det(A - \lambda E) = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{1n} \\ a_{21} & a_{22} - \lambda & a_{23} - \lambda & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & a_{nn} - \lambda \end{vmatrix} \quad (2)$$

Xarakteristik detirminant $\det(A - \lambda E)$ yoyib yozilganda u λ ga nisbatan $n - 1$ tartibli ko'phad bo'ladi, chunki bu determinantni hisoblaganda, uning bosh dioganalidagi elementlarning ko'paytmasi eng katta hadi $(-1)^n \lambda^n$ ga teng bo'lgan ko'phadni beradi, ya'ni

$$D(\lambda) \equiv (-1)^n [\lambda^n - \rho_1 \lambda^{n-1} + \rho_2 \lambda^{n-2} - \dots + (-1)^n \rho_n] \quad (3)$$

ko'phad (3) A matritsaning xarakteristik ko'phadi, uning ildizlari $\lambda_1, \lambda_2, \dots, \lambda_n$ (ular haqiqiy, yoki kompleks bo'lishi mumkin) esa

A matritsaning xarakteristik sonlari yoki xos qiymatlari deyiladi. Sonlar $\rho_1, \rho_2, \dots, \rho_n$ xarakteristik ko'phad (3) ning koeffitsientlari deyiladi. Nolga teng bo'lmagan $X = (x_1, x_2, \dots, x_n)$ vektor A matritsaning xos vektori deyiladi, agar A matritsa X vektorni λX vektorga o'tkazsa:

$$AX = \lambda X \quad (4)$$

bo'lsa, boshqacha qilib aytganda A matritsaning x vektorga ko'paytmasi va xarakteristik son λ ning x vektorga ko'paytmasi aynan bir vektor bo'lsa A matritsaning har bir xos qiymati λ_1 ga o'zining xos vektori X_i , ($i = 1, 2, \dots, n$) mos keladi.

Xos vektorning koordinatalarini topish uchun quyidagi tenglamani tuzamiz:

$$(A - \lambda E)X = 0$$

bu tenglama xarakteristik tenglama deyiladi. Uni ushbu ko'rinishda yozamiz:

$$\begin{pmatrix} a_{11} - \lambda & a_{12} & a_{1n} \\ a_{21} & a_{22} - \lambda & a_{2n} \\ \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{nn} - \lambda \end{pmatrix} * \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \dots \\ 0 \end{pmatrix}$$

bunda ko'paytirishni bajarib, bir jinsli tenglamalarni hosil qilamiz:

$$\begin{cases} (a_{11} - \lambda)x_1 + a_{12}x_2 + \dots + a_{1n}x_n = 0 \\ a_{21}x_1 + (a_{22} - \lambda)x_2 + \dots + a_{2n}x_n = 0 \\ \dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + (a_{nn} - \lambda)x_n = 0 \end{cases}$$

sistema (6) ning determinanti nolga teng bo'ladi, chunki ana shu shartdan A matritsaning xos qiymatlari aniqlangan edi. Bu sistemani yechib, xos vektor X ning barcha koordinatalarini topamiz. Sistema (6) ga ketma-ket $\lambda_1, \lambda_2, \dots, \lambda_n$ larni qo'yish natijasida n ta xos vektorlarni topamiz.

Misol. Quyidagi matritsani Xos qiymat va xos vektorlarni hisoblang.

$$A = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$$

Qaralayotgan A matritsa uchun xarakteristik determinantni tuzib, uni yechamiz:

$$\begin{vmatrix} 3 - \lambda & 1 \\ 2 & 4 - \lambda \end{vmatrix} = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

$$\lambda_{12} = \frac{7}{2} \pm \sqrt{\frac{49}{4} - 10} = \frac{7}{2} \pm \sqrt{\frac{49 - 40}{4}} = \frac{7}{2} \pm \sqrt{\frac{9}{4}} = \frac{7}{2} \pm \frac{3}{2},$$

$$\lambda_1 = 5, \lambda_2 = 2$$

$\lambda_1 = 5$ deb olamiz va sistema (6) ga mos sistemani tuzamiz. Natijada:

$$\begin{cases} (3 - \lambda_1)x_1 + x_2 = 0 \\ 2x_1 + (4 - \lambda_1)x_2 = 0 \end{cases}$$

sistemani hosil qilamiz. Bu tenglamalardan $x_2 = 2x_1$ tenglamaga keltiramiz, $x_1 = 1$ bo'lsa, $x_2 = 2$ bo'ladi. $\lambda_2 = 2$ deb olsak

$$\begin{cases} x_1 + x_2 = 0 \\ 2x_1 + 2x_2 = 0 \end{cases}$$

sistemani hosil qilamiz, bundan $x_1 = -x_2$, $x_1 = 1$ bo'lsa, $x_2 = -1$ bo'ladi.

Xarakteristik determinantni bevosita yoyish usuli. Dastlab, uchinchi tartibli matritsa misolida xarakteristik determinantni bevosita yoyish usuli orqali xarakteristik ko'phadning koefitsientlari qanday topilishini qaraymiz. So'ngra bu natijani n -tartibli matritsa uchun umumlashtiramiz. Matritsa A va xarakteristik determinantni qaraylik

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix},$$

$$\det(A - \lambda E) = \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix}$$

xarakteristik detirminantni uchburchaklar qoidasi bo'yicha yoyamiz:

$$\begin{aligned} \det(A - \lambda E) &= \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix} = \\ &= (-1)^3 [\lambda^3 - \lambda^2(a_{11} + a_{22} + a_{33}) + \\ &+ \lambda \left(\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} \right) - \\ &\quad - \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}] \end{aligned}$$

yoki

$$\text{Det}(A - \lambda E) = (-1)^3 (\lambda^3 - \rho_1 \lambda^2 + \rho_2 \lambda - \rho_3) = 0$$

bunda ρ_1 matritsaning diagonal elementlari yig'indisi, u matritsaning o'zi deyiladi va $S_p A$ orqali belgilanadi:

$$\rho_1 = S_p A = a_{11} + a_{22} + a_{33}.$$

ρ_2 koeffitsient A matritsaning barcha ikkinchi tartibli diagonal minorlarining yig'indisiga teng:

$$\rho_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

ikkinchi, uchinchi, n - tartibli diogonal minorlar deb, shunday minorlarga aytiladiki, ularning bosh diogonal elementlari A matritsa determinantining bosh diogonal elementlaridan iborat bo'ladi.

ρ_3 koeffitsient A matritsaning determinantiga teng bo'ladi.

$$\rho_3 = \det A = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Umuman olganda, n - tartibli determinant $\det(A - \lambda E)$ ni n - tartibli ko'phad ko'rinishida

$$D(\lambda)z(-1)^n [\lambda^n - \rho_1\lambda^{n-1} + \rho_2\lambda^{n-2} - \dots + (-1)^n \rho_n]$$

yo'zish talab qilingan bo'lsin, bu holda $\rho_1, \rho_2, \dots, \rho_n$ koeffitsientlar quyidagi qoida bo'yicha hisoblanadi:

$$\rho_1 = \sum_{\alpha < \beta} a_{ii} = S_p A$$

matritsaning barcha diogonal elementlarining yig'indisi;

$$\rho_2 = \sum_{\alpha < \beta} \begin{vmatrix} a_{\alpha\alpha} & a_{\alpha\beta} \\ a_{\beta\alpha} & a_{\beta\beta} \end{vmatrix}$$

matritsaning barcha ikkinchi tartibli diogonal elementlarining yig'indisi;

$$\rho_3 = \sum_{\alpha < \beta < \gamma} \begin{vmatrix} a_{\alpha\alpha} & a_{\alpha\beta} & a_{\alpha\gamma} \\ a_{\beta\alpha} & a_{\beta\beta} & a_{\beta\gamma} \\ a_{\gamma\alpha} & a_{\gamma\beta} & a_{\gamma\gamma} \end{vmatrix}$$

matritsaning barcha uchinchi tartibli diagonal minorlarining yig'indisi va hokazo.

Nihoyat, ρ_n koeffitsient A matritsaning determinantiga teng:

$$\rho_n = \det A.$$

A matritsaning k - tartibli minorlarining soni

$$C_n^k = \frac{n(n-1)(n-2)\dots(n-k+1)}{k!}, (k = 1, 2, 3, \dots, n)$$

formula bo'yicha aniqlanadi.

Misol. Quyidagi matritsaning xarakteristik ko'phadini bevosita yoyish usuli bilan topilsin.

$$A = \begin{bmatrix} -4 & -3 & 1 & 1 \\ 2 & 0 & 4 & -1 \\ 1 & 1 & 2 & -2 \\ 1 & 1 & -1 & -1 \end{bmatrix}$$

Yechish. Dastlab

$$\rho_1 = S_\rho A = a_{11} + a_{22} + a_{33} + a_{44} = -4 + 0 + 2 - 1 = -3$$

ni hisoblaymiz.

$$\rho_2 = \sum_{\alpha < \beta} \begin{vmatrix} a_{\alpha\alpha} & a_{\alpha\beta} \\ a_{\beta\alpha} & a_{\beta\beta} \end{vmatrix}$$

to'rtinchi tartibli matritsaning ikkinchi tartibli diaganol minorlari ($k = 2$) soni $C_4^2 = \frac{4 \cdot 3}{1 \cdot 2} = 6$ ga teng. A na shu minorlarni hisoblaymiz.

$$\begin{aligned} \rho_2 = & \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{14} \\ a_{41} & a_{44} \end{vmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + \\ & + \begin{vmatrix} a_{22} & a_{24} \\ a_{42} & a_{44} \end{vmatrix} + \begin{vmatrix} a_{33} & a_{34} \\ a_{43} & a_{44} \end{vmatrix}, \end{aligned}$$

$$\rho_2 = \begin{vmatrix} -4 & -3 \\ 2 & 0 \end{vmatrix} + \begin{vmatrix} -4 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} -4 & 1 \\ 1 & -1 \end{vmatrix} + \\ + \begin{vmatrix} 0 & 4 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 0 & -1 \\ 1 & -1 \end{vmatrix} + \begin{vmatrix} 2 & -2 \\ -1 & -1 \end{vmatrix}$$

ga teng.

$$\rho_3 = \det A = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

ekanligi ma'lum.

To'rtinchi tartibli matritsaning uchunchi tartibli ($k = 3$) diagonal minorlari soni:

$$C_4^2 = \frac{4 \cdot 3 \cdot 2}{1 \cdot 2 \cdot 3} = 4$$

$$\rho_3 = \begin{vmatrix} -4 & -3 & 1 \\ 2 & 0 & 4 \\ 1 & 1 & 2 \end{vmatrix} + \begin{vmatrix} -4 & -3 & 1 \\ 2 & 0 & -1 \\ 1 & 1 & -1 \end{vmatrix} + \\ + \begin{vmatrix} -4 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & -1 & -1 \end{vmatrix} + \begin{vmatrix} 0 & 4 & -1 \\ 1 & 2 & -2 \\ 1 & -1 & -1 \end{vmatrix} = 24$$

ekanligini aniqlaymiz va nihoyat:

$$\rho_4 = \det A = \begin{vmatrix} -4 & -3 & 1 & 1 \\ 2 & 0 & 4 & -1 \\ 1 & 1 & 2 & -2 \\ 1 & 1 & -1 & -1 \end{vmatrix} = -1, 5.$$

Shunday qilib,

$$D(\lambda) = \lambda^4 - \rho_1 \lambda^3 + \rho_2 \lambda^2 - \rho_3 \lambda + \rho_4 = \\ = \lambda^4 + 3\lambda^3 - 7\lambda^2 - 24\lambda - 15$$

ga ega bo'lamiz.

Mavzuni mustahkamlash uchun misollar.

Quyidagi matritsaning xos son va ularga mos vektorlarini bevosita yoyish usuli bilan topilsin.

$$1. A = \begin{pmatrix} 8 & -6 & 8 \\ 1 & 0 & 1 \\ -5 & 4 & -5 \end{pmatrix}.$$

$$2. A = \begin{pmatrix} 3 & -1 & 1 \\ 1 & 1 & 1 \\ -1 & 1 & 1 \end{pmatrix}.$$

$$3. A = \begin{pmatrix} -1 & 2 & 2 \\ -1 & 3 & 3 \\ 1 & -2 & -2 \end{pmatrix}.$$

$$4. A = \begin{pmatrix} 1 & 1 & -1 \\ -1 & 3 & -1 \\ 1 & -1 & 3 \end{pmatrix}.$$

$$5. A = \begin{pmatrix} 2 & -1 & 1 \\ -2 & 3 & 0 \\ -2 & 2 & 1 \end{pmatrix}.$$

$$6. A = \begin{pmatrix} -5 & 4 & -4 \\ -4 & 3 & -4 \\ 4 & -4 & 3 \end{pmatrix}.$$

$$7. A = \begin{pmatrix} 1 & 0 & 3 \\ -2 & 3 & -2 \\ 0 & 0 & -2 \end{pmatrix}.$$

$$8. A = \begin{pmatrix} 9 & 0 & -5 \\ -10 & -1 & 5 \\ 10 & 0 & -6 \end{pmatrix}.$$

$$9. A = \begin{pmatrix} 5 & -6 & -3 \\ 4 & -5 & -4 \\ -2 & 2 & 4 \end{pmatrix}.$$

$$10. A = \begin{pmatrix} 0 & 1 & -1 \\ -2 & 3 & -1 \\ 1 & -1 & 2 \end{pmatrix}.$$

$$11. A = \begin{pmatrix} 5 & -6 & -3 \\ -1 & 0 & 1 \\ 8 & -8 & -6 \end{pmatrix}.$$

$$12. A = \begin{pmatrix} 2 & 0 & 1 \\ 5 & -3 & 6 \\ 0 & 0 & 2 \end{pmatrix}.$$

$$13. A = \begin{pmatrix} -1 & 0 & 6 \\ -2 & 1 & 7 \\ 0 & 0 & 2 \end{pmatrix}.$$

$$14. A = \begin{pmatrix} -2 & 1 & -1 \\ -4 & 1 & -2 \\ 1 & -1 & 0 \end{pmatrix}.$$

$$15. A = \begin{pmatrix} -5 & 3 & -3 \\ 0 & 1 & 0 \\ 6 & -3 & 4 \end{pmatrix}.$$

$$16. A = \begin{pmatrix} -2 & 1 & 1 \\ -1 & 1 & -1 \\ -4 & 1 & 3 \end{pmatrix}.$$

$$17. A = \begin{pmatrix} 2,3 & 1 & 1,3 \\ 1 & 2,8 & 1,3 \\ 1,3 & 1,3 & 3,3 \end{pmatrix}.$$

$$18. A = \begin{pmatrix} 2,6 & 1 & 1,6 \\ 1 & 3,1 & 1,6 \\ 1,6 & 1,6 & 3,6 \end{pmatrix}.$$

$$19. A = \begin{pmatrix} 3,5 & 1 & 2,5 \\ 1 & 4 & 2,5 \\ 2,5 & 2,5 & 4,5 \end{pmatrix}.$$

$$20. A = \begin{pmatrix} 1,8 & 0,9 & 1 \\ 0,9 & 1,8 & 0,3 \\ 1 & 0,3 & 1,8 \end{pmatrix}.$$

$$21. A = \begin{pmatrix} 1,5 & 0,6 & 0,7 \\ 0,6 & 1,5 & 0,3 \\ 0,7 & 0,3 & 1,5 \end{pmatrix}.$$

$$22. A = \begin{pmatrix} 2,7 & 1 & 1,7 \\ 1 & 3,2 & 1,7 \\ 1,7 & 1,7 & 3,7 \end{pmatrix}.$$

$$23. A = \begin{pmatrix} 1,4 & 1,2 & -1,3 \\ 1,2 & 0,9 & 0,4 \\ -1,3 & 0,4 & 0,8 \end{pmatrix}.$$

$$24. A = \begin{pmatrix} 2 & 1,5 & 3,5 & 4,5 \\ 1,5 & 2 & 2 & 1,6 \\ 3,5 & 2 & 2 & 1,7 \\ 4,5 & 1,6 & 1,7 & 2 \end{pmatrix}.$$

$$25. A = \begin{pmatrix} 1 & 0,5 & 1,2 & -1 \\ 0,5 & 2 & -0,5 & 0 \\ 1,2 & -0,5 & -1 & 1,4 \\ -1 & 0 & 1,4 & 1 \end{pmatrix}.$$

$$26. A = \begin{pmatrix} 1,2 & 0,5 & 2 & 1 \\ 0,5 & 1 & 0,8 & 2' \\ 2 & 0,8 & 1 & 1 \\ 1 & 2 & 1 & 2 \end{pmatrix}.$$

$$27. A = \begin{pmatrix} 0,5 & 1,2 & 1 & 0,9 \\ 1,2 & 2 & 0,5 & 1,2 \\ 1 & 0,5 & 1 & 1 \\ 0,5 & 1,2 & 1 & 2,2 \end{pmatrix}.$$

$$28. A = \begin{pmatrix} 1,6 & 1 & 1,4 & 1 \\ 1 & 1 & 0,5 & 2 \\ 1,4 & 0,5 & 2 & 1,2 \\ 1 & 2 & 1,2 & 0,5 \end{pmatrix}$$

$$29. A = \begin{pmatrix} 2,4 & 0,5 & 2 & 1 \\ 0,5 & 1 & 0,8 & 2 \\ 2 & 0,8 & 1 & 0,5 \\ 1 & 2 & 0,5 & 1,2 \end{pmatrix}.$$

$$30. A = \begin{pmatrix} 0,5 & 1,2 & 2 & 1 \\ 1,2 & 2 & 0,5 & 1,2 \\ 2 & 0,5 & 1 & 0,5 \\ 1 & 1,2 & 0,5 & 1,6 \end{pmatrix}$$

5.2-§. Matritsalarining xos sonlari va xos vektorlari topishning A.N.Krilov usuli

Matritsalarining minimal ko‘phadlari. Avval chiziqli algebradan ayrim ta’rif va teoremlarni keltiramiz.

Agar A kvadrat matritsa uchun

$$f(A) \equiv a_0 A^m + a_1 A^{m-1} + \dots + a_{m-1} A + a_m E = 0$$

tenglik o‘rinli bo‘lsa, u holda

$$f(\lambda) = a_0\lambda^m + a_1\lambda^{m-1} + \dots + a_n$$

ko‘phad A matritsa uchun nolga aylantiruvchi ko‘hxad deyiladi. Faqat keltirilgan, ya‘ni bosh koeffitsienti birga teng bo‘lgan ko‘phadlarni qaraymiz. Bunday ko‘phadlarning to‘plami bo‘sh emas, Gamilton-Keli teoremasiga ko‘ra A matritsaning xos ko‘phadi $P(\lambda)$ uning nolga aylantiruvchi ko‘phadidir: $P(A) = 0$.

Demak, n -tartibli ixtiyoriy kvadrat matritsa uchun n - darajali nolga aylantiruvchi ko‘phad mavjud. Bunday ko‘phad yagona emas, chunki agar $P(\lambda)$ A matritsa uchun nolga aylantiruvchi ko‘phad bo‘lsa, u holda $P(\lambda)$ ga bo‘linadigan har qanay boshqa ko‘phad ham nolga aylantiruvchi ko‘phad bo‘ladi. A matritsani nolga aylantiruvchi ko‘phadlar orasida eng kichik darajaga ega bo‘lgan yagona $\varphi(\lambda)$ ko‘phad mavjud. Bu ko‘phad matritsaning minimal ko‘phadi deyiladi. Har qanday nolga aylantiruvchi ko‘phad, shu jumladan A matritsaning xos ko‘phadi $P(\lambda)$ ham minimal ko‘phadga bo‘linadi. Minimal ko‘phadning ildizlari xos ko‘phadning barcha bir-biridan farqli ildizlaridan iboratdir.

Yana quyidagi tushunchani kiritamiz. Faraz qilaylik, $\bar{c}$ biror vektor bo‘lsin. Ma‘lumki, n o‘lchovli fazoda n tadan ortiq chiziqli erkli vektor bo‘lishi mumkin emas. Shuning uchun

$$\bar{c}, A\bar{c}, A^2\bar{c}, \dots, A^n\bar{c} \quad (1)$$

vektorlar orasida chiziqli bog‘lanish mavjuddir. Xattoki, ixtiyoriy $\bar{c}$ vektor uchun ham

$$\varphi(A)\bar{c} = \bar{0} \quad (2)$$

chiziqli bog‘lanish mavjud. Demak, A matritsaning $\varphi(\lambda)$ minimal ko‘phadining darajasi n dan kichik bo‘lsa, (1) sistemada chiziqli erkli vektorlarning soni n dan kichikdir. Berilgan $\bar{c}$ vektor uchun

$$\psi(A)\bar{c} = \bar{0} \quad (3)$$

tenglikni qanoatlantiradigan $\psi(\lambda)$ ko'phadlar orasida bosh koefitsienti birga teng bo'lgan eng kichik darajali yagona $\varphi\bar{c}^{(\lambda)}$ ko'phad mavjudki, uning uchun

$$\varphi\bar{c}(\lambda)\bar{c} = \bar{0}$$

tenglik o'rinli bo'ladi. Bunday ko'phad $\bar{c}$ vektorning minimal ko'phadi deyiladi va u (3) tenglikni qanoatlantiruvchi $\psi(\lambda)$ ko'phadning bo'luvchisi bo'ladi. Xususiyl holda, ixtiyoriy $\bar{c}$ vektorning minimal ko'phadi $\varphi_{\bar{c}}(\lambda)A$ matritsa minimal ko'phadi $\varphi(\lambda)$ ning bo'luvchisi bo'ladi. Agar (1) sistemada $\bar{c}$, $A\bar{c}$, $A^2\bar{c}$, ..., $A^{m-1}\bar{c}$ vektorlar chiziqli erkli bo'lib, $A^m\bar{c}$ ularga chiziqli bog'liq bo'lsa,

$$A^m\bar{c} = q_m\bar{c} + q_{m-1}A\bar{c} + \dots + q_1A^{m-1}\bar{c}$$

u holda

$$\lambda^m - q_1\lambda^{m-1} - q_2\lambda^{m-2} - \dots - q_{m-1}\lambda - q_m = 0$$

ko'phad A matritsaning minimal ko'phadi $\varphi(\lambda)$ ga yoki uning bo'luvchisi $\varphi_{\bar{c}}(\lambda)$ ga teng.

Minimal ko'phadni topish. Endi A.N.Krilov metodini ko'rib chiqamiz. Ixtiyoriy noldan farqli $\bar{c}^{(0)} = (c_{01}, c_{02}, \dots, c_{0n})'$ vektorni olib,

$$c^{(i)} = A\bar{c}^{(i-1)} = (c_{i1}, c_{i2}, \dots, c_{in})' \quad (i = \overline{1, n}) \quad (4)$$

vektorlar ketma-ketligini tuzamiz. Yuqorida aytganimizdek, bu vektorlar orasida

$$q_1\bar{c}^{(n-1)} + q_2\bar{c}^{(n-2)} + \dots + q_n\bar{c}^{(0)} = \bar{c}^{(n)} \quad (5)$$

chiziqli kombinasiya mavjuddir. Agar buni koordinatalarda yozib olsak, $q_1, q_2, \dots, q_n$ larni topish uchun quyidagi chiziqli algebraik

tenglamalar sistemasiga ega bo‘lamiz:

$$\begin{cases} q_1 c_{n-1,1} + q_2 c_{n-2,1} + \dots + q_n c_{01} = c_{n1} \\ q_1 c_{n-1,2} + q_2 c_{n-2,2} + \dots + q_n c_{02} = c_{n2} \\ q_1 c_{n-1,n} + q_2 c_{n-2,n} + \dots + q_n c_{0n} = c_{nn} \end{cases} \quad (6)$$

Bu sistemaning determinanti

$$\Delta = |c_{n-1,1} \dots c_{01} \dots \dots \dots c_{n-1,n} \dots c_{0n}|$$

faqat $\bar{c}^{(n-1)}$, $\bar{c}^{(n-2)}$, ..., $\bar{c}^{(0)}$ vektorlar chiziqli erkli bo‘lgandagina noldan farqlidir, chunki bu determinantning ustunlari shu vektorlar koordinatalaridan tuzilgan.

Arap Gauss metodining to‘g‘ri yurishidagi barcha n qadam bajarilib, (6) sistema quyidagi

$$\begin{cases} q_1 + b_{12}q_2 + b_{13}q_3 + \dots + b_{1n}q_n = d_1 \\ q_2 + b_{23}q_3 + \dots + b_{2n}q_n = d_2 \\ \dots \cdot \dot{q} \cdot \dot{q}_n = \dot{d}_n \end{cases} \quad (7)$$

uchburchak shaklga keltirilsa, u holda $\Delta \neq 0$ bo‘lib, $\bar{c}^{(0)}$, $\bar{c}^{(1)}$ $\bar{c}^{(n-1)}$ vektorlar chiziqli erkli. U vaqtda (7) sistemadan qaralayotgan kombinatsiyaning koeffitsientlari q_n , q_{n-1} ..., q_1 ni topa olamiz.

Agar Gauss metodidagi to‘g‘ri yurishning faqat m ta qadami bajarilsa, u holda faqat avvalgi m ta $\bar{c}^{(0)}$, $\bar{c}^{(1)}$, ..., $\bar{c}^{(m-1)}$ vektorlar chiziqli erkli bo‘ladi. Kerakli

$$q_1 \bar{c}^{(m-1)} + q_2 \bar{c}^{(m-2)} + \dots + q_m \bar{c}^{(0)} = \bar{c}^{(m)}$$

chiziqli kombinatsiyaning kordinatalarda yozib olamiz:

$$\begin{cases} q_1 c_{m-1,1} + q_2 c_{m-2,1} + \dots + q_m c_{01} = c_{m1} \\ q_1 c_{m-1,2} + q_2 c_{m-2,2} + \dots + q_m c_{02} = c_{m2} \\ \dots \dots \dots \\ q_1 c_{m-1,n} + q_2 c_{m-2,n} + \dots + q_m c_{0n} = c_{mn} \end{cases} \quad (8)$$

Bu sistemadan Gauss metodi yordamida m ta chiziqli erkli tenglamalarni ajratib olib, $q_n, q_{n-1}, \dots, q_1$ koeffitsiyentlarni topamiz.

Shunday qilib, biz $m = n$ bo'lganda A matritsaning xos ko'phadini va $m < n$ bo'lganda uning bo'luvchisini topishimiz mumkin. Avval $m = n$ bo'lgan holni ko'raylik. Bu holda (2.5) chiziqli kobinatsiyaning $q_1, q_2, \dots, q_n$ koeffitsientlari

$$P(\lambda) = \lambda^n - p_1\lambda^{n-1} - \dots - p_n$$

xos ko'phadning mos ravishda $p_1, p_2, \dots, p_n$ koeffitsientlariga teng:

$$q_i = p_i (i = 1, 2, \dots, n).$$

Xaqiqatan xam, Gamilton-Keli teoremasiga ko'ra

$$P(A) \equiv A^n - p_1A^{n-1} - \dots - p_nE = 0.$$

Bu tenglikni $\bar{c}^{(0)}$ vektorga ko'paytirib va

$$A^i \bar{c}^{(0)} = \bar{c}^{(i)} \quad (i = 1, 2, \dots, n).$$

larni hisobga olib,

$$p_1 \bar{c}^{(n-1)} + p_2 \bar{c}^{(n-2)} + \dots + p_n \bar{c}^{(0)} = \bar{c}^{(n)}$$

ga ega bo'lamiz. Bu tenglikni (5) dan ayirib,

$$(q_1 - p_1) \bar{c}^{(n-1)} + (q_2 - p_2) \bar{c}^{(n-2)} + \dots + (q_n - p_n) \bar{c}^{(0)} = 0$$

ni xosil qilamiz.

$\bar{c}^{(0)}, \bar{c}^{(1)}, \dots, \bar{c}^{(n-1)}$ vektirlar chiziqli erkli bo'lganligi uchun (2.9) tenglik faqat $p_i = q_i (i = 1, 2, \dots, n)$ bo'lgandagina bajariladi.

Demak, $m = n$ bo'lganda qurilgan chiziqli kombinatsiyaning ko'rinishiga qarab, A matritsaning $P(\lambda)$ xos ko'phadini yozish

mumkin. $P(\lambda) = 0$ tenglamani yechib matritsaning barcha xos sonlarini topamiz. Agar $m < n$ bo'lsa, qurilgan chiziqli kombinatiya

$$q_1 \bar{c}^{(m-1)} + q_2 \bar{c}^{(m-2)} + \dots + q_m \bar{c}^{(0)} = \bar{c}^{(m)} \quad (10)$$

ko'rinishga ega bo'ladi. Endi $\bar{c}^{(l)} = A^i \bar{c}^{(0)}$ ($i = 1, 2, \dots, m$) larni hisobga olib (10) tenglikni

$$(A^m - q_1 A^{m-1} - q_2 A^{m-2} - \dots - q_m E) \bar{c}(0) = 0$$

yoki

$$\varphi_{\bar{c}(0)}(A) \bar{c}^{(0)} = 0$$

ko'rinishda yozib olamiz. Bu yerda

$$\varphi_{\bar{c}}(0)(\lambda) = \lambda^m - q_1 \lambda^{m-1} - q_2 \lambda^{m-2} - \dots - q_m$$

Demak, izlanayotgan kombinatsiyaning koeffitsientlari $q_1, q_2, \dots, q_m$ $\bar{c}^{(0)}$ vektorning minimal ko'phadi $\varphi_{\bar{c}}^{(0)}(\lambda)$ ning koeffitsientlaridir. Bunday ko'phad $\bar{c}^{(0)}, \bar{c}^{(1)}, \dots, \bar{c}^{(m-1)}$ vektorlar chiziqli erkli bo'lganligi uchun yagonadir.

Shunday qilib, $m < n$ bo'lganda biz $P(\lambda)$ ning $\varphi_{\bar{c}}^{(0)}(\lambda)$ bo'luvchisini topamiz va $\varphi_{\bar{c}}^{(0)} \lambda = 0$ tenglamani yechib, matritsaning bir qism xos sonlarini topamiz. Dastlabki $\bar{c}^{(0)}$ vektorni boshqacha tanlab, qolgan xos sonlarni ham topish mumkin. Shu bilan birga yangi tanlangan vektor oldin aniqlangan vektorlarning chiziqli kombinatsiyasi bo'lmashligi kerak.

Matritsaning xos vektorlarini topish. Endi xos vektorlarni topish masalasiga o'tamiz.

Faraz qilaylik, λ_i

$$\varphi_{\bar{c}}(0)(\lambda) = \lambda^m - q_1 \lambda^{m-1} - q_2 \lambda^{m-2} - \dots - q_m$$

minimal ko'phadning ildizi bo'lsin (keyingi muloxazalar $m = n$ va $m < n$ xollar uchun bir xil). A matritsaning λ_i xos soniga

mos keladigan $\overline{x^{(l)}}$ xos vektorini oldingi punktda topilgan $\bar{c}^{(0)}, \bar{c}^{(1)}, \dots, \bar{c}^{(m-1)}$ vektorlarning chiziqli kombinatsiyasi shaklida izlaymiz:

$$\bar{x}^{(l)} = \beta_{i1}\bar{c}^{(0)} + \beta_{i2}\bar{c}^{(1)} + \dots + \beta_{lm}\bar{c}^{(m-1)} \quad (2.11)$$

Bu tenglikni A ga ko'paytirib va $\bar{c}^{(j)} = A\bar{c}^{(j-1)}$ xamda $A\bar{x}^{(i)} = \lambda_i\bar{x}^{(i)}$ tengliklarni hisobga olib,

$$\begin{aligned} \lambda_i (\beta_{i1}\bar{c}^{(0)} + \beta_{i2}\bar{c}^{(1)} + \dots + \beta_{im}\bar{c}^{(m-1)}) = \\ = \beta_{i1}\bar{c}^{(1)} + \beta_{i2}\bar{c}^{(2)} + \dots + \beta_{im}\bar{c}^{(m)} \end{aligned} \quad (2.12)$$

ga ega bo'lamiz. Bundan ta shqari, yana

$$\varphi_{\bar{c}^{(0)}}(A)\bar{c}^{(0)} = \bar{c}^{(m)} - q_1\bar{c}^{(m-1)} - q_2\bar{c}^{(m-2)} - \dots - q_m\bar{c}^{(0)} = 0$$

ni hisobga olsak, u holda (2.12) ni

$$\begin{aligned} \lambda_i (\beta_{i1}\bar{c}^{(0)} + \beta_{i2}\bar{c}^{(1)} + \dots + \beta_{lm}\bar{c}^{(m-1)}) = \\ = \beta_{i1}\bar{c}^{(1)} + \beta_{i2}\bar{c}^{(2)} + \dots + \beta_{l,m-2}\bar{c}^{(m-1)} + \\ + \beta_{lm} (q_1\bar{c}^{(m-1)} + q_2\bar{c}^{(m-2)} + \dots + q_{m-1}\bar{c}^{(1)} + q_m\bar{c}^{(0)}) \end{aligned}$$

yoki

$$\begin{aligned} (\lambda_i\beta_{i1} - \beta_{im}q_m)\bar{c}^{(0)} + (\lambda_i\beta_{i2} - \beta_{i1} - \beta_{im}q_{m-1})\bar{c}^{(1)} + \\ + \dots + (\lambda_i\beta_{i,m-1} - \beta_{i,m-2} - q_2\beta_{im})\bar{c}^{(m-2)} + \\ + (\lambda_i\beta_{im} - \beta_{t,m-1} - q_1\beta_{im})\bar{c}^{(m-1)} = 0 \end{aligned}$$

ko'rinishda yozib olishimiz mumkin. Bundan $\bar{c}^{(0)}, \bar{c}^{(1)}, \dots, \bar{c}^{(m-1)}$ vektorlarning chiziqli erkliligini hisobga olsak,

$$\lambda_i\beta_{i1} - \beta_{tm}q_m = 0$$

$$\lambda_{if i2} - \beta_{i1} - \beta_{im}q_{m-1} = 0,$$

$$\lambda_t \beta_{l,m-1} - \beta_{l,m-2} - \beta_{lm} q_2 = 0,$$

$$\lambda_i \beta_{lm} - \beta_{l,m-1} - \beta_{im} q_1 = 0$$

tengliklar kelib chiqadi. Oxirgi tenglikdan boshlab, ketma-ket β_{ik} larni topamiz:

$$\beta_{i,m-1} = (\lambda_l - q_1) \beta_{lm},$$

$$\beta_{l,m-2} = (\lambda_i^2 - q_1 \lambda_l - q_2) \beta_{lm},$$

$$\beta_{l1} = (\lambda_l^{m-1} l + q_1 \lambda_l^{m-2} - \dots - q_{m-1}) \beta_{lm},$$

$$(\lambda_l^m - q_1 \lambda_l^{m-1} - \dots - q_m) \beta_{lm} = 0.$$

Oxirgi tenglik barcha β_{im} lar uchun o'rinlidir, chunki

$$\varphi_c(\lambda_i) = \lambda_l^m - q_1 \lambda_l^{m-1} - \dots - q_m = 0$$

tenglikdan hisoblashni kontrol qilish uchun foydalanish mumkin. Hisoblashni soddalashtirish maqsadida $\beta_{in} = 1$ deb olishimiz mumkin. Unda qolganlari quyidagicha topiladi:

$$\left\{ \begin{array}{l} \beta_{im} = 1 \\ \beta_{i,m-1} = \lambda_i - q_1 \\ \beta_{i,m-2} = \lambda_i^2 - q_1 \lambda_i - q_2 \\ \dots\dots\dots \\ \beta_{i1} = i_i^{m-1} - q_1 \lambda_i^{m-2} - \dots - q_{m-1} \end{array} \right. \quad (2.13)$$

Bularni hisoblashda Gerner sxemasidan foydalanish ma'quldir. Agar berilgan λ_i xos songa A matritsaning bir necha xos vektori mos kelsa, u holda ularni izlash uchun boshqa dastlabki vektorni tanlab olib, hisoblash jarayonini takrorlash mumkin.

1-misol. A.N.Krilov metodi bilan quyidagi

$$A = \begin{bmatrix} -1 & 2 & 2 & 0 \\ 2 & 1 & 3 & 2 \\ 2 & -1 & 2 & 3 \\ 0 & 2 & 1 & -2 \end{bmatrix}$$

matritsaning xarakteristik ko'phadi topilsin.

Yechish. Dastlabki $\bar{c}^{(0)}$ vektor sifatida $(1, 0, 0, 0)'$ ni olib, $\bar{c}^{(1)}$, $\bar{c}^{(2)}$, $\bar{c}^{(3)}$, $\bar{c}^{(4)}$ larni topamiz:

$$\bar{c}^{(1)} = \begin{bmatrix} -1 & 2 & 2 & 0 \\ 2 & 1 & 3 & 2 \\ 2 & -1 & 2 & 3 \\ 0 & 2 & 1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

$$\bar{c}^{(2)} = \begin{bmatrix} 9 \\ 6 \\ 0 \\ 6 \end{bmatrix}, \quad \bar{c}^{(3)} = \begin{bmatrix} 3 \\ 36 \\ 30 \\ 0 \end{bmatrix}, \quad \bar{c}^{(4)} = \begin{bmatrix} 129 \\ 132 \\ 30 \\ 102 \end{bmatrix}.$$

Bu vektorlar yordamida (2.6) sistemani tuzamiz:

$$\begin{cases} 3q_1 + 9q_3 - q_3 + q_4 = 129 \\ 36q_1 + 6q_2 + 2q_3 = 132 \\ 30q_1 + 2q_3 = 30 \\ 6q_2 = 102 \end{cases}$$

Bu sistemani Gauss metodi bilan yechamiz:

$$q_1 = 0, \quad q_2 = 17, \quad q_3 = 15, \quad q_4 = -9$$

Demak, A matritsaning xarakteristik ko'phadi

$$\varphi(\lambda) = \lambda^4 - 17\lambda^2 - 15\lambda + 9$$

ekan.

2- misol. Quyidagi

$$A = \begin{bmatrix} 5 & 30 & -48 \\ 3 & 14 & -24 \\ 3 & 15 & -25 \end{bmatrix}$$

matridaning xos sonlari va xos vektorlari tolilsin.

Yechish. 1- misoldagidek, $\bar{c}^{(j)}$ vektorlarni topamiz:

$$\bar{c}^{(0)} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \bar{c}^{(1)} = \begin{bmatrix} 5 \\ 3 \\ 3 \end{bmatrix}, \quad \bar{c}^{(2)} = \begin{bmatrix} -29 \\ -15 \\ -15 \end{bmatrix}, \quad \bar{c}^{(3)} = \begin{bmatrix} 125 \\ 63 \\ 63 \end{bmatrix}$$

(2.6) sistema kuyidagi ko‘rinishga ega:

$$\begin{cases} -29q_1 + 5q_2 + q_3 = 125 \\ -15q_1 + 3q_2 = 63 \\ -15q_1 + 3q_2 = 63 \end{cases}$$

Bu sistemani Gauss metodi bilan yechganda to‘g‘ri yurishning uchinchi qadami bajarilmaydi, chunki uchinchi tenglama ikkinchi bilan bir xil. Shuning uchun xam (2.8) sistemani tuzamiz:

$$\begin{cases} 5q_1 + q_2 = -29 \\ 3q_1 = -15 \end{cases}$$

Bundan $q_1 = -5$, $q_2 = -4$ va $\varphi_c^-(\lambda) = \lambda^2 + 5\lambda + 4$. Shunday kilib, $\lambda_1 = -4$, $\lambda_2 = -1$. Endi uchinchi xos sonni topish uchun $a_{11} + a_{22} + a_{33} = \lambda_1 + \lambda_2 + \lambda_3$ tenglikdan foydalanamiz: $5 + 14 - 25 = -1 - 4 + \lambda_3$. Demak, $\lambda_3 = -1$. Shunday qilib, $\lambda_2 = \lambda_3 = -1$ ekan. Endi bu xos sonlarga mos keladigan xos vektorlarni topamiz. Buning uchun (2.11) - (2.13) va $\beta_{i2} = 1$ dan foydalanib, quyidagilarni yoza olamiz:

$$\bar{x}^{(l)} = \beta_{i1}\bar{e}^{(0)} + \beta_{i2}\bar{c}^{(1)} = \begin{bmatrix} 5(\lambda_i - q_1) + 1 \\ 3(\lambda_i - q_1) \\ 3(\lambda_i - q_1) \end{bmatrix}$$

Bundan esa

$$\bar{x}^{(1)} = (6, 3, 3)', \quad \bar{x}^{(2)} = (21, 12, 12)'$$

ni topamiz. Uchinchi vektorni topish uchun dastlabki vektorni boshqacha tanlash kerak.

Mavzuni mustahkamlash uchun misollar.

Quyidagi matritsalarini A.Krilov usuli yordamida xos son va vektorlarini toping.

$$1. A = \begin{pmatrix} 4 & -5 & 2 \\ 5 & -7 & 3 \\ 6 & -9 & 4 \end{pmatrix}.$$

$$2. A = \begin{pmatrix} 3 & 1 & 0 \\ -4 & -1 & 0 \\ 4 & -8 & -2 \end{pmatrix}.$$

$$3. A = \begin{pmatrix} 1 & 2 & -4 \\ 2 & -2 & -2 \\ -4 & -2 & 1 \end{pmatrix}.$$

$$4. A = \begin{pmatrix} 4 & -1 & -2 \\ 2 & 1 & -2 \\ 1 & -1 & 1 \end{pmatrix}.$$

$$5. A = \begin{pmatrix} -3 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{pmatrix}.$$

$$6. A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$7. A = \begin{pmatrix} 1 & -3 & 4 \\ 4 & -7 & 8 \\ 6 & -7 & 7 \end{pmatrix}.$$

$$8. A = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 4 & 1 \\ 0 & 1 & 4 \end{pmatrix}.$$

$$9. A = \begin{pmatrix} 7 & -12 & 6 \\ 10 & -19 & 10 \\ 12 & -24 & 13 \end{pmatrix}.$$

$$10. A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix}.$$

$$11. A = \begin{pmatrix} 2 & -5 & -3 \\ -1 & -2 & -3 \\ 3 & 15 & 12 \end{pmatrix}.$$

$$12. A = \begin{pmatrix} 4 & -4 & 2 \\ 2 & -2 & 1 \\ -4 & 4 & -2 \end{pmatrix}.$$

$$13. A = \begin{pmatrix} -1 & 3 & -1 \\ -3 & 5 & -1 \\ -3 & 3 & 1 \end{pmatrix}.$$

$$14. A = \begin{pmatrix} 0 & 1 & 0 \\ -4 & 4 & 0 \\ -2 & 1 & 2 \end{pmatrix}.$$

$$15. A = \begin{pmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{pmatrix}.$$

$$16. A = \begin{pmatrix} 5 & 6 & -3 \\ -1 & 0 & 1 \\ 1 & 2 & 1 \end{pmatrix}.$$

$$17. A = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 8 \end{pmatrix}.$$

$$18. A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ -1 & 1 & 3 \end{pmatrix}.$$

$$19. A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$20. A = \begin{pmatrix} 7 & -4 & -2 \\ -2 & 5 & -2 \\ 0 & 0 & 9 \end{pmatrix}.$$

$$21. A = \begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & 3 & -1 \end{pmatrix}.$$

$$22. A = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & 16 \end{pmatrix}.$$

$$23. A = \begin{pmatrix} 3 & 0 & 4 \\ 0 & 1 & 0 \\ 4 & 0 & 3 \end{pmatrix}.$$

$$24. A = \begin{pmatrix} -1 & -2 & 0 \\ -2 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

$$25. A = \begin{pmatrix} -6 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & -2 & 1 \end{pmatrix}.$$

$$26. A = \begin{pmatrix} 3 & 0 & -1 \\ 0 & 3 & 0 \\ -1 & 0 & 3 \end{pmatrix}.$$

$$27. A = \begin{pmatrix} 1,7 & 2,8 & 0,3 \\ 2,8 & 1,2 & 0,6 \\ 0,3 & 0,6 & 1,5 \end{pmatrix}.$$

$$28. A = \begin{pmatrix} 1,7 & 0,4 & 2,8 \\ 0,4 & 3,2 & 1,2 \\ 2,8 & 1,2 & 0,5 \end{pmatrix}.$$

$$29. A = \begin{pmatrix} 2,3 & 1,4 & 0,6 \\ 1,4 & 1,7 & 0,5 \\ 0,6 & 0,5 & 1,3 \end{pmatrix}.$$

$$30. A = \begin{pmatrix} 2,3 & 3,5 & 1,4 \\ 3,5 & 0,4 & 0,6 \\ 1,4 & 0,6 & 1,3 \end{pmatrix}.$$

VI BOB.

FUNKSIYALARNI INTERPOLYATSIYASLASH

Funksiyalarni yaqinlashtirish masalasi qo'yilgan shartga qarab turlicha bo'ladi. Hisoblash matematikasida keng qo'llaniladigan usullardan ba'zilarini eslatib o'tamiz. Xususan, funksiyalarga yaqinlashtirish interpolyatsiyalash, o'rtacha kvadratik ma'noda yaqinlashtirish, tekis yaqinlashish va splayn yaqinlashish. Interpolyatsiyalash funksiyalari yaqinlashtirish nazariyasida olingan natijalarni funksiya jadvalini zichlashtirish, sonli differensiallash va integrallash, matematik fizika masalalarining to'rdagi analogini qurishda keng qo'llaniladi.

6.1-§. Lagranj interpolyatsion ko'phadi.

$[a, b]$ kesmada turli $n+1$ ta $x_k, k = 0, 1, \dots, n$ nuqtalarda $f(x)$ funksiyaning qiymatlari $f(x_k), k = 0, 1, \dots, n$ berilgan bo'lsin. Darajasi n ga teng shunday

$$L_n(x) = a_0 + a_1x + \dots + a_nx^n \quad (1)$$

algebraik ko'phad qurilsinki, u

$$a_0 + a_1x_k + \dots + a_nx_k^n = f(x_k), k = 0, 1, \dots, n \quad (2)$$

shartlarni qanoatlantirsin.

(1) chiziqli algebraik tenglamalar sistemasining determinanti Vandermond determinantidir, u noldan farqli, chunki $x_k, k = 0, 1, \dots, n$ lar turli. Demak, (1) ko'phadning koeffitsiyentlari (2) dan bir qiymatli ko'rinishda topiladi.

$$L_n(x_i) = f(x_i), i = 0, 1, \dots, n \quad (3)$$

shartni qanoatlantiradigan $L_n(x)$ ko'phad $f(x)$ funksiyaning $\{x_i\}_{i=0}^n$ tugun nuqtalar yordamida qurilgan interpolatsion ko'phad deyiladi.

Bu interpolatsion ko'phadning ko'proq ishlatiladigan ko'rinishi - Lagranj formulasini keltiramiz. Uni quyidagi ko'rinishda yozamiz:

$$L_n(x) = \sum_{k=0}^n b_k(x) f(x_k) \quad (4)$$

(3) ga asosan

$$\sum_{k=0}^n b_k(x_i) f(x_k) = f(x_i), \quad i = 0, 1, \dots, n$$

bo'ladi.

Bu munosabatlar o'rinli bo'lishi uchun $b_k(x)$ funksiyalar quyidagi shartlarni bajarishi kerak:

$$b_k(x_i) = \begin{cases} 0, & i \neq k \\ 1, & i = k, i = 0, 1, \dots, n \end{cases}$$

Bulardan ko'rinib turibdiki, har bir $b_k(x)[a, b]$ da n tadan kam emas nolga ega bo'ladi. Biroq $L_n(x)$ darajasi n ga teng bo'lgan algebraik ko'phad bo'lganligi uchun $b_k(x)$ ning darajasini n ga teng ko'phad ko'rinishida izlash maqsadga muvofiqdir. Uni quyidagicha yozamiz:

$$b_k(x) = A_k(x - x_0)(x - x_1)\dots(x - x_{k-1})(x - x_{k+1})\dots(x - x_n)$$

$$b_k(x_k) = 1 \text{ shartdan}$$

$$A_k^{-1} = (x_k - x_0)(x_k - x_1)\dots(x_k - x_{k-1})(x_k - x_{k+1})\dots(x_k - x_n)$$

kelib chiqadi. Agar

$$\omega'_{n+1}(x_k) = A_k^{-1}$$

deb olsak, (4) ko'phad

$$L_n(x) \sum_{k=0}^n \frac{\omega_{n+1}(x)}{(x - x_k) \cdot \omega'_{n+1}(x_k)} f(x_k) \quad (5)$$

ko'rinishga ega bo'ladi va uni Lagranj interpolatsion ko'phadi deyiladi.

(5) dagi

$$\frac{\omega_{n+1}(x)}{(x - x_k) \cdot \omega'_{n+1}(x_k)}$$

ko'phad interpolatsiyalashning fundamental ko'phadlari deyiladi, ba'zan uni x_k nuqtaning ta'sir etuvchi ko'phadi deb ham aytiladi.

Misol. Jadval bilan berilgan funksiyaning Lagranj ko'phadi bilan yaqinlashtiring:

x	0	1	2	5
$f(x)$	2	3	12	147

Yechish.

$$\begin{aligned} P_3(x) &= \sum_{i=0}^3 \frac{\omega_4(x)}{(x - x_i) \omega'_4(x_i)} f(x_i) = \\ &= \frac{(x-1)(x-2)(x-5)}{(-1)(-2)(-5)} \cdot 0 + \frac{x(x-2)(x-5)}{1 \cdot (-1)(-4)} \cdot 3 + \\ &+ \frac{x(x-1)(x-5)}{2 \cdot 1(-3)} \cdot 12 + \frac{x(x-1)(x-2)}{5 \cdot 43} \cdot 147 = \\ &= x^3 + x^2 - x + 2. \end{aligned}$$

Tugun nuqtalarda funksiyaning qiymatlari berilgan bo'lsa, Lagranj interpolatsion ko'phadidan foydalanib berilgan nuqtada funksiyaning qiymatini hisoblash dasturi.

```
class Program
{
    public static void Main(string[] args)
    {
        double S=0;
        int j,i;
        double[] x=new double[100];
        double[] f=new double[100];
        Console.Write("a=");
        double a=double.Parse(Console.ReadLine());
        Console.Write("n=");
        int n=int.Parse(Console.ReadLine());
        for(i=0; i<n; i++)
        {Console.Write("x["+i+"]=");
        x[i]=double.Parse(Console.ReadLine());}
        for(i=0; i<n; i++)
        {Console.Write("f["+i+"]=");
        f[i]=double.Parse(Console.ReadLine());
        }
        for(j=0;j<n; j++)
        {
            double p=1;
            for(i=0; i<n;i++)
            if(i!=j)
                p=p*(a-x[i])/(x[j]-x[i]);
            S=S+f[j]*p;
        }
        Console.WriteLine(S);
        Console.ReadKey(true);
    }
}
```

Mavzuni mustahkamlash uchun misollar

Lagranj interpolatsion ko'phadini ishlatib argumentning berilgan qiymatida funksiya qiymatini taqriban toping.

1.	x_i	y_i	2.	x_i	y_i	3.	x_i	y_i
0	1.25	4.82835	0	13.5	4.90583	0	0.145	4.97674
1	1.27	4.84418	1	13.7	4.92007	1	0.147	4.99043
2	1.29	4.85989	2	13.9	4.93459	2	0.149	5.00391
3	1.31	4.87523	3	14.1	4.94882	3	0.151	5.01730
4	1.33	4.86331	4	14.3	4.96571	4	0.153	5.03207

4.	x_i	y_i	5.	x_i	y_i	6.	x_i	y_i
	0.451	0.43587		0.724	0.9		0.349	0.34196
	0.452	0.43677		0.725	0.89957		0.350	0.3429
	0.453	0.43766		0.726	0.89914		0.351	0.34384
	0.454	0.43856		0.727	0.89870		0.352	0.34478
	0.455	0.43945		0.728	0.89825		0.353	0.34488

7.	x_i	y_i	8.	x_i	y_i	9.	x_i	y_i
	0.725	0.66314		0.251	0.24837		0.341	2.19280
	0.727	0.66463		0.254	0.25128		0.343	2.20293
	0.729	0.66612		0.257	0.25418		0.345	2.21309
	0.731	0.66761		0.260	0.25708		0.347	2.22331
	0.733	0.66769		0.263	0.25718		0.349	2.23347

10.	x_i	y_i	11.	x_i	y_i	12.	x_i	y_i
	0.112	1.2942		0.119	1.3152		0.441	2.7605
	0.113	1.2972		0.121	1.3213		0.445	2.7861
	0.114	1.3002		0.123	1.3273		0.449	2.8119
	0.115	1.3032		0.125	1.3335		0.453	2.8379
	0.116	1.3062		0.127	1.3397		0.457	2.8622

13.	x_i	y_i	14.	x_i	y_i	15.	x_i	y_i
	0.117	1.3092		0.750	5.6234		0.261	1.8239
	0.122	1.3243		0.751	5.6364		0.263	1.8323
	0.127	1.3397		0.752	5.6494		0.265	1.8408
	0.132	1.3552		0.753	5.6624		0.267	1.8493
	0.137	1.3707		0.754	5.6754		0.269	1.8578

16.	x_i	y_i	17.	x_i	y_i	18.	x_i	y_i
	0.271	1.8664		0.281	1.9099		0.291	1.9643
	0.273	1.874		0.283	1.9187		0.293	1.9634
	0.275	1.8837		0.285	1.9275		0.295	1.9724
	0.277	1.8924		0.287	1.9364		0.297	1.9816
	0.279	1.9017		0.289	1.9442		0.299	1.9904

19.	x_i	y_i	20.	x_i	y_i	21.	x_i	y_i
	0.501	3.1695		0.511	3.2434		0.521	3.3189
	0.503	3.1842		0.513	3.3342		0.523	3.3342
	0.505	3.1988		0.515	3.2734		0.525	3.3496
	0.507	3.2136		0.517	3.2885		0.527	3.3651
	0.509	3.2282		0.519	3.2957		0.529	3.3785

22.	x_i	y_i	23.	x_i	y_i	24.	x_i	y_i
	0.531	3.3962		0.541	3.4753		0.551	3.5563
	0.533	3.4119		0.543	3.4914		0.553	3.5737
	0.535	3.4276		0.545	3.5075		0.555	3.5892
	0.537	3.4435		0.547	3.5237		0.557	3.6058
	0.539	3.4591		0.549	3.539		0.559	3.6059

25.	x_i	y_i	26.	x_i	y_i	27.	x_i	y_i
	0.561	3.6391		0.351	11.8069		0.432	4.7223
	0.563	3.6559		0.353	11.8157		0.434	4.7346
	0.565	3.6728		0.355	11.8245		0.436	4.7469
	0.567	3.6897		0.357	11.8344		0.438	4.7682
	0.569	3.7065		0.359	11.8412		0.440	4.7805

28.	x_i	y_i	29.	x_i	y_i	30.	x_i	y_i
	0.125	0.6432		0.931	2.2484		0.341	4.4928
	0.127	0.6447		0.934	2.2513		0.343	4.5029
	0.129	0.6462		0.937	2.2542		0.345	4.513
	0.131	0.6477		0.940	2.2571		0.347	4.5234
	0.133	0.6479		0.943	2.2575		0.349	4.5335

6.2-§. Nyuton interpolatsion ko'phadi

Bizga $[a, b]$ da aniqlangan $f(x)$ fimksiyaning $[a, b]$ ga tegishli turli $\{x_k\}_{k=0}^n$ nuqtalarda qiymatlari ma'lum bo'lsin.

Quyidagicha aniqlangan

$$f(x_i, x_{i+1}) = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}, i = 0, 1, \dots, n-1$$

miqdorlar birinchi tartibli ayirmalar nisbati deyiladi, ular yordamida aniqlangan

$$f(x_i, x_{i+1}, x_{i+2}) = \frac{f(x_{i+1}, x_{i+2}) - f(x_i, x_{i+1})}{x_{i+2} - x_i}$$

miqdorlar ikkinchi tartibli ayirmalar nisbati deyiladi.

Yuqori tartibli ayirmalar nisbati ham shunday aniqlanadi, masalan, k -tartibli $f(x_i, x_{i+1}, x_{i+k+1})$ va $f(x_{i+1}, x_{i+2}, \dots, x_{i+k+1})$ ayirmalar nisbati ma'lum bo'lsa, $(k+1)$ -tartibli ayirmalar nisbati

$$f(x_i, x_{i+1}, x_{i+k+1}) =$$

$$= \frac{f(x_{i+1}, x_{i+2}, \dots, x_{i+k+1}) - f(x_i, x_{i+1}, \dots, x_{i+k})}{x_{i+k+1} - x_i}$$

aniqlanadi, $i = 0, 1, \dots, n - k - 1$

Ayirmalar nisbati quyidagi xossalarga ega.

1-xossa. Algebraik yig'indidan olingan ayirmalar nisbati qo'shiluvchilardan olingan ayirmalar nisbatlarining yig'indisiga teng.

2-xossa. O'zgarmasni ayirmalar nisbati belgisidan tashqariga chiqarish mumkin.

3-xossa. Ayirmalar nisbati o'z argumentlariga nisbatan simmetrik funksiyadir.

4-xossa. n -darajali algebraik ko'phaddan olingan k -tartibli ayirmalar nisbati, agar $k > m$ bo'lsa nolga, $k = m$ da o'zgarmasga va $k < m$ bo'lsa argumentlariga nisbatan $(m-k)$ -darajali simmetrik biqinsli ko'phadga teng.

Interpolyatsion ko'phad $L_n(x)$ ning boshqa formasini chiqaramiz. Bining uchun Lagranj interpolatsion ko'phadi $L_n(x)$ dan x_0 nuqtani ishtirok ettirib, birinchi tartibli ayirmalar nisbatini olamiz:

$$L_n(x, x_0) = \frac{L_n(x) - L_n(x_0)}{x - x_0}$$

Bundan

$$L_n(x) = L_n(x_0) + L_n(x, x_0)(x - x_0) \quad (1)$$

hosil bo'ladi.

Endi x_1 nuqtani qo'llab,

$$L_n(x, x_0, x_1) = \frac{L_n(x, x_0) - L_n(x_0, x_1)}{x - x_1}$$

ni hosil qilamiz. Bundan

$$L_n(x, x_0) = L_n(x_0, x_1) + L_n(x, x_0, x_1)(x - x_1) \quad (2)$$

bo'ladi. (2) va (1) dan esa

$$\begin{aligned} L_n(x) &= L_n(x_0) + L_n(x_0, x_1)(x - x_0) + \\ &+ L_n(x, x_0, x_1)(x - x_0)(x - x_1) \end{aligned} \quad (3)$$

hosil bo'ladi.

Mana shu yo'l bilan ketma-ket hamma nuqtalami qo'llab $L_n(x)$ dan ayirmalar nisbatini topamiz va natijada quyidagiga ega bo'lamiz:

$$\begin{aligned} L_n(x) &= L_n(x_0) + L_n(x_0, x_1)(x - x_0) + \\ &+ L_n(x_0, x_1, x_2)(x - x_0)(x - x_1) + \\ &\dots\dots\dots \\ &+ L_n(x_0, x_1, \dots, x_n)(x - x_0)(x - x_1)\dots(x - x_{n-1}) + \\ &+ L_n(x_0, x_1, \dots, x_n)(x - x_0)(x - x_1)\dots(x - x_n) \end{aligned} \quad (4)$$

$L_n(x)$ ko'phadning darajasi n ga teng bo'lganligi uchun 4-xossaga asosan

$$L_n(x_0, x_1, \dots, x_n) = 0$$

bo'ladi va $L_n(x_k) = f(x_k)$, $k = 0, 1, \dots, n$ ekanligini e'tiborga olsak (4) ifoda quyidagi ko'rinishga keladi:

$$\begin{aligned} L_n(x) &= f_n(x_0) + f_n(x_0, x_1)(x - x_0) + \\ &+ f(x_0, x_1, x_2)(x - x_0)(x - x_1) + \dots + \\ &+ f_n(x_0, x_1, \dots, x_n)(x - x_0)(x - x_1)\dots(x - x_{n-1}) \end{aligned} \quad (5)$$

Agar $x_0 < x_1 < \dots < x_n$ bo'lsa, (5) ifodani Nyutonning oldiga interpolatsiyalash formulasi deyiladi.

Agar interpolatsiyalashga tugun nuqtalami $x_n > x_{n-1} > \dots > x_1 > x_0$ tartibda jalb qilsak, Nyutonning ortga interpolatsiyalash deb ataluvchi

$$\begin{aligned} L_n(x) = & f_n(x_n) + f_n(x_n, x_{n-1})(x - x_n) + \\ & + f_n(x_n, x_{n-1}, x_{n-2})(x - x_n)(x - x_{n-1}) + \dots \\ & + f_n(x_n, x_{n-1}, \dots, x_0)(x - x_1) \dots (x - x_n) \end{aligned} \quad (6)$$

formulasiga ega bo'lamiz.

Tugun nuqtalarda funksiyaning qiymatlari berilgan bo'lsa, Nyuton interpolatsion ko'phadidan foydalanib berilgan nuqtada funksiyaning qiymatini hisoblash dasturi.

Misol. $y = f(x)$ funksiyaning quyidagi

x	0	5	10	12	13	15	16
$f(x)$	1	151	1051	17897	2263	3451	4177

jadvalda berilgan qiymatlaridan foydalanib, uning $x = 12,5$ dagi qiymatini toping.

Yechish. Bo'lingan ayirmalar jadvalini tuzimiz:

0	1			
5	151	30	15	
10	1051	180	27	1
12	<u>1789</u>	369	35	1
13	2263	<u>474</u>	<u>40</u>	1
15	3451	594	44	
16	4177	726		

Uchinchi tartibli bo'lingan ayirma o'zgarmas bo'lganligi uchun $y = f(x)$ funksiya 3-darajali ko'phad bo'ladi. Berilgan $x = 12,5$ qiymat jadvaldagi $x = 12$ va $x = 13$ qiymatlar orasida bo'lganligi

uchun, osti chizilgan bo'lingan ayirmalardan foydalanib, Nyuton-ning interpolatsion formulasini tuzamiz:

$$f(x) = 1789 + 474(x - 12) + 40(x - 12)(x - 13) + \\ + (x - 12)(x - 13)(x - 15).$$

Bundan

$$f(12,5) = 1789 + 474 \cdot 0,5 - 40 \cdot 0,5 \cdot 0,5 + \\ + 0,5 \cdot 0,5 \cdot 0,25 = 2016,625$$

Mavzuni mustahkamlash uchun misollar

Nyuton interpolatsion ko'phadini ishlatib argumentning berilgan qiymatida funksiya ko'phad ko'rinishini toping.

1.	x_i	y_i	2.	x_i	y_i	3.	x_i	y_i	4.	x_i	y_i	5.	x_i	y_i
0	-1	-3	0	2	4	0	0	-1	0	7	2	0	-3	7
1	0	5	1	3	1	1	2	-4	1	9	-2	1	-1	-1
2	3	2	2	5	7	2	3	2	2	13	3	2	3	4
3	5	7	3	6	9	3	5	6	3	10	3	3	4	7
4	7	9	4	7	12	4	6	7	4	17	3	4	6	9
5	8	8	5	9	15	5	8	10	5	20	3	5	9	12

6.	x_i	y_i	7.	x_i	y_i	8.	x_i	y_i	9.	x_i	y_i	10.	x_i	y_i
0	1	-3	0	-2	4	0	2	9	0	-4	2	0	-1	4
1	2	-7	1	-1	9	1	4	-3	1	-2	8	1	5	-7
2	4	2	2	2	1	2	5	6	2	0	5	2	3	1
3	6	5	3	5	8	3	7	12	3	8	12	3	9	25
4	7	9	4	9	12	4	10	15	4	12	20	4	12	30
5	9	12	5	14	15	5	15	20	5	15	25	5	15	35

11		12	13	14	15	16	17	18
x_i	y_i	y_i	y_i	y_i	y_i	y_i	y_i	y_i
1,1	0,05	0,07	0,24	2,02	3,02	-0,45	3,12	2,62
1,2	0,1	0,13	0,29	2,05	3,05	-0,41	3,15	2,64
1,3	0,14	0,19	0,34	2,08	3,07	0,36	3,17	2,66
1,4	0,19	0,25	0,39	2,09	3,09	0,32	3,19	2,68
1,5	0,22	0,31	0,45	2,11	3,11	0,28	3,21	2,7
1,6	0,27	0,37	0,5	2,12	3,12	0,24	3,22	2,72

19		20	21		22		23	
x_i	y_i	y_i	x_i	y_i	x_i	y_i	x_i	y_i
1,1	3,62	1,23	0,43	1,63	0,02	1,02	0,35	2,73
1,2	3,64	1,28	0,48	1,73	0,08	1,09	0,41	2,3
1,3	3,66	1,33	0,55	1,87	0,12	1,14	0,47	1,96
1,4	3,68	1,39	0,62	2,03	0,17	1,21	0,51	1,78
1,5	3,7	1,44	0,7	2,22	0,23	1,3	0,56	1,59

24		24		25		26	
x_i	y_i	x_i	y_i	x_i	y_i	x_i	y_i
0,43	1,63	0,41	2,57	0,02	1,02	0,35	2,73
0,48	1,73	0,46	2,32	0,08	1,09	0,41	2,3
0,55	1,87	0,52	2,09	0,12	1,14	0,51	1,78
0,62	2,03	0,6	1,86	0,17	1,21	0,6	1,86
0,7	2,22	0,65	1,74	0,23	1,30	0,56	1,59

26		27		28		29		30	
x_i	y_i	x_i	y_i	x_i	y_i	x_i	y_i	x_i	y_i
0,41	2,57	0,68	0,81	0,11	9,06	0,15	5,12	3	9
0,46	2,32	0,73	0,89	0,15	6,61	0,2	4,21	7	12
0,47	1,96	0,52	2,09	0,16	1,02	0,21	4,691	9	20
0,88	1,2	0,54	3,35	0,2	2,21	0,3	6,1	10	15
0,65	1,74	0,6	1,34	0,24	2,74	0,4	7,24	11	25

6.3-§. Splayn-yaqinlashtirish. Splayn interpolyasiya.

Ta’rif. Silliqligi yuqori bo’lmagan funktsiyalar uchun ko’phadlar yaqinlashish apparati sifatida qator noqulayliklarga ega. Bulardan eng asosiysi shundan iboratki, bunday funktsiyalar-ning biror nuqta atrofidagi holati, ularning to’la xolati bilan uzviy bog’liqdir. Bundan tashqari interpolatsion ko’phadlarning nuqsoni sifatida ularning har doim ham interpolatsiyalanuvchi funktsiyaga yaqinlashmasligidir. Eng yaxshi tekis yaqinlashuvchi ko’phadlarning kamchiligi sifatida shuni ko’rsatish mumkinki, ularni qurish juda qiyin va odatda bunday ko’phadning darajasi ortishi bilan koeffitsientlari ham tez o’sib boradi.

Oxirgi vaqtlarda shu nuqssondan holi bo’lgan boshqa yaqinlashish apparatlari ishlab chikilmoqda. Nazariy tadqiqot va tatbiqlarda yaxshi natija beradigan apparat – splayn-funktsiyalar apparatidir. Splaynning ta’rifi bilan tanishaylik. Xaqiqiy o’qdagi $[a, b]$ oraliqda ushbu

$$\Delta_n : a = x_0 < x_1 < \dots < x_n = b$$

to’r berilgan bo’lsin. Faraz qilaylik, $H_m(P)$ darajasi m dan ortmaydigan ko’phadlar to’plami, $C^{(k)} = C^{(k)}[a, b]$ o’zi va k tartibgacha xosilalari $[a, b]$ oraliqda uzluksiz bo’lgan funktsiyalar to’plami bo’lsin.

Ta’rif. Quyidagi ikkita shartni qanoatlantiruvchi ushbu

$$S_m(x) = S'_m(x, \Delta_n)$$

funktsiya defekti 1 ga teng bo’lgan m - darajali polinomial splayn deyiladi:

1. Xar bir $[x_i, x_{i+1}]$ ($i = \overline{0, n}$) oraliqda $S_m(x) \in H_m(P)$;
2. $S_m(x) \in C^{(m-1)}[a, b]$.

Bu yerdagi $\{x_i\}$ nuqtalar splayn tugunlari deyiladi. $S_m(x)$ splaynning m - xosilasi $[a, b]$ oraliqda uzilishga ega bo'lishi xam mumkin.

Agar $k = 0, 1, \dots, m$ lar uchun

$$S_m^{(k)}(a+0) = S_m^{(k)}(b-a)$$

tengliklar bajarilsa, $S_m(x)$ splayn $b-a$ davrli davriy splayn deyiladi.

Ta'rifni qanoatlantiruvchi splaynlar bilan bir qatorda shunday splaynlar ham qaraladiki, ularning silliqligi Δ_n to'ring turli qismlarida turlichadir. Bunday splaynlar $[a, b]$ oraliqning turli qismlarida turli sillqlikka ega bo'lgan funktsiyalarni yaqinlashtirishda foydalaniladi.

Odatda, splayn yagona ravishda aniqlanishi uchun $[a, b]$ oraliqning chetki a va b nuqtalarida chegaraviy shartlar deb ataluvchi qo'shimcha shartlar qo'yiladi. Amalda uchinchi darajali, ya'ni kubik splaynlar keng qo'llaniladi.

Splaynlarning hisoblash matematikasida keng qo'llanilayotganligi sabablaridan yana biri ularning qiymatlarini kompyuterlarda hisoblashning qulayligi va ular yordamida interpolatsiyaslash kabi jarayonlarning keng sinfdagi to'rlar uchun yaxshi -yaqinlashishligidadir (yuqorida aytilgandek ko'phad bilan interpolatsiyaslash bunday emas).

Bundan buyon biz interpolatsion kubik va

$$S_3''(x) = S_3''(b) = 0$$

chegaraviy shartlarni qanoatlantiruvchi splaynlar bilan shug'ullanamiz.

Interpolatsion kubik splaynlarni qurish. Oldingi punktda aytilgandan so'ng quyidagi ta'rifni bera olamiz.

Ta'rif. Quyidagi to'rt shartni qanoatlantiruvchi ushbu

$$S(f, x) = S_3(f, x, \Delta_n)$$

funktsiya interpolyasiyon kubik splayn deyiladi:

1. Har bir $[x_i, x_{i+1}]$ ($i = \overline{0, n}$) oralikda $S(f, x) \in H_3(P)$;
2. $S(f, x) \in C^2[a, b]$
3. To'rtinchi x_k ($k = \overline{0, n}$) tugunlarida $S(f, x_k) = f_k$ tenglik o'rinli;
4. $S''(f, x)$ uchun $S''(f, a) = S''(f, b) = 0$

chegaraviy shartlar bajariladi. Bu to'rt shartni qanoatlantiruvchi yagona $S(f, x)$ splayn mavjudligini ko'rsatamiz. Buning uchun avval quyidagi yordamchi faktlarni keltiramiz.

Lemma. Faraz qilaylik, $A = [a_{ij}]$ n - tartibli kvadrat matritsaning elementlari

$$\min_{1 \leq i \leq n} \left\{ |a_{il}| - \sum_{i \neq l} |a_{ij}| \right\} = q > 0 \quad (9.2)$$

shartni qanoatlantirsin. U holda $\bar{A} = \bar{b}$ sistema yagona yechimga ega bo'lib, uning yechimi

$$\max_{1 \leq k \leq n} |x_k| \leq q^{-1} \max_{1 \leq k \leq n} |b_k| \quad (9.3)$$

tengsizlikni qanoatlantiradi.

Eslatma. Ko'pincha $x = a$ va $x = b$ nuqtalarda $f(x)$ funksiya haqida qo'shimcha ma'lumotga ham ega bo'lishimiz mumkin. Masalan, splayn tuzishdan asosiy maqsad

$$f(a) + \alpha f'(a) = A, \quad f(b) + \beta f'(b) = B$$

chegaraviy shartlarni qanoatlantiruvchi differentsial tenglamani yechishdan iborat bo'lishi mumkin. Bunday holda, splayn tuzishda $M_0 = M_n = 0$ chegaraviy shart o'rniga yuqoridagi shartni olish kerak.

Endi kubik splayn funksiyani qurish bilar shug'illanamiz:

$S_3'''(x) [x_{t-1}, x_i]$ kesmada uzluksiz bo'lganligidan $x_{t-1} \leq x \leq x_i$ da ushbu

$$S_3''(x) = M_{i-1} \frac{x_i - x}{h_i} + M_i \frac{x - x_{i-1}}{h_i}$$

tenglikni yozish mumkin. Bu yerda $h_i = x_i - x_{i-1}$,

$$M_i = S_3''(x_i). \quad (2)$$

tenglikni ikki marta integrallaymiz:

$$\begin{aligned} S_3(x) = & M_{i-1} \frac{(x_i - x)^3}{6h_i} + M_i \frac{(x - x_{i-1})^3}{6h_i} + \\ & + A_i \frac{(x_i - x)}{h_i} + B_i \frac{(x - x_{t-1})}{h_i}, \end{aligned}$$

bunda A_i va B_i integrallash doimiylari bo'lib, ular ta'rifning uchinchi shartidan topiladi, ya'ni (3) da $x = x_{i-1}$, $x = x_i$ deb, mos ravishda

$$M_{i-1} \frac{h_i^2}{6} + A_i = f_{i-1}, \quad M_i \frac{h_i^2}{6} + B_i = f_i$$

larni hosil qilamiz. Bundan A_i va B_i ni topib (3)ga qo'yib, quyidagiga bo'lish mumkin:

$$\begin{aligned} S_3(x) = & M_{i-1} \frac{(x_1 - x)^3}{6h_i} + M_i \frac{(x - x_{i-1})^3}{6h_i} + \\ & + \left(f_{i-1} - \frac{M_{i-1}h_i^2}{6} \right) \frac{(x_i - x)}{h_i} + \left(f_i - \frac{M_i h_i^2}{6} \right) \frac{(x - x_{i-1})}{h_i} \end{aligned}$$

(4) dan hosila olamiz:

$$S_3'(x) = -M_{i-1} \frac{(x_i - x)^2}{2h_i} + M_i \frac{(x - x_{i-1})^2}{2h_i} +$$

$$+\frac{f_i - f_{i-1}}{h_i} - \frac{M_i - M_{i-1}}{6}h_i$$

(5) ni $[x_i, x_{i+1}]$ kesma uchun yozamiz:

$$\begin{aligned} S'_3(x) = & -M_i \frac{(x_{i+1} - x)^2}{2h_{i+1}} + M_{i+1} \frac{(x - x_i)^2}{2h_{i+1}} + \\ & + \frac{f_{i-1} - f_i}{h_{i+1}} - \frac{M_{i+1} - M_i}{6}h_{i+1} \end{aligned}$$

Ta'rifning ikkinchi shartiga ko'ra $S'_3(x)$ va $S''_3(x)$ funksiyalar $[a, b]$ da uzluksiz. $S'_3(x)$ ning $x_1, x_2, \dots, x_{n-1}$ nuqtalarda uzluksizligidan foydalanib, (5) va (6) tengliklardan quyidagi $n - 1$ ta tenglamaga ega bo'lamiz:

$$\frac{h_i}{6}M_{i-1} + \frac{h_i + h_{i+1}}{3}M_i + \frac{h_{i+1}}{6}M_{i+1} = \frac{f_{i+1} - f_i}{h_{i+1}} - \frac{f_i - f_{i-1}}{h_i}.$$

Bu tenglamalarni (1) bilan to'ldirib hamda

$$a_i = \frac{h_i}{6}, \quad b_i = \frac{h_i + h_{i+1}}{3}, \quad c_i = \frac{h_{i+1}}{6}, \quad d_i = \frac{f_{i+1} - f_i}{h_{i+1}} - \frac{f_i - f_{i-1}}{h_i}$$

belgilashlarni kiritdik, u holda $M_1, M_2, \dots, M_{n-1}$ noma'lumlarni topish uchun

$$\left. \begin{aligned} b_1 M_1 + c_1 M_2 &= d_1 \\ a_2 M_1 + b_2 M_2 + c_2 M_3 &= d_2 \\ a_3 M_2 + b_3 M_3 + c_3 M_4 &= d_3 \\ \dots \quad \dots \quad \dots \quad \dots & \\ \dots \quad \dots & \\ a_{n-2} M_{n-3} + b_{n-2} M_{n-2} + c_{n-2} M_{n-1} &= d_{n-2} \\ a_{n-1} M_{n-2} + b_{n-1} M_{n-1} &= d_{n-1} \end{aligned} \right\}$$

tenglamalar sistemasini hosil qilamiz. Bu tenglamalar sistemasining matritsasi (8) ga ko'ra salmoqli bosh diagonalga ega bo'lganligidan

ixtiyoriy $f_i(i = \overline{0, n})$ uchun yagona yechimga ega [1]. (9) tenglamalar sistemasi haydash usuli bilan yechiladi. Uni quyida keltiramiz:

$$p_k = a_k q_{k-1} + b_k (q_0 = 0), \quad q_k = -\frac{c_k}{p_k},$$

$$u_k = \frac{d_k - a_k u_{k-1}}{p_k}, \quad u_0 = 0 (k = \overline{1, n-1})$$

yordamchi miqdorlarni hosil qilamiz. So'ng

$$M_k = q_k M_{k+1} + u_k (k = \overline{n-2, 1}), \quad M_{n-1} = u_{n-1}$$

dan $M_{n-1}, M_{n-2}, \dots, M_1$ larni aniqlaymiz.

Misol. Quyida $f(x)$ funksiya jadval ko'rinishida berilgan. Interpolyatsion kubik splayn quring va funksiyaning $x = 0, 22; 0, 25; 0, 28; 0, 31; 0, 35$ dagi qiymatlarini hisoblang.

i	0	1	2	3	4	5
x_i	0,2	0,24	0,27	0,30	0,32	0,38
$f(x_i)$	1,2214	1,2710	1,3100	1,3499	1,3771	1,4623

$$2M_0 - 0,1M_1 = 2,5699, \quad 0,3M_4 + 2M_5 = 3,3378.$$

Bu yerda $M_i = S_3''(x_i)$

Yechish. $h_1 = 0,04; h_2 = 0,03; h_3 = 0,02; h_4 = 0,02; h_5 = 0,06$ ekanligi jadvaldan ko'rinib turibdi. $M_i(i = \overline{0, 5})$ larni qiymatlarini haydash usuli bilan topamiz: $M_0 = 1,387; M_1 = 2,032; M_2 = 0,670; M_3 = 1,289; M_4 = 1,550; M_5 = 1,436$.

Endi har bir oraliq uchun kubik splaynlarni quramiz.

1. $i = 1$, ya'ni $[0, 2; 0, 24]$ oraliq uchun:

$$S_3(x) = M_0 \frac{(x_1 - x)^3}{6h_1} + M_1 \frac{(x - x_0)^3}{6h_1} +$$

$$+ \left(f_0 - \frac{M_0 h_1^2}{6} \right) \frac{x_1 - x}{h_1} + \left(f_1 - \frac{M_1 h_1^2}{6} \right) \frac{x - x_0}{h_4} \approx$$

$$\begin{aligned} & \frac{1,387}{0,24}(0,24-x)^3 + \frac{2,032}{0,24}(x-0,2)^3 + \\ & + \left(1,2214 - \frac{1,387 \cdot 0,04^2}{6}\right) \frac{0,24-x}{0,04} + \\ & + \left(1,2712 - \frac{2,032 \cdot 0,04^2}{6}\right) \frac{x-0,2}{0,04}, \end{aligned}$$

$$S_3(x) \approx 2,688x^3 - 0,919x^2 + 1,257x + 0,985.$$

$$S_3(0,22) \approx 2,688 \cdot 0,22^3 - 0,919 \cdot 0,22^2 + 1,257 \cdot 0,22 + 0,985.$$

$$S_3(0,22) \approx 0,029 - 0,044 + 0,277 + 0,985 \approx 1,24654.$$

2. $i = 2$, ya'ni $[0,24; 0,27]$ oraliq uchun:

$$\begin{aligned} S_3(x) &= M_1 \frac{(x_2 - x)^3}{6h_2} + M_2 \frac{(x - x_1)^3}{6h_2} + \\ &+ \left(f_1 - \frac{M_1 h_2^2}{6}\right) \frac{x_2 - x}{h_2} + \left(f_2 - \frac{M_2 h_2^2}{6}\right) \frac{x - x_1}{h_2} \approx \\ &\approx 11,289 \cdot (0,27 - x)^3 + 3,722 \cdot (x - 0,24)^3 + \\ &+ 42,363 \cdot (0,27 - x) + 43,663 \cdot (x - 0,24) \\ S_3(x) &\approx -7,567x^3 + 6,464x^2 - 0,526x + 1,13 \end{aligned}$$

3. $i = 3$, ya'ni $[0,27; 0,30]$ oraliq uchun:

$$\begin{aligned} S_3(x) &= M_2 \frac{(x_3 - x)^3}{6h_3} + M_3 \frac{(x - x_2)^3}{6h_3} + \\ &+ \left(f_2 - \frac{M_2 h_3^2}{6}\right) \frac{x_3 - x}{h_3} + \left(f_3 - \frac{M_3 h_3^2}{6}\right) \frac{x - x_2}{h_3} \approx \\ &\approx \frac{0,67}{60,03}(0,3 - x)^3 + \frac{1,289}{60,03}(x - 0,27)^3 + \end{aligned}$$

$$+ \left(1, 31 - \frac{0,67 \cdot 0,03^2}{6}\right) \frac{0,3 - x}{0,03} +$$

$$+ \left(1, 3499 - \frac{1,289 \cdot 0,03^2}{6}\right) \frac{x - 0,27}{0,03}$$

$$S_3(x) \approx 3,439x^3 - 2,45x^2 + 1,888x + 0,911.$$

$$S_3(0,28) \approx 3,439 \cdot 0,28^3 - 2,45 \cdot 0,28^2 + 1,888 \cdot 0,28 + 0,911$$

$$S_3(0,28) \approx 0,075 - 0,192 + 0,529 + 0,911 \approx 1,3226$$

4. $i = 4$, ya'ni $[0, 3; 0, 32]$ oraliq uchun:

$$S_3(x) = M_3 \frac{(x_4 - x)^3}{6h_4} + M_4 \frac{(x - x_3)^3}{6h_4} +$$

$$+ \left(f_3 - \frac{M_3 h_4^2}{6}\right) \frac{x_4 - x}{h_4} + \left(f_4 - \frac{M_4 h_4^2}{6}\right) \frac{x - x_3}{h_4} \approx$$

$$\approx \frac{1,289}{60,02} (0,32 - x)^3 + \frac{1,55}{60,02} (x - 0,3)^3 +$$

$$+ \left(1,3499 - \frac{1,289 \cdot 0,02^2}{6}\right) \frac{0,32 - x}{0,02} +$$

$$+ \left(1,3771 - \frac{1,55 \cdot 0,02^2}{6}\right) \frac{x - 0,3}{0,02}.$$

$$S_3(x) \approx 2,175 \cdot x^3 - 1,313 \cdot x^2 + 1,547 \cdot x + 0,945$$

$$S_3(0,31) \approx 2,175 \cdot 0,31^3 - 1,313 \cdot 0,31^2 + 1,547 \cdot 0,31 + 0,945$$

$$S_3(0,31) \approx 0,065 - 0,126 + 0,48 + 0,945 \approx 1,3636$$

5. $i = 5$, ya'ni $[0, 32; 0, 38]$ oraliq uchun:

$$S_3(x) = M_4 \frac{(x_5 - x)^3}{6h_5} + M_5 \frac{(x - x_4)^3}{6h_5} +$$

$$\begin{aligned}
& \left(f_4 - \frac{M_4 h_5^2}{6} \right) \frac{x_5 - x}{h_5} + \left(f_5 - \frac{M_5 h_5^2}{6} \right) \frac{x - x_4}{h_5} \approx \\
& \approx \frac{1,55}{6 \cdot 0,06} (0,38 - x)^3 + \frac{1,436}{60,06} (x - 0,32)^3 + \\
& + \left(1,3771 - \frac{1,550,06^2}{6} \right) \frac{0,38 - x}{0,06} + \\
& + \left(1,4623 - \frac{1,4360,06^2}{6} \right) \frac{x - 0,32}{0,06},
\end{aligned}$$

$$S_3(x) \approx -0,317 \cdot x^3 + 1,08 \cdot x^2 + 0,781 \cdot x + 1,027$$

$$S_3(0,35) \approx -0,317 \cdot 0,35^3 + 1,08 \cdot 0,35^2 + 0,781 \cdot 0,35 + 1,027,$$

$$S_3(0,35) \approx -0,014 + 0,132 + 0,273 + 1,027 \approx 1,4184.$$

Mavzuni mustahkamlash uchun misollar

$f(x)$ funksiyaning qiymatlari quyidagi jadvallarda berilgan. Kubik interpolyatsion splayn quring va ko'rsatilgan nuqtalarda funksiyaning qiymatini hisoblang. Hisoblashlarda $M_i = S_3''(x_i)$ deb hisoblang.

1.	i	0	1	2	3	4	5
	x_i	0,3	0,12	0,21	0,27	0,33	0,35
	$f(x_i)$	1,12	1,16	1,2	1,25	1,32	0,34

$$2M_0 + M_1 = 3,22, \quad 0,5M_4 + 2M_5 = 3,36, \quad x = 0,1.$$

2.	i	0	1	2	3	4	5
	x_i	0,1	0,15	0,19	0,25	0,28	0,30
	$f(x_i)$	1,11	1,16	1,21	1,28	1,32	0,35

$$2M_0 + M_1 = 3,3722, \quad 0,5M_4 + 2M_5 = 3,3614, \quad x = 0,20.$$

3.

i	0	1	2	3	4	5
x_i	0,2	0,24	0,26	0,29	0,32	0,38
$f(x_i)$	1,22	1,27	1,3	1,37	1,38	1,46

$$2M_0 + 0,1M_1 = 2,5699, \quad 0,3M_4 + 2M_5 = 3,3378, \quad x = 0,31$$

4.

i	0	1	2	3	4	5
x_i	0,3	0,32	0,4	0,45	0,5	0,6
$f(x_i)$	1,14	1,12	1,69	1,64	1,75	1,46

$$2M_0 + 0,1M_1 = 2,5, \quad 0,3M_4 + 2M_5 = 3,5, \quad x = 0,5$$

5.

i	0	1	2	3	4	5
x_i	0,1	0,13	0,17	0,20	0,25	0,28
$f(x_i)$	0,1	0,13	0,17	0,2	0,25	0,28

$$2M_0 + 0,5M_1 = -0,26; \quad 0,4M_4 + 2M_5 = -0,66; \quad x = 0,15$$

6.

i	0	1	2	3	4	5
x_i	0,1	0,15	0,17	0,2	0,25	0,28
$f(x_i)$	0,1	0,12	0,16	0,2	0,24	0,64

$$2M_0 + 0,5M_1 = -0,44, \quad 0,4M_4 + 2M_5 = -0,8, \quad x = 0,15$$

7.

i	0	1	2	3	4	5
x_i	0,1	0,15	0,18	0,22	0,28	0,30
$f(x_i)$	1,11	1,16	1,2	1,25	1,32	0,35

$$2M_0 + M_1 = 3,3722, \quad 0,5M_4 + 2M_5 = 3,3614, \quad x = 0,16$$

8.

i	0	1	2	3	4	5
x_i	0,3	0,35	0,4	0,45	0,5	0,6
$f(x_i)$	1,52	1,16	1,72	1,24	1,32	0,34

$$2M_0 + M_1 = 3,22, \quad 0,5M_4 + 2M_5 = 3,36, \quad x = 0,55$$

9.

i	0	1	2	3	4	5
x_i	0,2	0,24	0,27	0,30	0,32	0,38
$f(x_i)$	1,22	1,27	1,31	1,35	1,38	1,46

$$2M_0 + 0,1M_1 = 2,56, \quad 0,3M_4 + 2M_5 = 3,34. \quad x = 0,26$$

10.

i	0	1	2	3	4	5
x_i	0,3	0,34	0,37	0,4	0,42	0,48
$f(x_i)$	1,14	1,27	1,3	1,34	1,7	1,46

$$2M_0 + 0,1M_1 = 2,56, \quad 0,3M_4 + 2M_5 = 3,78. \quad x = 0,45$$

11.

i	0	1	2	3	4	5
x_i	0,5	0,55	0,6	0,7	0,75	0,8
$f(x_i)$	1,2	1,3	1,4	1,5	1,32	0,34

$$2M_0 + M_1 = 3, \quad 0,5M_4 + 2M_5 = 3, \quad x = 0,65.$$

12.

i	0	1	2	3	4	5
x_i	1	1,5	2	3	3,5	4
$f(x_i)$	5	6	7	2	8	3

$$2M_0 + M_1 = 2, \quad 0,5M_4 + 2M_5 = 1, \quad x = 2,5.$$

13.

i	0	1	2	3	4	5
x_i	0,2	0,24	0,26	0,29	0,32	0,38
$f(x_i)$	1,22	1,27	1,27	1,34	1,38	1,46

$$2M_0 + 0,1M_1 = 2,5699, \quad 0,3M_4 + 2M_5 = 3,3378, \quad x = 0,31$$

14.

i	0	1	2	3	4	5
x_i	3	3,2	4	4,5	5	6
$f(x_i)$	1	12	69	64	75	46

$$2M_0 + 0,1M_1 = 5, \quad 0,3M_4 + 2M_5 = 3, \quad x = 6,5$$

15.

i	0	1	2	3	4	5
x_i	3	5	6	7	9	10
$f(x_i)$	8	12	16	7	24	64

$$2M_0 + 5M_1 = 2, \quad 4M_4 + 2M_5 = -3, \quad x = 4$$

16.

i	0	1	2	3	4	5
x_i	0.10	0.30	0.40	0.60	0.70	0.80
$f(x_i)$	0.25	0.50	0.65	0.55	0.42	0.30

$$2M_0 + M_1 = 4, \quad 4M_4 + M_5 = 8, \quad x = 0,5$$

17.

i	0	1	2	3	4	5
x_i	-2.00	-1.80	-1.70	-1.60	-1.40	-1.30
$f(x_i)$	5.10	4.00	3.20	3.90	4.80	6.10

$$M_0 + M_1 = 3, \quad 5M_4 + M_5 = 1, \quad x = -1,5$$

18.

i	0	1	2	3	4	5
x_i	1.30	1.40	1.60	1.70	2.00	2.10
$f(x_i)$	2.40	1.80	1.20	1.40	2.30	2.90

$$2M_0 + M_1 = 0, \quad M_4 + M_5 = 1, \quad x = 1,8$$

19.

i	0	1	2	3	4	5
x_i	0.40	0.70	0.90	1.10	1.40	1.60
$f(x_i)$	0.15	0.83	1.65	1.52	0.90	0.31

$$M_0 + M_1 = 2, \quad 3M_4 + 2M_5 = 3. \quad x = 1,5$$

20.

i	0	1	2	3	4	5
x_i	2.00	2.50	2.70	2.90	3.20	3.40
$f(x_i)$	-0.11	-0.81	-1.05	-0.90	-0.23	-0.05

$$2M_0 + M_1 = 2, \quad M_4 + 2M_5 = 3. \quad x = 3$$

21.

i	0	1	2	3	4	5
x_i	1.10	2.00	2.50	2.90	3.50	4.00
$f(x_i)$	0.32	0.05	-0.10	-0.12	0.12	0.27

$$M_0 + M_1 = 2, \quad M_4 + 2M_5 = 1. \quad x = 3$$

22.

i	0	1	2	3	4	5
x_i	0,3	0,5	0,8	0,9	1,2	1,4
$f(x_i)$	1,1	0,6	0,4	0,38	0,65	0,9

$$M_0 + M_1 = 1, \quad M_4 + 2M_5 = 0. \quad x = 1$$

23.

i	0	1	2	3	4	5
x_i	-0,4	-0,1	0,1	0,2	0,5	0,7
$f(x_i)$	1,3	3,5	4,2	4	2,8	1,6

$$2M_0 + M_1 = 0, \quad M_4 + 2M_5 = 1. \quad x = 0,3$$

24.

i	0	1	2	3	4	5
x_i	1.2	1.4	1.5	1.6	1.8	2.1
$f(x_i)$	0.9	3.3	4.1	3.9	2.8	1.1

$$M_0 + 2M_1 = 0, \quad 2M_4 + M_5 = 0. \quad x = 2$$

25.

i	0	1	2	3	4	5
x_i	-0.9	-0.8	-0.5	-0.4	-0.2	-0.1
$f(x_i)$	0.15	0.61	1.2	1.1	0.7	0.22

$$M_0 + M_1 = -1, \quad M_4 - 2M_5 = -3. \quad x = -0,6$$

26.

i	0	1	2	3	4	5
x_i	-1	-0.8	-0.7	-0.4	-0.3	-0.2
$f(x_i)$	1.4	0.9	0.65	0.51	0.78	1.3

$$M_0 - M_1 = 1, \quad M_4 - 2M_5 = 0. \quad x = -0.5$$

27.

i	0	1	2	3	4	5
x_i	0.2	0.3	0.5	0.7	0.9	1.2
$f(x_i)$	-2.1	-0.5	1.15	1.3	-0.6	-2.7

$$M_0 - M_1 = 2, \quad M_4 + M_5 = 1. \quad x = 1$$

28.

i	0	1	2	3	4	5
x_i	2.2	2.5	2.6	2.8	3.1	3.2
$f(x_i)$	1.7	0.8	0.52	0.3	0.91	1.5

$$M_0 - 2M_1 = -3, \quad 2M_4 - M_5 = 0. \quad x = 3$$

29.

i	0	1	2	3	4	5
x_i	-0.3	-0.1	0.2	0.3	0.7	0.9
$f(x_i)$	-2.1	1.3	3	2.4	-2.3	-8

$$2M_0 - M_1 = 2, \quad M_4 - M_5 = -3. \quad x = 0.5$$

30.

i	0	1	2	3	4	5
x_i	1	2	3	7	6	9
$f(x_i)$	11	8	5	9	2	5

$$M_0 + M_1 = 4, \quad M_4 + M_5 = 2. \quad x = 4$$

VII BOB.

ANIQ INTEGRALLARNI TAQRIBIY HISOBLASH

Ma'lumki, ko'pgina funksiyalarning boshlang'ich funksiyalarini natijaviy ko'rinishda yozish imkoni hamma vagt ham mavjud bo'lavermaydi. Shuning uchun bu funksiyalarning aniq integralarini ba'zan taqribiy usullar bilan hisoblash zarurati tug'iladi.

Aniq integrallarni taqribiy hisoblash uchun to'g'ri to'rtburchaklar, trapetsiyalar hamda parabolalar(Simpson) usullari mavjud bo'lib, ular egri chiziqli trapetsiya yuzi haqidagi masalaning geometrik yechimi bilan uzviy bog'liq.

$$\int_a^b f(x)dx$$

aniq integralning taqribiy qiymatini topish kerak bo'ladi.

AB egri chiziqli trapetsiyaning yuzini berilgan integralning geometrik ifodasi sifatida qarab, bu yuzni topamiz.

7.1-§. Aniq integralni to'g'ri to'rtburchak usuli bilan taqribiy hisoblash.

Eng sodda kvadratur formulalar: to'g'ri to'rtburchak, trapetsiya va Simpson formulalari. Eng sodda kvadratur formulalarni oddiy muloxazalar asosida qurish mumkin. Aytaylik,

$$\int_a^b f(x)dx$$

integralni hisoblash talab qilinsin. Agar qaralayotgan oraliqda $f(x) \approx \text{const}$ bo'lsa, u vaqtda

$$\int_a^b f(x)dx \approx (b-a)f\left(\frac{a+b}{2}\right)$$

deb olishimiz mumkin. Bu formula to'g'ri to'rtburchaklar formulasi deyiladi.

To'g'ri to'rtburchaklar usuli algoritmi:

1. $h = (b-a)/n$ orqali bo'linish qadamini hisoblaymiz;
2. $x_0 = a + h/2$, $x_k = x_{k-1} + h$ ($k = 1, 2, 3, \dots$) nuqtalarni belgilab olamiz;
3. $x_1, x_2, \dots, x_n$ nuqtalardan chegaraviy egri chiziqli bilan kesishgunga qadar perpendikulyar o'tkazamiz va kesishish nuqtalarning ordinatalarini quyidagicha

$$y_1 = f(x_1), y_2 = f(x_2), \dots, y_n = f(x_n)$$

belgilaymiz;

4. $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ nuqtalardan chapga qarab gorizontali ravishda mos holda $(x_0, y_1), (x_1, y_2), \dots, (x_{n-1}, y_n)$ nuqtalargacha bo'lgan kesma o'tkazamiz va hosil qilingan har bir to'g'ri to'rtburchak yuzini topamiz:

$$y_1 h, y_2 h, \dots, y_n h$$

5. n ta to'g'ri to'rtburchak yuzini qo'shamiz:

$$S = h(y_1 + y_2 + \dots + y_n)$$

Demak,

$$\int_a^b f(x)dx = h(y_1 + y_2 + \dots + y_n)$$

Bu to'g'ri to'rtburchaklar formulasidir.

To'g'ri to'rtburchaklar formulasining o'ng tomonidagi ifoda berilgan integralga intiladi. Biroq, fiksirlangan h da uning har biri berilgan integraldan $R_n(f)$ kattalikka farq qiladi. Berilgan absolyut xatodan n parameter tanlanadi va shuningdek, h qadam $|R_n(f)| < \varepsilon$ tengsizlikdan topiladi.

$R_n(f)$ kattalik

$$R_n(f) = \frac{b-a}{2n} f'(\zeta) h^2, \zeta \in [a, b]$$

tenglik orqali xarakterlanadi.

Misol. $\int_0^1 \frac{x}{x+1} dx$ integralni $n = 10$ bo'lganda hisoblang.

Yechish.

1. $[0, 1]$ kesmani $h = 0,1$ qadam bilan 10 ta bo'lakka bo'lib olamiz;

2. Quyidagilarni hisoblaymiz:

$$x_0 = 0.05, \quad x_1 = 0.15, \quad \dots \quad x_{10} = 1.05.$$

3. Quyidagilarni hisoblaymiz:

$$y_1 = f(x_1) = 0,130435,$$

$$y_2 = f(x_2) = 0,2,$$

$$\dots,$$

$$y_{10} = f(x_{10}) = 0,512195$$

$$\int_0^1 \frac{x}{x+1} dx = h(y_1 + y_2 + \dots + y_n) =$$

$$= 0.1 * 3,53622247 = 0,1$$

$$\int_a^b f(x)dx$$

aniq integralni to'g'ri to'rtburchaklar usuli yordamida hisoblashning C#dagi dasturi:

```
public static void Main(string[] args)
double h,p,s;
Console.Write("a=");
double a=double.Parse(Console.ReadLine());
Console.Write("b=");
double b=double.Parse(Console.ReadLine());
Console.Write("n=");
int n=int.Parse(Console.ReadLine());
double e=double.Parse(Console.ReadLine());
double [] x=new double[11];
double [] p=new double[10];
h=(b-a)/n;x[0]=a-h/2;p=0;
for (int i=1; i<=n; i++)
\{
x[i]=x[i-1]+h; p+=f(x);
Console.WriteLine("x["+i+"]="+x[i]);
\}
s=h*p;
Console.WriteLine(s);
Console.ReadKey(true);
}
}
}
```

Izoh. $f(x)$ funksiyani C# dagi yozuvini yozing.

Mavzuni mustahkamlash uchun misollar.

Quyidagi integrallarni to'g'ri to'rtburchak usuli bilan $\varepsilon = 0,01$ aniqlikda taqribiy hisoblang.

$$1. \int_{0,5}^{1,5} \frac{\sqrt{x^2 + 0,5}}{\sqrt{x^2 - 1 + 4x}} dx, \quad (n = 10).$$

$$2. \int_2^9 \sqrt{6x^2 - 3x + 4} dx, \quad (n = 8).$$

$$3. \int_{1,3}^{2,3} \frac{\sqrt{0,5x^2 + 1,5}}{\sqrt{0,6x^2 + 1,3 + 1,2}} dx, \quad (n = 20).$$

$$4. \int_{0,4}^{1,4} \frac{\sqrt{3,1x^2 + 4}}{\sqrt{0,5x^3 + 1 + 1,4}} dx, \quad (n = 6).$$

$$5. \int_{0,7}^{1,7} \frac{2x^2 + 4}{\sqrt{x + 2 + 3x^2}} dx, \quad (n = 8).$$

$$6. \int_{0,5}^{2,5} \frac{\sqrt{x^2 + 2x}}{\sqrt{4x + 1 + 2x^2}} dx, \quad (n = 10).$$

$$7. \int_0^1 \frac{1}{\sqrt{2x^3 - 3x^2 + 1}} dx, \quad (n = 10).$$

$$8. \int_{0,3}^{1,3} \frac{1}{\sqrt{2x^2 + 1,8x + 1}} dx, \quad (n = 12).$$

$$9. \int_{1,2}^{2,2} \frac{\sqrt{x+1}}{\sqrt{2x^2+1+2}} dx, \quad (n=10).$$

$$10. \int_1^2 \frac{3x+2,1}{\sqrt{2x^2+4}} dx, \quad (n=1).$$

$$11. \int_{1,1}^{2,1} \frac{1}{\sqrt{2x^2+0,6}} dx, \quad (n=10).$$

$$12. \int_{0,4}^{1,4} \frac{\sin(x+1)}{2+\cos(x^2+1)} dx, \quad (n=10).$$

$$13. \int_{0,4}^{1,2} \frac{\cos(x^2+0,4)}{0,3+\sin(x+1)} dx, \quad (n=8).$$

$$14. \int_{0,2}^{0,8} \frac{\sin(0,4x^2+0,1)}{0,2+\sin(x+0,5)} dx, \quad (n=6).$$

$$15. \int_{0,5}^{1,5} \frac{\cos(0,4x+0,6)}{0,4+3\sin(0,5x+1)} dx, \quad (n=10).$$

$$16. \int_{0,2}^{1,2} (3x+1) \sin x dx, \quad (n=10).$$

$$17. \int_{0,4}^{0,8} \frac{\sin(x^2 + 0,5)}{2x^2 + 1} dx, \quad (n = 6).$$

$$18. \int_{0,5}^{1,3} \frac{1}{\cos x + \sqrt{x^2 + 1}} dx, \quad (n = 8).$$

$$19. \int_{0,5}^{1,4} \sqrt{x + 1} \cos x dx, \quad (n = 12).$$

$$20. \int_{0,4}^{1,8} (2 + 3x) \sin(x^2 + 1) dx, \quad (n = 14).$$

$$21. \int_{0,5}^{2,4} x^2 \lg x dx, \quad (n = 10).$$

$$22. \int_{1,2}^{1,4} \frac{\ln(x^2 + 3)}{x^2 + 1} dx, \quad (n = 8)$$

$$23. \int_{0,5}^{1,5} \frac{\operatorname{tg}(x^2 + 1)}{x + 1} dx, \quad (n = 6).$$

$$24. \int_0^{\frac{\pi}{2}} \sqrt{1 - \frac{1}{2} \sin x^2} dx, \quad (n = 12).$$

$$25. \int_{0,5}^2 (\lg x^2 + 2, 8) dx, \quad (n = 10).$$

$$26. \int_{0,5}^2 \left(\frac{x}{4} + 1 \right) \cos \frac{x}{4} dx, \quad (n = 14).$$

$$27. \int_{1,2}^{2,4} \frac{x}{4} \ln \left(\frac{x}{4} + 2 \right) dx, \quad (n = 12).$$

$$28. \int_{0,6}^{1,6} (\sqrt{x} + 1) \sin 2x dx, \quad (n = 10).$$

$$29. \int_{0,5}^{1,5} \frac{\sqrt[3]{x+1} e^{x+1}}{\sin(x^2+1)} dx, \quad (n = 12).$$

$$30. \int_{0,5}^{1,3} \frac{\sqrt{x^2+1}}{\sin x} dx, \quad (n = 10)$$

7.2-§. Aniq integrallarni trapetsiyalar usuli bilan taqribiy hisoblash

Faraz qilaylik, $f(x)$ funktsiya chiziqli funktsiyaga yaqin bo'lsin, u holda tabiiy ravishda integralni balandligi $b - a$ ga va asoslari $f(a)$ va $f(b)$ ga teng bo'lgan trapetsiya yuzi bilan almashtirish mumkin, u holda

$$\int_a^b f(x) dx \approx \frac{b-a}{2} (f(a) + f(b))$$

deb olishimiz mumkin. Bu formula trapetsiya formulasi deyiladi.

Trapetsiyalar usuli algoritmi $[a, b]$ kesmani

$$a = x_0, x_1, x_2, \dots, x_n = b$$

nuqtalar bilan n ta teng bo'lakka bo'lamiz. Har bir qo'shni bo'luvchi nuqtalar orasidagi masofa $h = (b - a)/n$;

$[a, b]$ kesmani bo'luvchi nuqtalardan chegaraviy egri chiziq bilan keishgunga qadar perpendikulyar o'tkazamiz. Egri chiziq mos nuqtalarining ordinatalari quyidagicha bo'ladi:

$$y_0 = f(x_0), y_1 = f(x_1), y_2 = f(x_2), \dots, y_n = f(x_n)$$

perpendikulyarlarning $y = f(x)$ chiziq bilan kesishgan qo'shni nuqtalarini vatarlar bilan birlashtiramiz va hosil qilingan har bir to'g'ri chiziqli trapetsiyalarning yuzini topamiz:

$$\frac{y_0 + y_1}{2} h, \frac{y_1 + y_2}{2} h, \dots, \frac{y_n + y_{n-1}}{2} h$$

Barcha n ta trapetsiya yuzini qo'shamiz:

$$S = h \left(\frac{y_0}{2} + y_1 + y_2 + \dots + \frac{y_n}{2} \right)$$

bo'linish qadami $h = (b - a)/n$ ekanligi va hosil bo'lgan yig'indi inobatga olinsa, egri chiziqli trapetsiyaning yuzini taqriban quyidagicha yozish mumkin:

$$\int_a^b f(x) dx \approx h \left(\frac{y_0}{2} + y_1 + y_2 + \dots + \frac{y_n}{2} \right)$$

Bu trapetsiya formulasidir.

Trapetsiya formulasining o'ng tomonidagi ifoda integral yig'indisidir va $h \rightarrow 0$ da berilgan integralga intiladi. Biroq, fiksirlangan h da uning har biri berilgan integraldan $R_n(f)$ kattalikka farq qiladi.

Berilgan $\varepsilon > 0$ absolyut xatodan n parameter tanlanadi va shuningdek, h qadam $|R_n(f)k| < \varepsilon$ tengsizlikdan topiladi.

$R_n(f)$ kattalik

$$R_n(f) = \frac{b-a}{12} f''(\xi) h^2, \xi \in [a, b]$$

tenglik orqali xarakterlanadi.

Misol. $\int_0^1 (x+1)dx$ integralni trapetsiyalar usulida ($n = 10$)

taqribiy hisoblang.

Yechish.

1. $[0, 1]$ kesmani $h = 0,1$ qadam bilan 10 ta bo'lakka bo'lib olamiz;

2. Quyidagilarni hisoblaymiz:

$$x_0 = 0, \quad x_1 = 0.1, \quad \dots \quad x_{10} = 1.$$

3. Quyidagilarni hisoblaymiz:

$$y_0 = f(x_0) = 1, \quad y_1 = f(x_1) = 1.1,$$

$$y_2 = f(x_2) = 1.2, \quad \dots, \quad y_{10} = f(x_{10}) = 2$$

$$\int_0^1 (x+1)dx = h \left(\frac{y_0 + y_n}{2} + y_1 + \dots + y_{n-1} \right) =$$

$$= 0.1 * 15 = 1.5$$

$\int_a^b f(x)dx$ aniq integralni trapetsiya usuli yordamida hisoblashning C# dagi dasturi.

```
using System;
namespace Integrallarnitaqribiyhisoblash
\{
class Program
\{
public static void Main(string[] args)
\{
double h,p,m,k,s;
Console.Write("a=");
double a=double.Parse(Console.ReadLine());
Console.Write("b=");
double b=$ double.Parse(Console.ReadLine());
Console.Write(" ${n}=$ ");
int n=int.Parse(Console.ReadLine());
double e=double.Parse(Console.ReadLine());
double [] x=new double[10];
double [] p=new double[10];
${h}=(b-a) / n ; x[0]=a ; p=0$;
for (int i=1 ; i<=n ; i++)$
\{
x[i]=x[i-1]+h ; p+=f(x) ;
Console.WriteLine(x[i]);
\}
m=1 / (1+a) ; k=1 / (1+b) ;$
s=h*(((m+k) / 2+p)) ;$
Console.WriteLine(s);
Console.ReadKey(true);
```

Izoh. $f(x)$ funksiyaning C# dagi ko'rishini yozing

Mavzuni mustahkamlash uchun misollar.

Quyidagi integrallarni trapetsiya usuli bilan integrallash oraliqini n ta teng bo'lakka bo'lib, taqribiy hisoblang.

$$1. \int_{0,5}^1 \frac{1,2 - x^2}{\sqrt{0,4x^2 + 2x}} dx, \quad (n = 10).$$

$$2. \int_{1,4}^{2,6} \frac{\sqrt{1,5x^2 + 3}}{\sqrt{0,5x^2 + x}} dx, \quad (n = 12).$$

$$3. \int_{0,6}^{2,6} \frac{\sqrt{3x + 1,4}}{\sqrt{1,2x^2 + 1,4}} dx, \quad (n = 20).$$

$$4. \int_{0,4}^{1,8} \frac{x^2 + 4x + 1}{\sqrt{x^2 + x}} dx, \quad (n = 16).$$

$$5. \int_{0,5}^{2,5} \frac{x^2 + 1}{\sqrt{4x + 1}} dx, \quad (n = 20).$$

$$6. \int_{0,2}^{1,4} \frac{\sqrt{x + 1}}{\sqrt{x^2 + 1}} dx, \quad (n = 12).$$

$$7. \int_{0,9}^{1,3} \frac{1}{\sqrt{3x^2 + 4}} dx, \quad (n = 10)$$

$$8. \int_{0,1}^{2,1} \frac{2x + 1}{\sqrt[3]{3x^2 + 1}} dx, \quad (n = 20).$$

$$9. \int_0^1 \frac{3x+1}{\sqrt[4]{2x^2+4}} dx, \quad (n=10).$$

$$10. \int_{0,4}^{1,6} \frac{2x^2+1}{\sqrt[3]{2x+1}} dx, \quad (n=14).$$

$$11. \int_{0,2}^{1,2} \frac{\sin(0,4x+0,6)}{1,2+\cos(x^2+0,4)} dx \quad (n=10).$$

$$12. \int_{0,4}^{2,4} \frac{\cos(x+2,6)}{1,4+\sin(x^2+4,2)} dx \quad (n=10).$$

$$13. \int_{0,6}^{2,6} \frac{\sin(x+1)+1,2}{\cos(x^2+4,2)} dx \quad (n=20).$$

$$14. \int_{0,4}^{1,4} \frac{\sin(0,5x^2+4,2)}{4,2+2\cos(x^2+1)} dx \quad (n=10).$$

$$15. \int_{1,2}^{2,4} \frac{1}{(x+1,2)\sin(2x^2+4)} dx \quad (n=12).$$

$$16. \int_{1,2}^{1,7} (x^2+1)\cos x dx \quad (n=10).$$

$$17. \int_{3,4}^{4,6} \frac{\sin x}{\sqrt{x^2 + 3}} dx \quad (n = 12).$$

$$18. \int_{1,2}^{2,6} (2, 4x + 1, 4) \lg(x + 2) dx \quad (n = 14).$$

$$19. \int_{0,2}^{1,4} (x^2 + 1) \cos(x^3 + 0, 4) dx \quad (n = 10).$$

$$20. \int_{1,4}^{2,6} \frac{\lg(x^2 + 3)}{2x + 1} dx \quad (n = 12).$$

$$21. \int_{0,12}^{0,64} \sqrt{x + 1} \sin(x + 3) dx \quad (n = 12).$$

$$22. \int_{1,5}^{2,5} \frac{\operatorname{tg}(x^2 + 1)}{x + 1} dx \quad (n = 10).$$

$$23. \int_{2,2}^{3,4} \frac{\sqrt{x + 1}}{\sin(2x^2 - 2)} dx \quad (n = 12).$$

$$24. \int_{1,2}^{2,4} (x^2 + 1) \ln(x + 1) dx \quad (n = 12).$$

$$25. \int_{0,6}^{1,4} (x+1,2)e^{x+1} dx \quad (n=8).$$

$$26. \int_{0,12}^{0,34} \frac{x^2+1}{\sqrt{x^2+4}} dx \quad (n=20).$$

$$27. \int_{0,4}^{0,8} \frac{1}{\sin(x^2+1)e^x} dx \quad (n=12).$$

$$28. \int_0^{0,6} \left[\frac{x}{2} + 2(\lg x^2 + 4,2) \right] dx \quad (n=6).$$

$$29. \int_{0,4}^{1,4} \frac{x}{2} \operatorname{tg}(x^2+2,4x) dx \quad (n=10).$$

$$30. \int_0^{1,4} (x^2+1) \sin(x+3,6) dx \quad (n=12).$$

7.3-§. Aniq integralni parabolalar (Simpson) usuli bilan taqribiy hisoblash

Nixoyat, $f(x)$ funktsiya $[a, b]$ oraliqda kvadratik funktsiyaga yaqin bo'lsin, u holda $\int_a^b f(x) dx$ ni taqribiy ravishda Ox o'qi va $x = a$, $x = b$ to'g'ri chiziqlar hamda $y = f(x)$ funktsiya

grafining abstsissalari $x = a$, $x = \frac{a+b}{2}$ va $x = b$ bo'lgan nuqtalaridan o'tuvchi ikkinchi tartibli parabola orqali chegaralangan yuza bilan almashtirish mumkin, u holda quyidagiga ega bo'lamiz:

$$\int_a^b f(x)dx \approx \frac{b-a}{2} \left\{ f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right\}.$$

$[a, b]$ kesma uzunligi $h = (b-a)/(2n)$ bo'lgan $2n$ ta juft bo'lakka $x_1, x_2, \dots, x_{2n-1}$ nuqtalar orqali ajratamiz;

Uzunligi $(b-a)/(2n)$ bo'lgan kichik kesmalarni qaraymiz:

$$[x_0; x_2], [x_2; x_4], \dots, [x_{2n-2}; x_{2n}] \quad (a = x_0, \quad b = x_{2n})$$

Bu kesmalarning o'rtalari mos ravishda $x_1, x_3, \dots, x_{2n-1}$ nuqtalar bo'ladi.

$$\int_a^b f(x)dx \text{ integralni bir nechta integral yig'indiga ajratamiz.}$$

$$\int_a^b f(x)dx \approx \int_{x_0}^{x_2} f(x)dx + \int_{x_2}^{x_4} f(x)dx + \dots + \int_{x_{2n-2}}^{x_{2n}} f(x)dx$$

$$[x_{2i}; x_{2i+2}], i = \overline{0, 2n-1} \text{ kesmalarda}$$

$$(x_{2i}, y_{2i}), (x_{2i+1}, y_{2i+1}), (x_{2i+2}, y_{2i+2})$$

nuqtalar orqali parabolalar o'tkazamiz. Mazkur uch nuqta orqali hamma vaqt parabola o'tkazish mumkin, shu bilan birga bunday parabola (x_{2i}, y_{2i+1}) kesmada faqat bitta bo'ladi. Yordamchi parabola bilan chegaralangan egri chiziqli trapetsiya yuzi taqriban

berilgan egri chiziqli trapetsiya yuziga teng.

$$\int_{x_{2i}}^{x_{2i+2}} f(x) dx = \int_{x_{2i}}^{x_{2i+2}} (Ax^2 + Bx + C) dx$$

$$\int_{x_{2i}}^{x_{2i+2}} (Ax^2 + Bx + C) dx = \frac{x_{2i+2} - x_{2i}}{2} (y_{chap} + 4y_{o'rta} + y_{o'ng})$$

Demak, shunga asosan

$$\int_{x_0}^{x_2} f(x) dx = \frac{b-a}{6} (y_0 + 4y_1 + y_2)$$

$$\int_{x_2}^{x_4} f(x) dx = \frac{b-a}{6} (y_2 + 4y_3 + y_4)$$

.....

$$\int_{x_{2n-2}}^{x_{2n}} f(x) dx = \frac{b-a}{6} (y_{2n-2} + 4y_{2n-1} + y_{2n})$$

hosil qilingan munosabatlarning chap va o'ng tomonlarini mos ravishda qo'shamiz:

$$\int_a^b f(x) dx \approx \frac{b-a}{6n} [(y_0 + y_{2n}) +$$

$$+ 2(y_2 + y_4 + \dots + y_{2n-2}) + 4(y_1 + y_3 + \dots + y_{2n-1})]$$

Oxirgi formula parabolalar yoki Simpson formulasi deyiladi. Simpson formulasining o'ng tomonidagi ifoda integral yig'indidir

va $h \Rightarrow 0$ da berilgan integralga intiladi. Biroq, fiksirlangan h da uning har biri berilgan integraldan $R_n(f)$ kattalikka farq qiladi. Berilgan $\varepsilon > 0$ absolyut xatodan n parameter tanlanadiva shuningdek, h qadam $|R_n(f)| < \varepsilon$ tengsizlikda topiladi.

$R_n(f)$ kattalik

$$R_n(f) = -\frac{b-a}{180} f^{(IV)}(\xi) h^4, \quad \xi \in [a, b]$$

tenglik orqali xarakterlanadi.

Misol. $\int_0^1 \frac{2x}{1+x^2} dx$ integralni $n = 10$ bo'lganda Simson usuli

lida yeching.

Yechish.

1. $[0, 1]$ kesmani $h = 0.05$ qadam bilan 20 ta bo'lakka bo'lib olamiz;

2. Quyidagilarni hisoblaymiz:

$$x_0 = 0, x_1 = 0.05, \dots, x_{20} = 1.$$

3. Quyidagilarni hisoblaymiz:

$$y_0 = f(x_0) = 0,$$

$$y_1 = f(x_1) = 0.09975,$$

$$y_2 = f(x_2) = 1.198,$$

$$\dots,$$

$$y_{20} = f(x_{20}) = 1$$

$$\int_0^1 \frac{2x}{1+x^2} dx \approx \frac{h}{3} [y_0 + y_n + 2(y_2 + y_4 + \dots + y_{2n-2}) +$$

$$+4(y_1 + y_2 + \dots + y_{2n-1})] = 0.693148$$

$\int_a^b f(x)dx$ aniq integralni Simson usuli yordamida hisoblashning C# dagi dasturi.

```
public static void Main(string[] args)
{
    double h,m,k,l,t,s;
    Console.Write("a=");
    double a=double.Parse(Console.ReadLine());
    Console.Write("b=");
    double b=double.Parse(Console.ReadLine());
    Console.Write("n=");
    int n=int.Parse(Console.ReadLine());
    //double e=double.Parse(Console.ReadLine());
    double [] x=new double[21];
    double [] y=new double[21];
    h=(b-a)/(2*n);x[0]=a;l=0;t=0;
    for (int i=1;i<(2*n);i++)
    {
        x[i]=x[i-1]+h; //p+=1/(1+x[i]);
        Console.WriteLine(x[i]);
    }
    for(int i=2; i < 20; i = i + 2)
    {
        l+=1/(1+x[i]);
    }
    for(int i=1; i <20; i = i + 2)
    {
        t+=1/(1+x[i]);
    }
}
```

```

m=1/(1+a); k=1/(1+b);
s=((b-a)/(6*n))*(m+k+2*1+4*t);
Console.WriteLine(s);
Console.ReadKey(true);
}

```

Mavzuni mustahkamlash uchun misollar.

Quyidagi integrallarni Parabolalar (Simpson) usuli bilan integrallash oraliq'ini $2n$ ta teng bo'lakka bo'lib taqribiy hisoblang.

$$1. \int_{0,2}^{1,2} \frac{\sqrt{x+1,2}}{\sqrt{x^2+1,2x+2,4}} dx, \quad (n=10).$$

$$2. \int_{1,4}^{1,4} \frac{(x^2+1)}{\sqrt{x+1}+2} dx, \quad (n=20).$$

$$3. \int_{0,6}^{1,8} \frac{\sqrt{x+1,4}}{\sqrt{x^2+0,6x+2}} dx, \quad (n=12).$$

$$4. \int_{0,6}^{1,6} \frac{0,5x^2+1}{\sqrt{0,4x^2+1,3x+1,4}} dx, \quad (n=10).$$

$$5. \int_{0,5}^{1,3} \frac{\sqrt{1,2x^2+0,4}}{\sqrt{0,4x^2+1,6x+1,4}} dx, \quad (n=8).$$

$$6. \int_{0,4}^{1,4} \frac{(0,4x^2+1,2)}{\sqrt{0,4x^2+1,6x+1,4}} dx, \quad (n=10).$$

$$7. \int_{1,5}^{2,1} \frac{1,2x^2 + 0,4}{\sqrt{2x^2 + 0,5x + 0,8}} dx, \quad (n = 6).$$

$$8. \int_{0,2}^{1,2} \frac{x^2 + 2}{\sqrt[3]{x^2 + 4}} dx, \quad (n = 10).$$

$$9. \int_{0,2}^{1,4} \frac{x + 4}{\sqrt{x^2 + 1}} dx, \quad (n = 16).$$

$$10. \int_{-2,5}^{1,5} \frac{x}{5\sqrt{x^2 + 1,4}} dx, \quad (n = 10).$$

$$11. \int_{0,4}^{1,4} \frac{x^2}{\sqrt{x + 1,6}} dx, \quad (n = 20).$$

$$12. \int_{0,4}^{2,8} \frac{5 - x}{\sqrt{x^3 + 1}} dx, \quad (n = 16).$$

$$13. \int_{0,6}^{2,6} \frac{\sqrt[3]{x^2 + 1}}{x^2 + 3} dx, \quad (n = 15).$$

$$14. \int_{1,4}^{2,6} \frac{3x + 0,5}{\sin x} dx, \quad (n = 12).$$

$$15. \int_{1,2}^{3,6} \frac{\operatorname{ctg}(x^2 + 1, 5)}{1 + 3x^3} dx, \quad (n = 14).$$

$$16. \int_{2,4}^3 \frac{\sin^2 x}{\sqrt{2x^2 + 1}} dx, \quad (n = 10).$$

$$17. \int_{0,2}^{1,2} \frac{\cos^2 x}{x^2 + 1} dx, \quad (n = 10).$$

$$18. \int_{0,2}^{1,2} (2x + 0, 5) \cos x dx, \quad (n = 20).$$

$$19. \int_{1,14}^{2,24} \frac{1 + \operatorname{tg}^2 x}{\sqrt{3x^2 + 1}} dx, \quad (n = 12).$$

$$20. \int_{0,18}^{0,88} \frac{\sin^2 x}{x^2 + 1} dx, \quad (n = 8).$$

$$21. \int_{0,12}^{0,64} \frac{x \ln x}{x^2 + 1} dx, \quad (n = 10).$$

$$22. \int_0^{\pi/2} \frac{\ln \sin(x + 1)}{x^2 + 1} dx, \quad (n = 8).$$

$$23. \int_0^{\pi/2} \sqrt{1 - \frac{1}{4} \sin 4x} dx, \quad (n = 10).$$

$$24. \int_0^1 e^{-x^2} \sin x dx, \quad (n = 20).$$

$$25. \int_{1,2}^{2,4} \ln(1 + x^2) \sin^2 x dx, \quad (n = 12).$$

$$26. \int_0^{\pi/2} \sqrt{2 + \sin^2 x} \operatorname{tg} x dx, \quad (n = 16).$$

$$27. \int_0^{\pi/2} \frac{\sin x}{x + 1} dx, \quad (n = 8).$$

$$28. \int_0^{\pi/2} \sqrt{x + 1} \ln(x + 1) dx, \quad (n = 10).$$

$$29. \int_{0,3}^{1,3} \sin x^2 \lg(x^2 + 1) dx, \quad (n = 10).$$

$$30. \int_0^1 e^{x^2} \cos^2 x dx, \quad (n = 20).$$

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MUNDARIJA

SO‘Z BOSHI.....	3
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I BOB. XATOLIKLAR NAZARIYASI

1.1-§. Aniq va taqribiy sonlar haqida tushuncha	4
1.2-§. Hisoblash xatosi	7
1.3-§. Funksiya xatoligi.....	11

II BOB. ALGEBRAIK VA TRANSENDENT TENGLAMALARNI YECHISH USULLARI

2.1-§. Umumiy mulohazalar	18
2.2-§. Algebraik va transcendent tenglamalarni oddiy iteratsiya usuli bilan yechish	23
2.3-§. Algebraik va transcendent tenglamalarni urinmalar (Nyuton) usuli bilan yechish	29
2.4-§. Algebraik va transcendent tenglamalarni yarim (teng ikkiga) bo‘lish usuli bilan yechish.....	34

III BOB. CHIZIQLI ALGEBRAIK TENGLAMALAR SISTEMASINI YECHISH USULLARI

3.1-§. Umumiy mulohazalar	40
3.2-§. Chiziqli algebraik tenglamalar sistemasini Gauss usuli bilan yechish	41
3.3-§. Chiziqli tenglamalar sistemasini iteratsiya va Zeydel usuli bilan yechish	49

IV BOB. CHIZIQLI BO‘LMAGAN ALGEBRAIK TENGLAMALAR SISTEMASINI TAQRIBIY HISOBLASH

4.1-§.	Chiziqli bo'lmagan algebraik tenglamalar sistemasini oddiy iteratsiya usuli bilan taqribiy hisoblash.....	60
4.2-§.	Chiziqli bo'lmagan algebraik tenglamalar sistemasini oddiy Nyuton usuli bilan taqribiy hisoblash.....	66

V BOB. MATRITSALARNING XOS SONLARI VA XOS VEKTORLARI TOPISH

5.1-§.	Xarakteristik ko'phad	71
5.2-§.	Matritsalarining xos sonlari va xos vektorlari topishning A.N.Krilov usuli.....	81

VI BOB. FUNKSIYALARNI INTERPOLYATSIIYASLASH

6.1-§.	Lagranj interpolyatsion ko'phadi.....	95
6.2-§.	Nyuton interpolyatsion ko'phadi.....	101
6.3-§.	Splayn-yaqinlashtirish. Splayn interpolyasiya.....	107

VII BOB.ANIQ INTEGRALLARNI TAQRIBIY HISOBLASH

7.1-§.	Aniq integralni to'g'ri to'rtburchak usuli bilan taqribiy hisoblash	121
7.2-§.	Aniq integrallarni trapetsiyalar usuli bilan taqribiy hisoblash	128
7.3-§.	Aniq integralni parabolalar (Simpson) usuli bilan taqribiy hisoblash	135

FOYDALANILGAN ADABIYOTLAR	144
--	------------

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